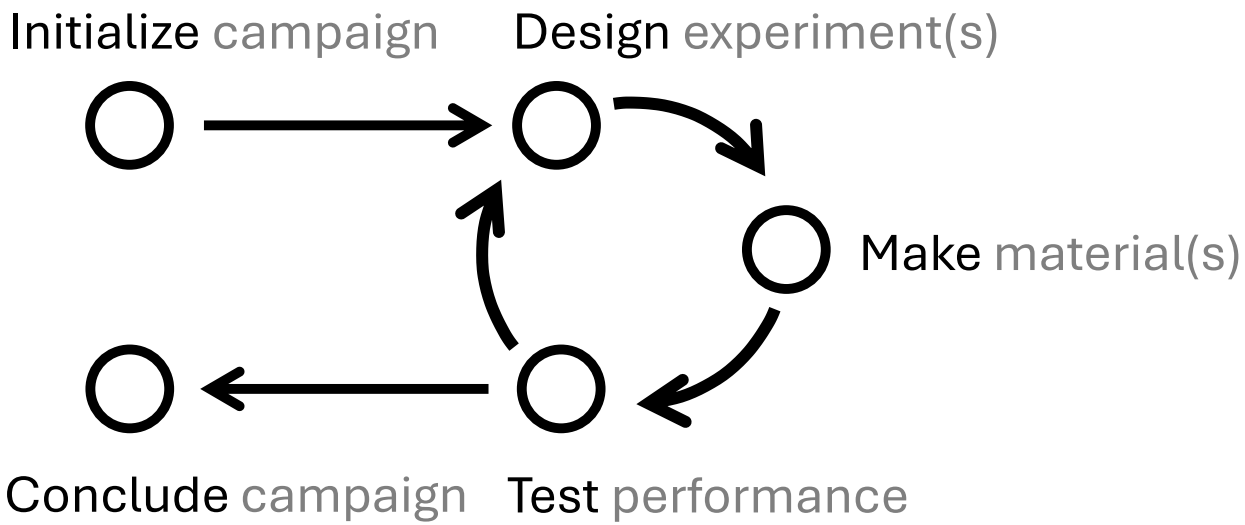
So, you want to build a self-driving lab?

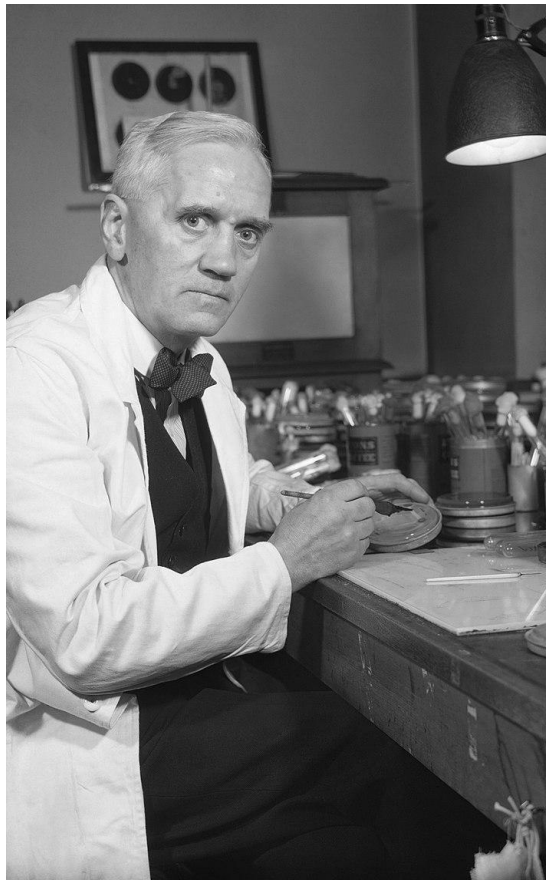


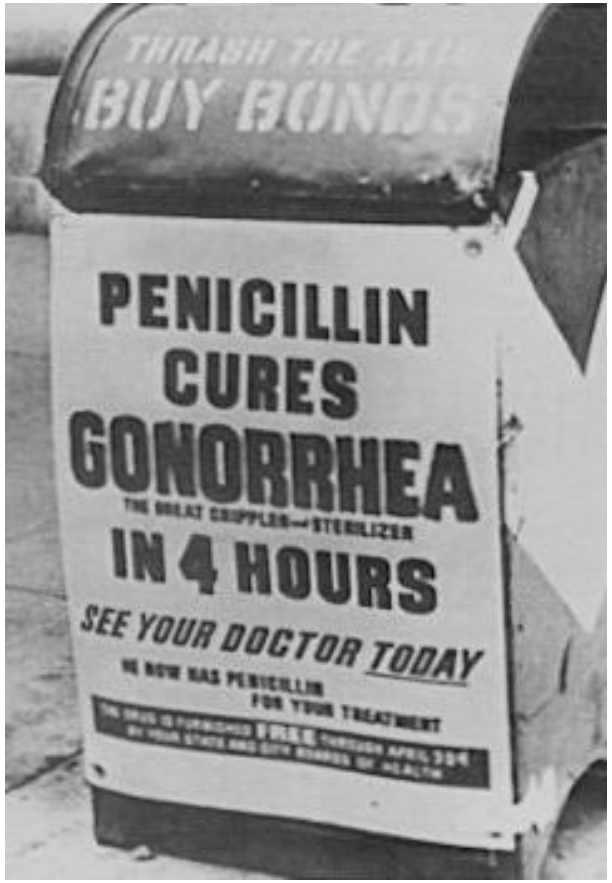
Sterling Baird, Ph.D.
Staff Scientist

Acceleration Consortium

SDL101 Breakfast
Accelerate 2025

In 1928, Alexander Fleming discovered penicillin
after forgetting about a petri dish before leaving for holiday







In 1952, Donald Stookey discovered thermal shock-resistant ceramic glass

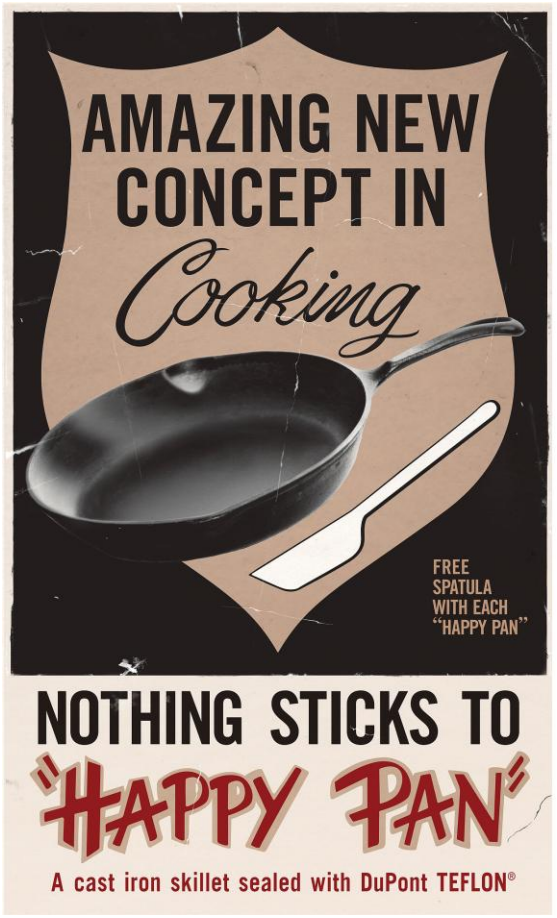
via a furnace accident.. “The thing bounced and didn’t break”

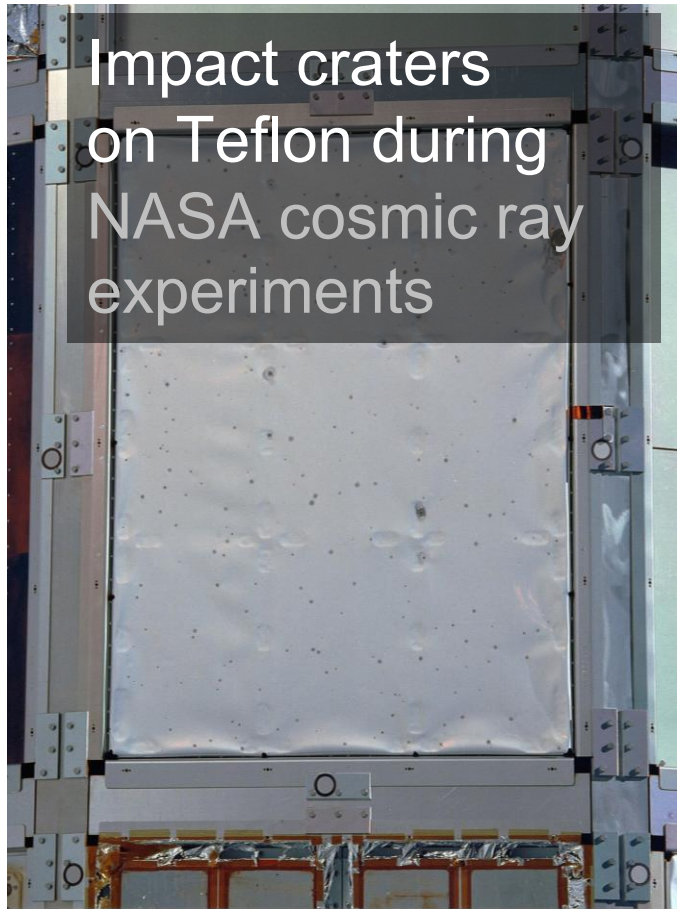




In 1938, Roy Plunkett discovered polytetrafluoroethylene (Teflon) when he noticed a waxy coating in the refrigerant container







In 1964, Stephanie Kwolek discovered Kevlar while studying Nylon when she noticed polymer crystals in a solution

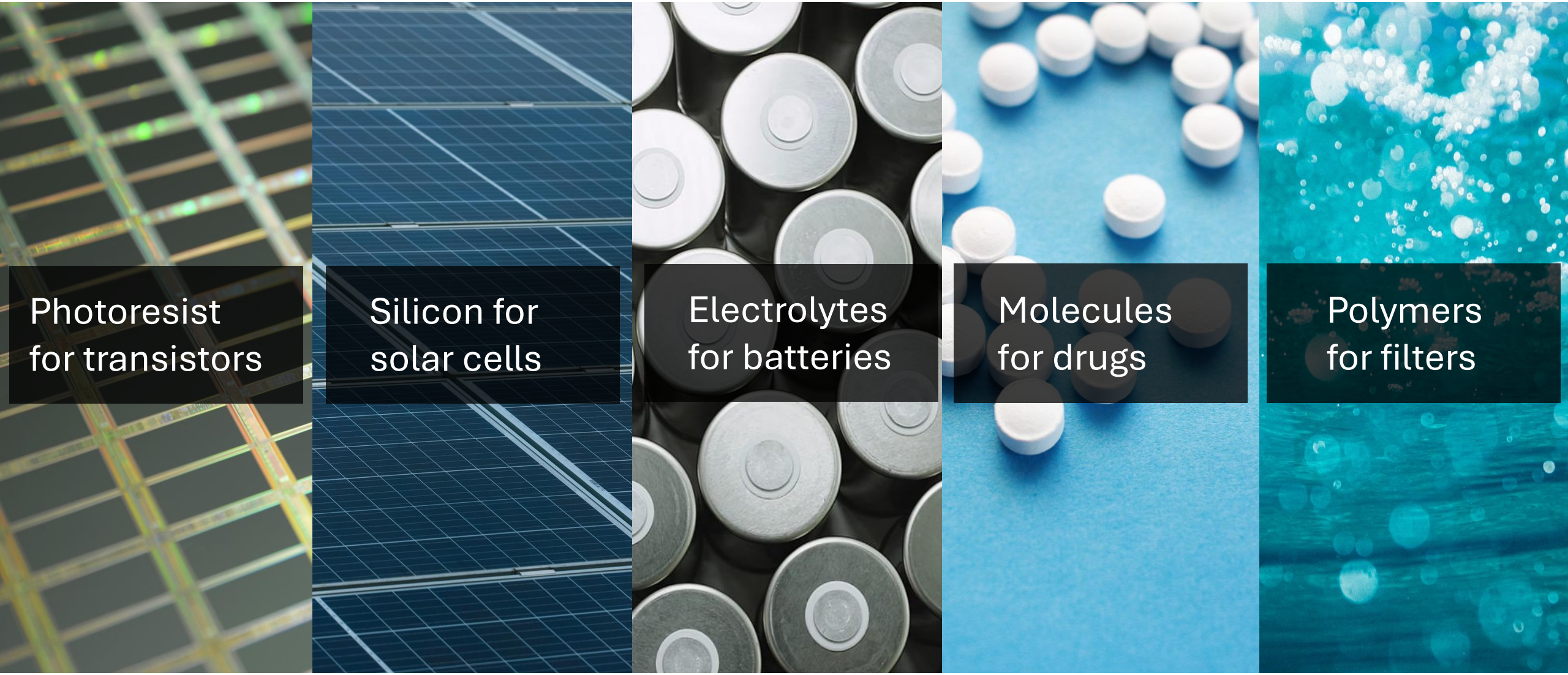




Materials are discovered through a variety of methods



We need even better materials



Combining AI and automation accelerates scientific discovery,
improving efficiency, time to results, and reproducibility



Strieth-Kalthoff et al. *Science* (2024)

This organic laser platform made 500 molecules in 3 months within a large combinatorial search space (190k molecules)

This included the 10 best performing

By comparison:

10 molecules in 3 years (manual)

40 molecules in 1 year (automated)



Strieth-Kalthoff et al. *Science* (2024)

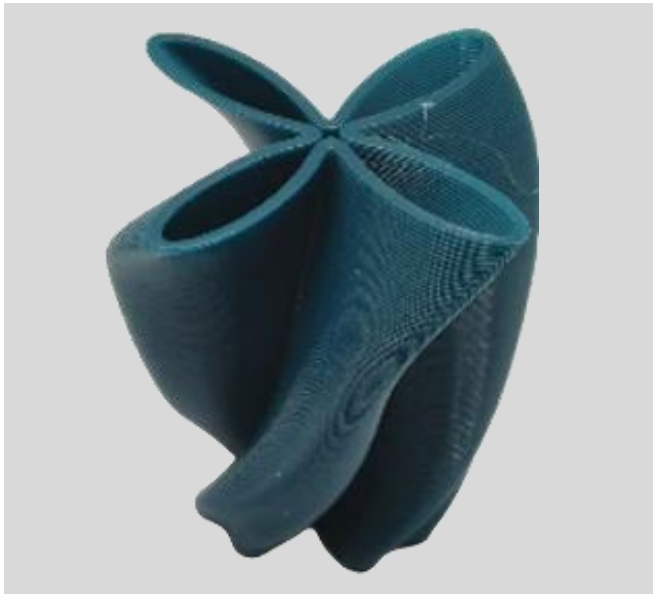
Combining AI and automation accelerates scientific discovery,
improving efficiency, time to results, and reproducibility



Strieth et al. *Science* (2024)



Ament et al. *NeurIPS* (2023)



Snapp et al. *Nat. Commun.* (2023)

Combining AI and automation accelerates scientific discovery, improving efficiency, time to results, and reproducibility

AI

Maximize "value" of experiments by

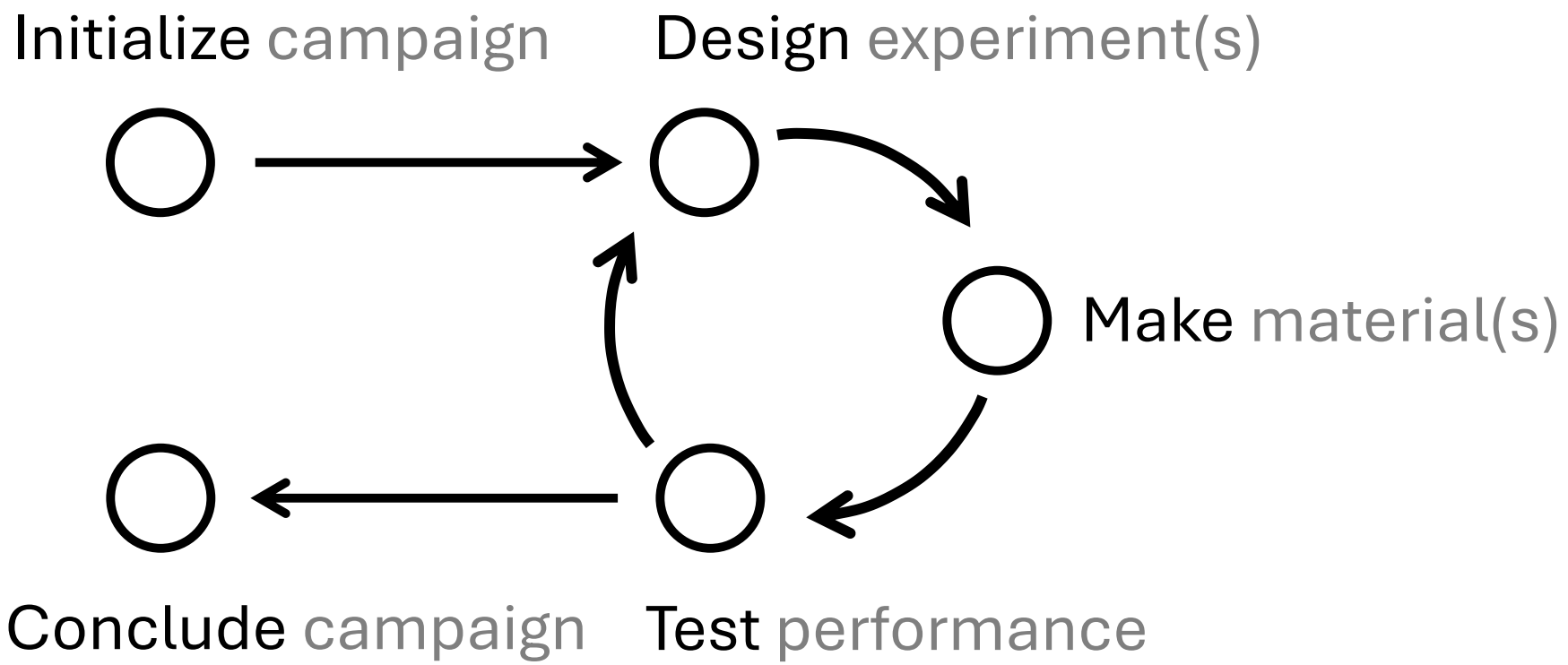
- reducing the number of required experiments
- choosing experiments with most information gain
- predicting properties and quantify uncertainty

Automation

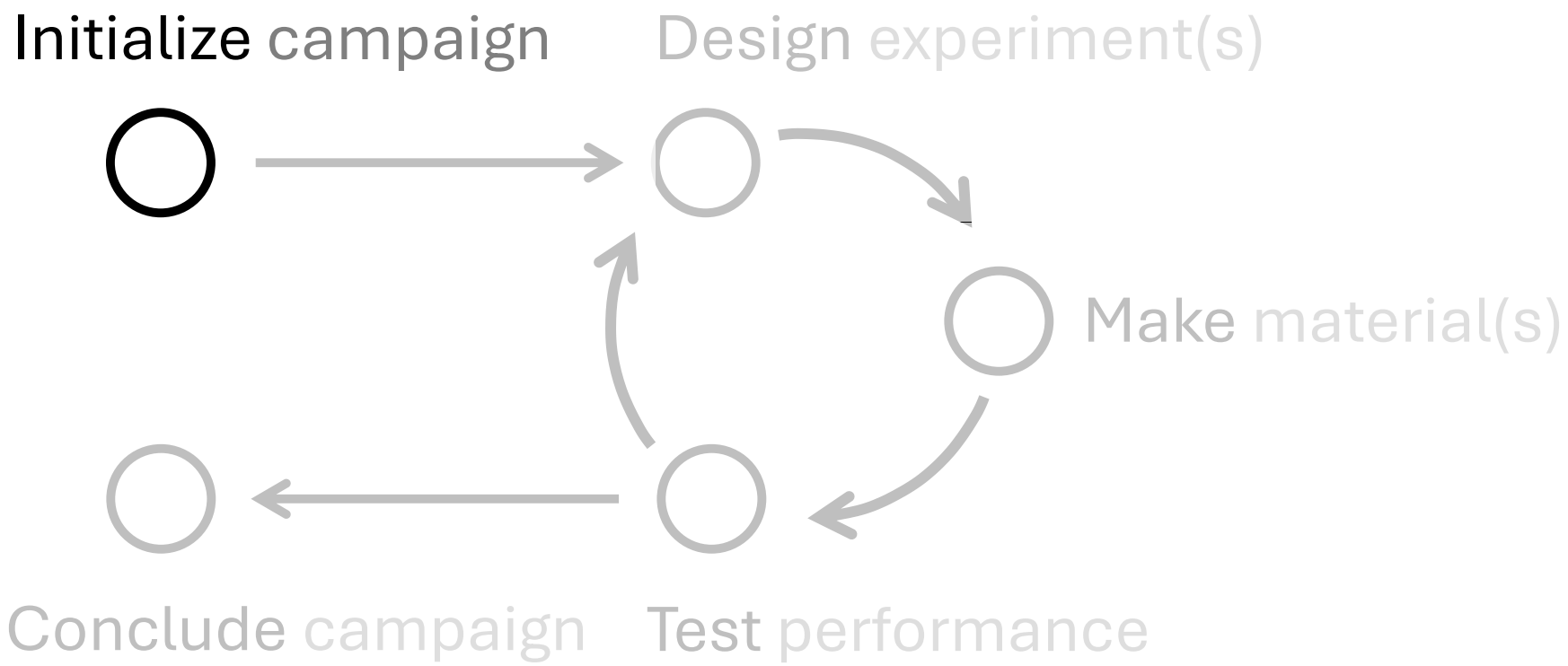
Minimize "burden" of experiments by

- reducing number of tedious, manual steps
- increasing reliability, repeatability, and replicability
- increasing speed and throughput

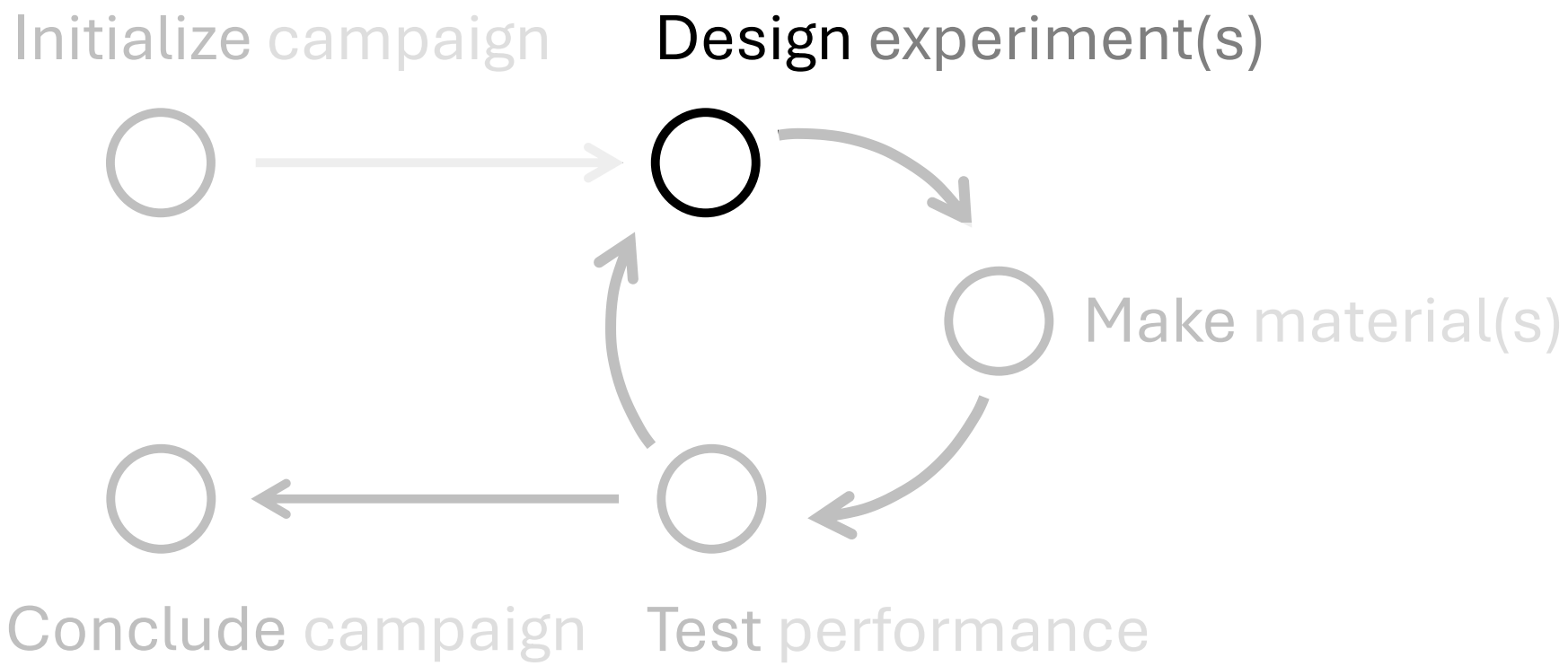
A self-driving lab (SDL) closes the loop of scientific discovery through design-make-test cycles



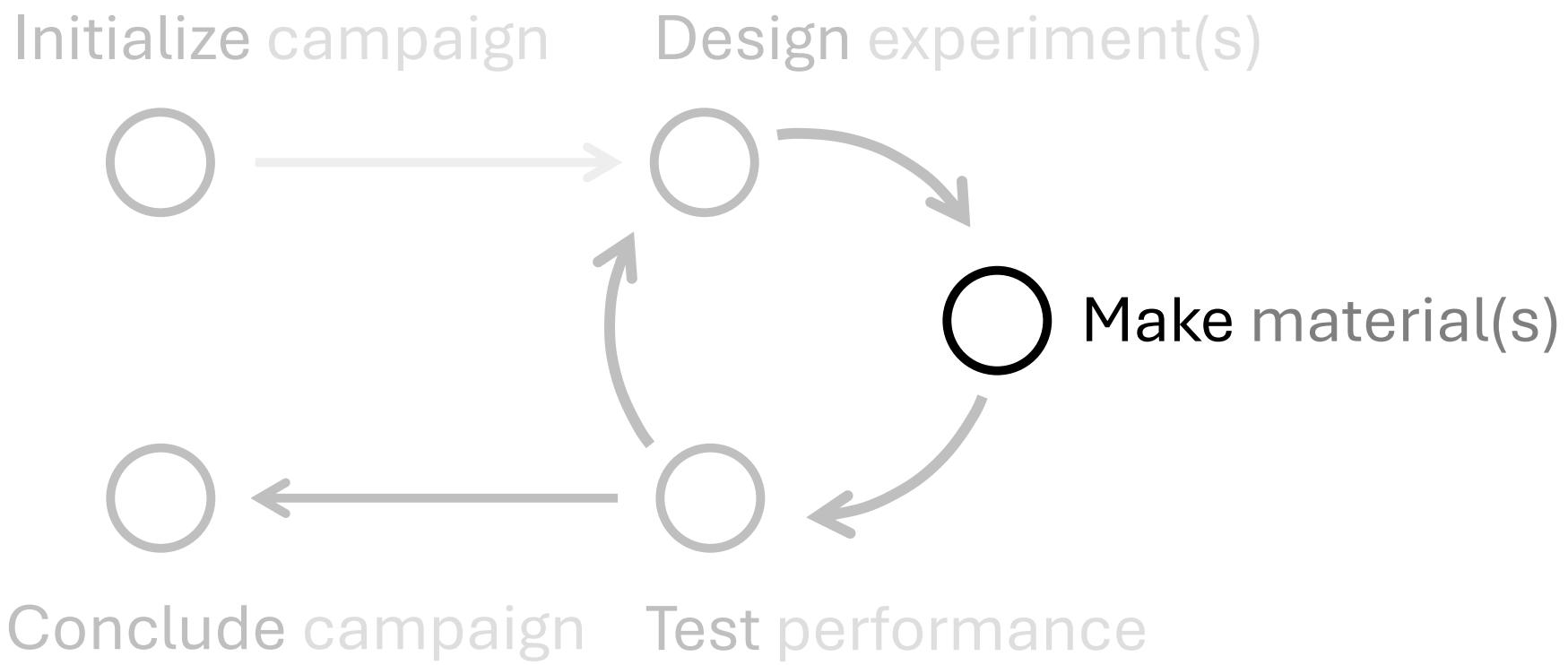
To initialize means to define your materials discovery task
via literature searches, discussions, data aggregation, etc.



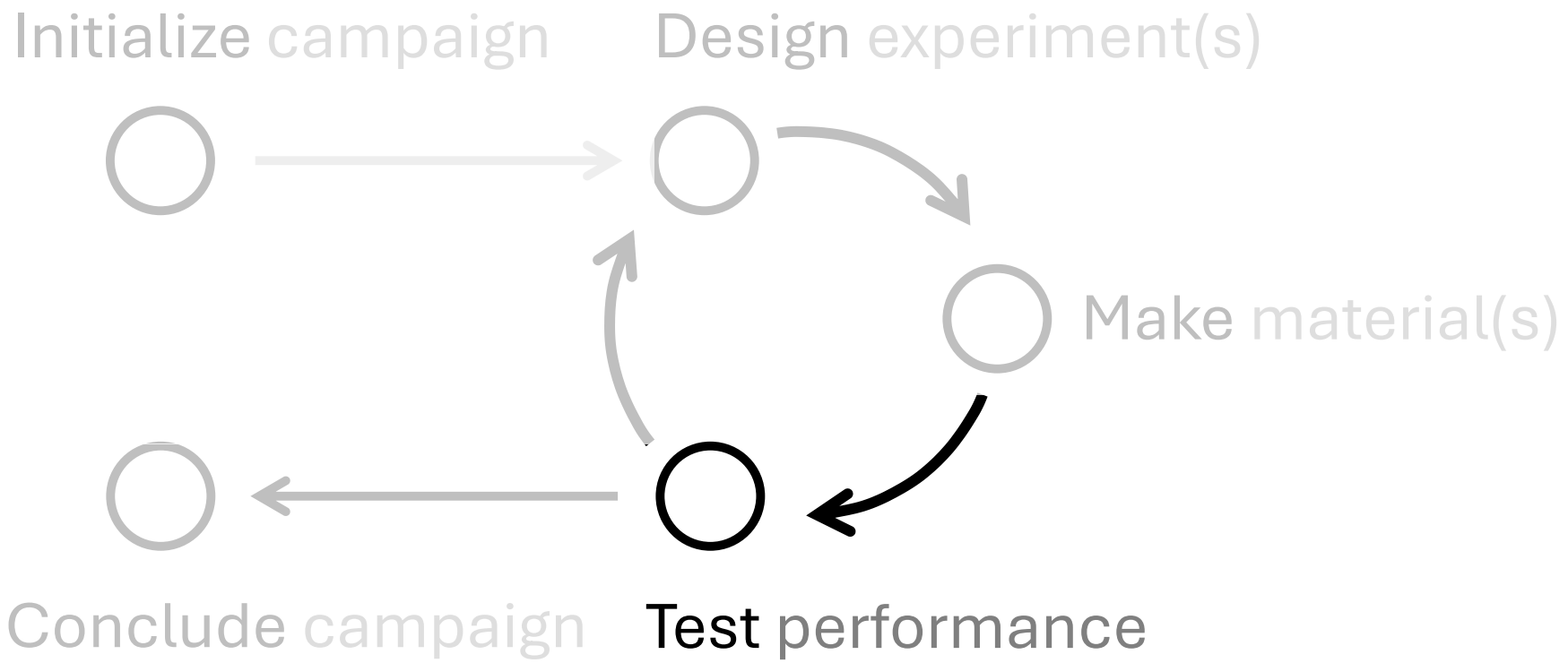
During the design phase, one “acquires” new “designs” to run
(e.g., the next molecule recipe or the next alloy composition)



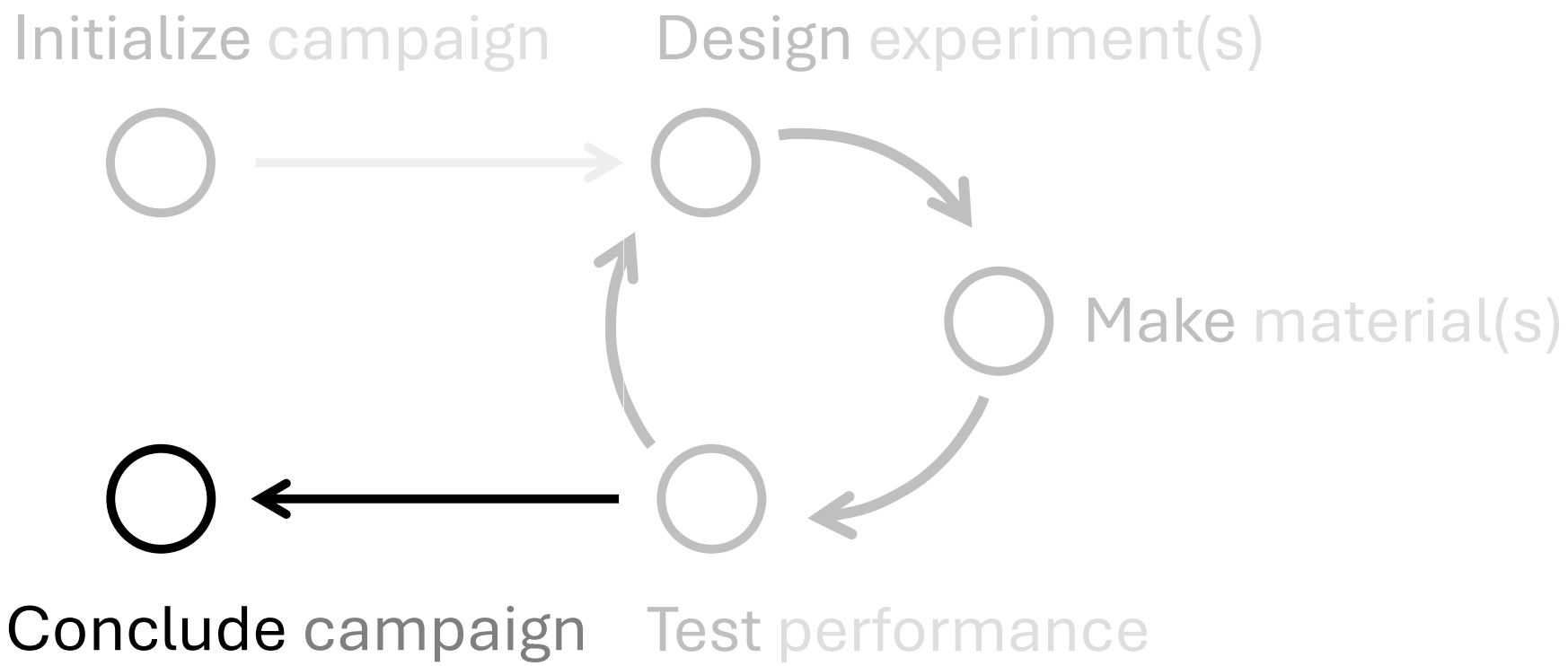
To make means to perform all synthesis and processing steps
(e.g., mixing, stirring, sonicating, heating)



The test phase is where you measure performance properties
(e.g., measuring purity of a drug or tensile strength of a metal alloy)



A campaign is concluded when you have reached your goal, exhausted your budget, or confirmed/rejected your hypothesis



Fully self-driving labs (SDLs) are autonomous systems that perform many iterations without human intervention

Automation

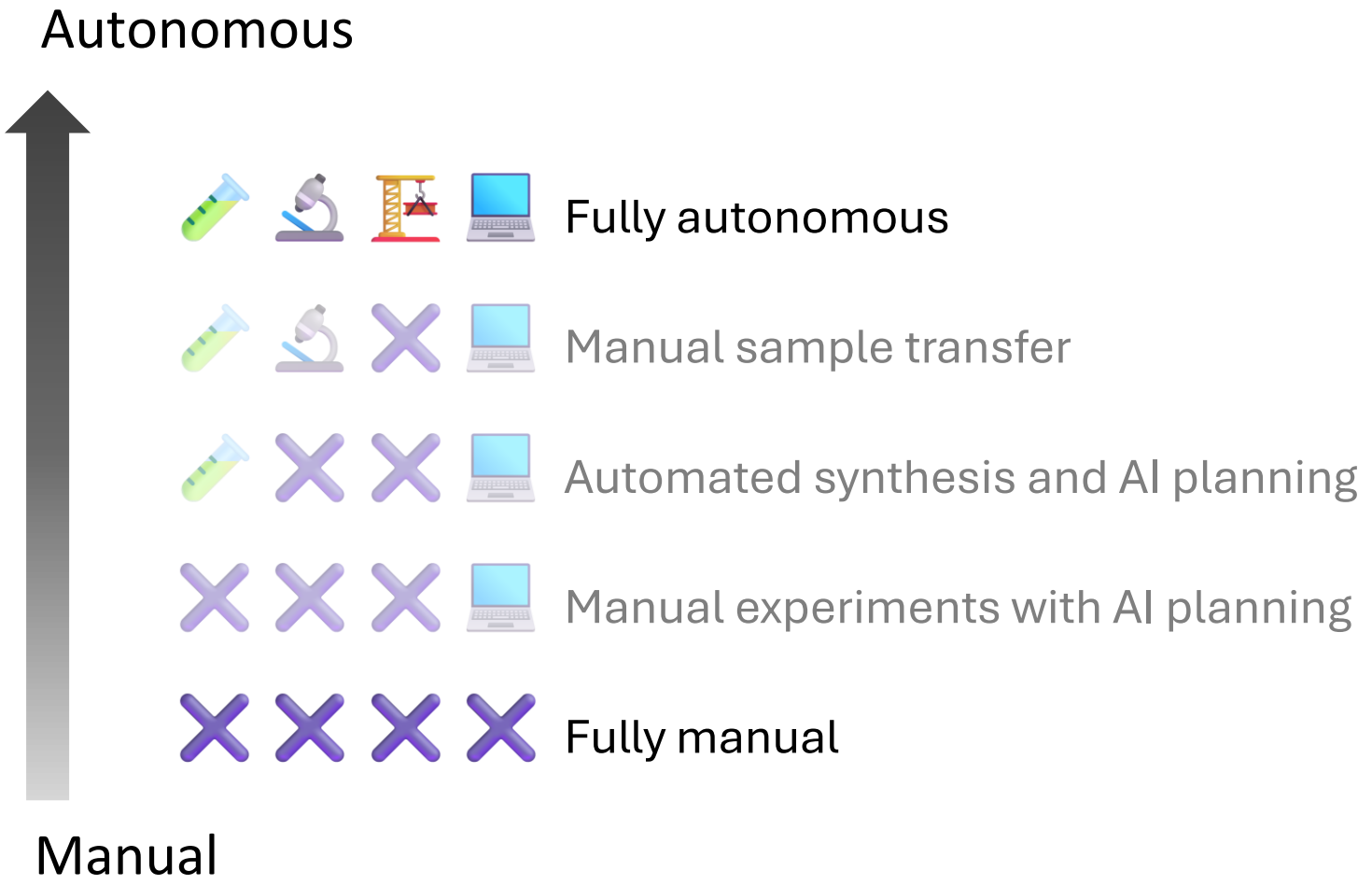
AI

	Manual	Task	Workflow	Lab
Human idea	0	1	2	2
Single Iteration	1	2	3	3
Multiple iterations	2	3	4	4
Generative	2	3	4	5

Tom et al. *Chem. Rev.* (2024)

SDLs automate and connect hardware and software

Category	Emoji
Synthesis	🧪
Characterization	🔬
Sample transfer	🏗️
Experiment planning	💻
Manual intervention	❌



Lo & Baird et al. *Digital Discovery* 2024



1

1. **ADA** (Curtis Berlinguette, Jason Hein, Alán Aspuru-Guzik); 2. **Artificial Chemist** (Milad Abolhasani); 3. **Reconfigurable flow chemistry** (Connor Coley);
4. **Chemputer** (Lee Cronin); 5. **Mobile robot chemist** (Andy Cooper)



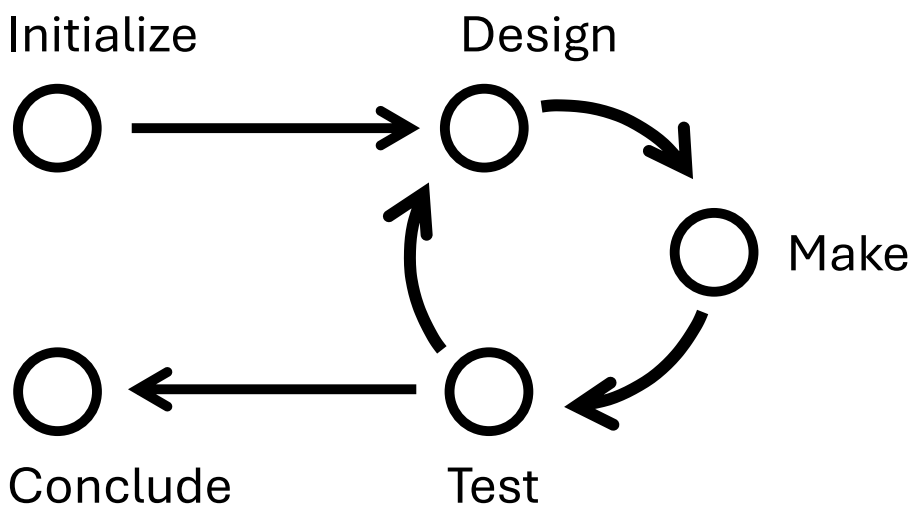
Lunt et al.
Chem. Sci. (2024)

Building a self-driving lab can be daunting



Learn vocabulary and concepts
Avoid all-or-nothing thinking
Leverage community resources

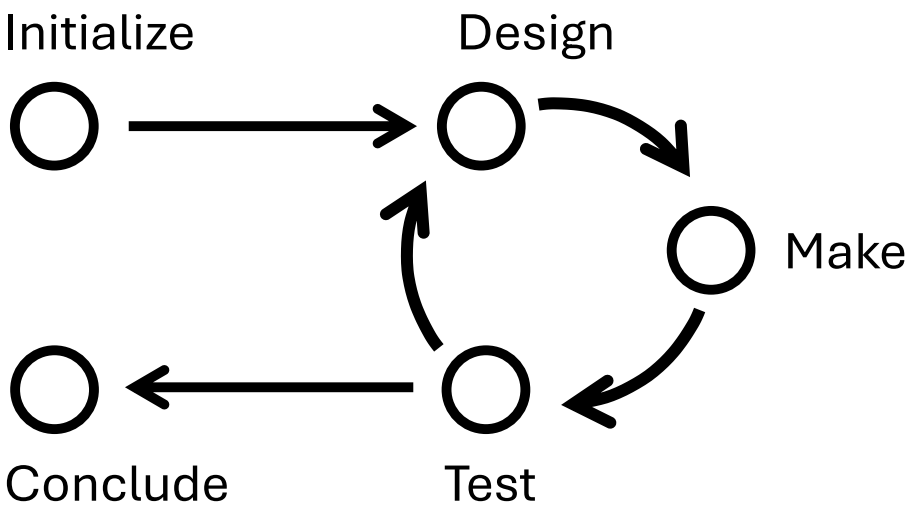
So, you want to build a self-driving lab?



1. AI
   to maximize value
2. Automation
   to minimize burden
3. Integration
   to scale quickly



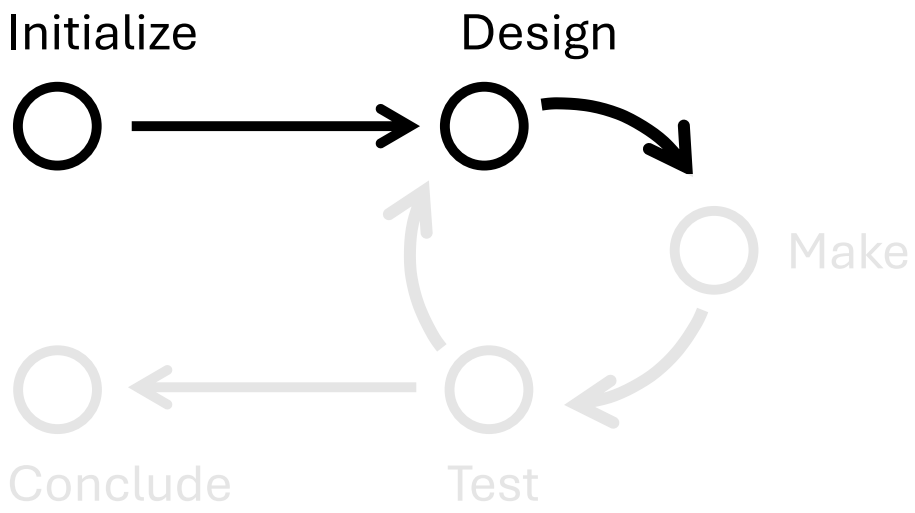
So, you want to build a self-driving lab?



1. AI
   to maximize value
2. Automation
   to minimize burden
3. Integration
   to scale quickly



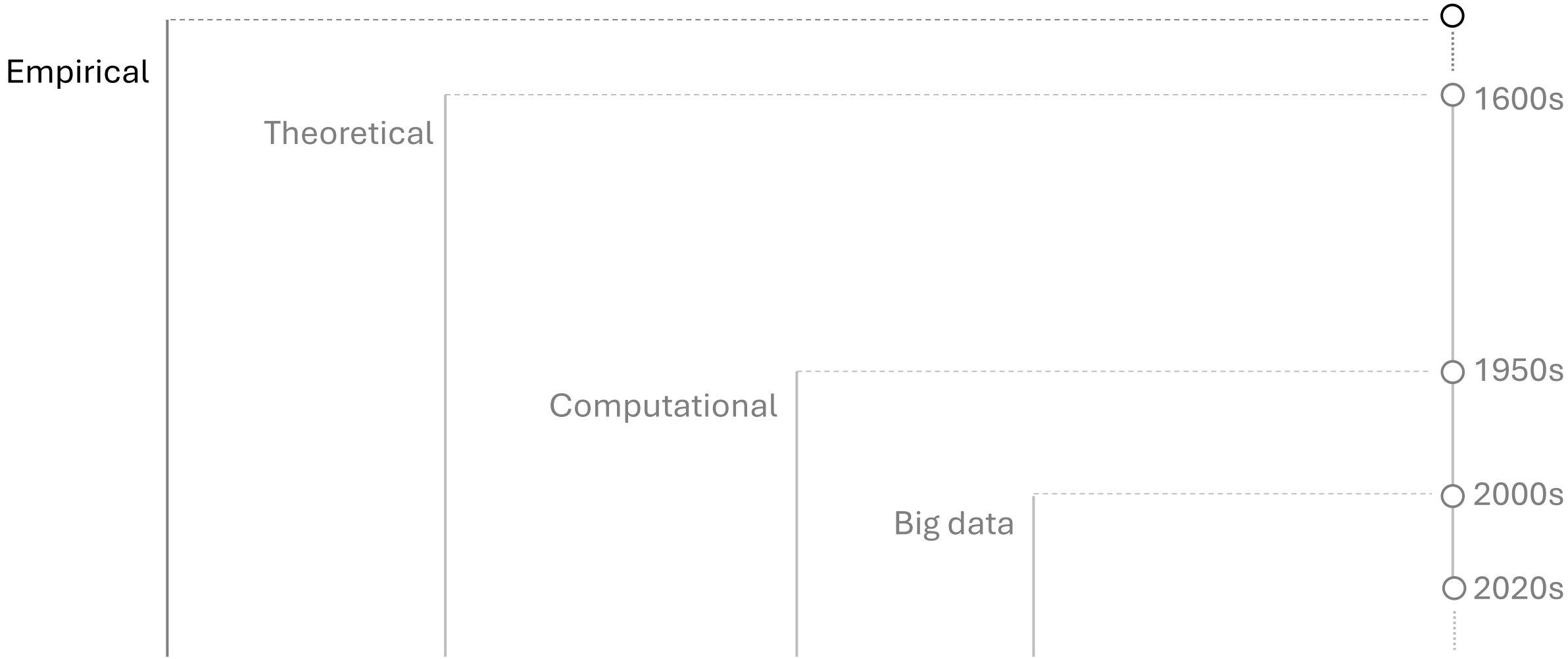
So, you want to build a self-driving lab?



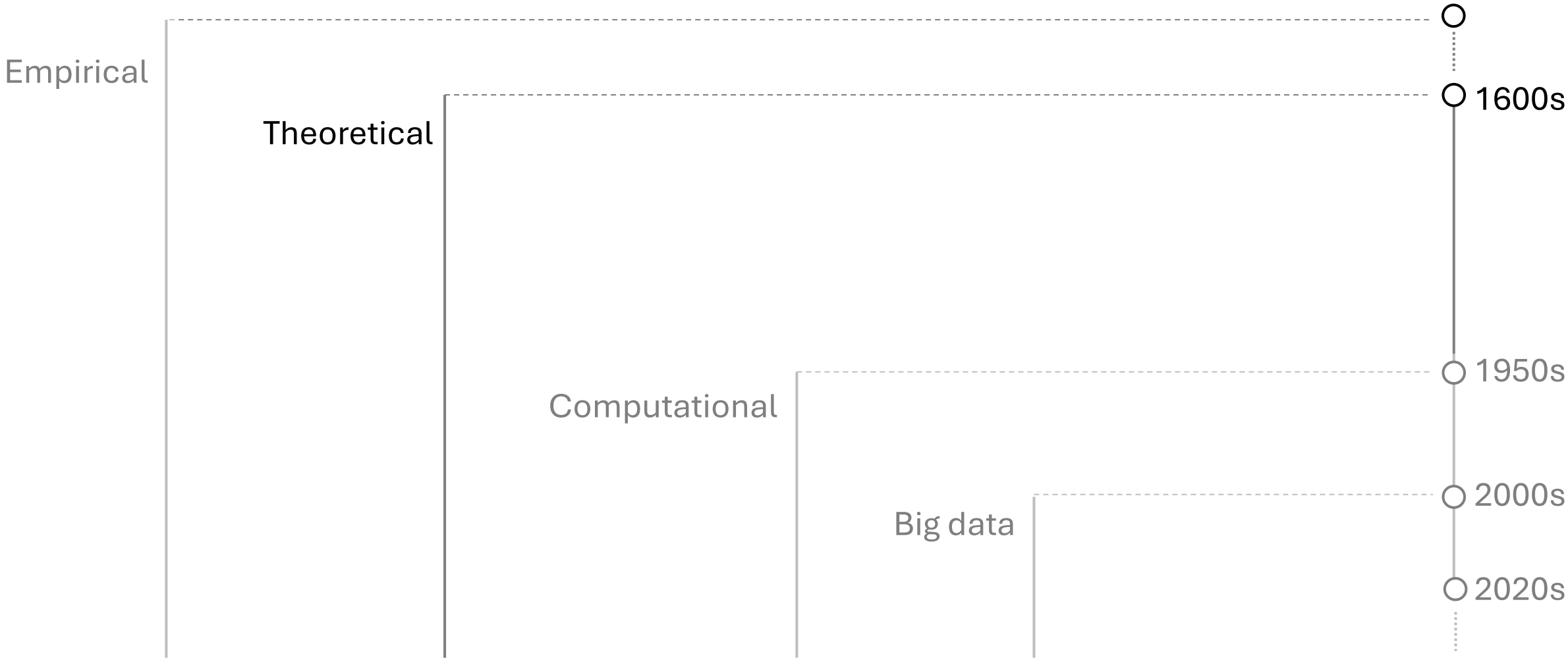
1. AI
   to maximize value
2. Automation
   to minimize burden
3. Integration
   to scale quickly



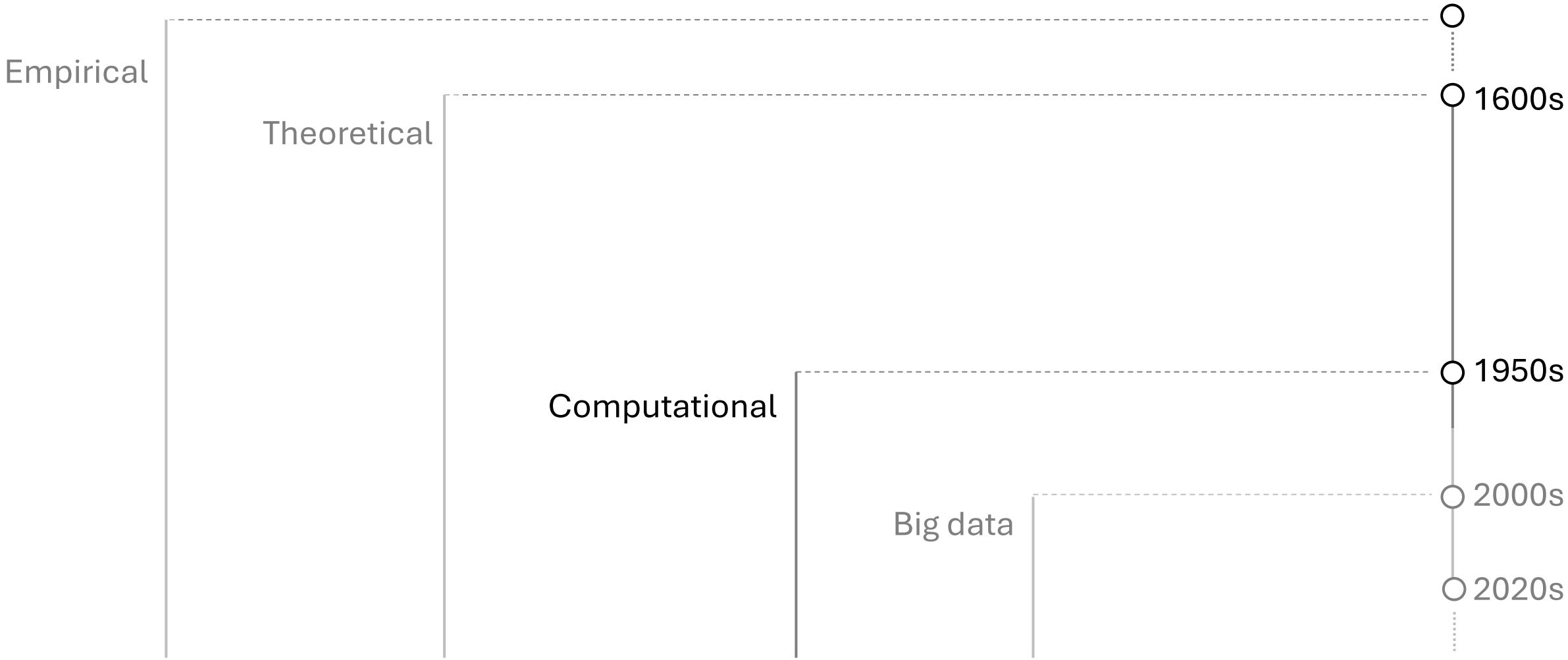
In the paradigms of science, **empirical science** has relied on observation and experimentation



In the 1600s, scientific laws generalized empirical observations (e.g., Newton's three laws of motion)



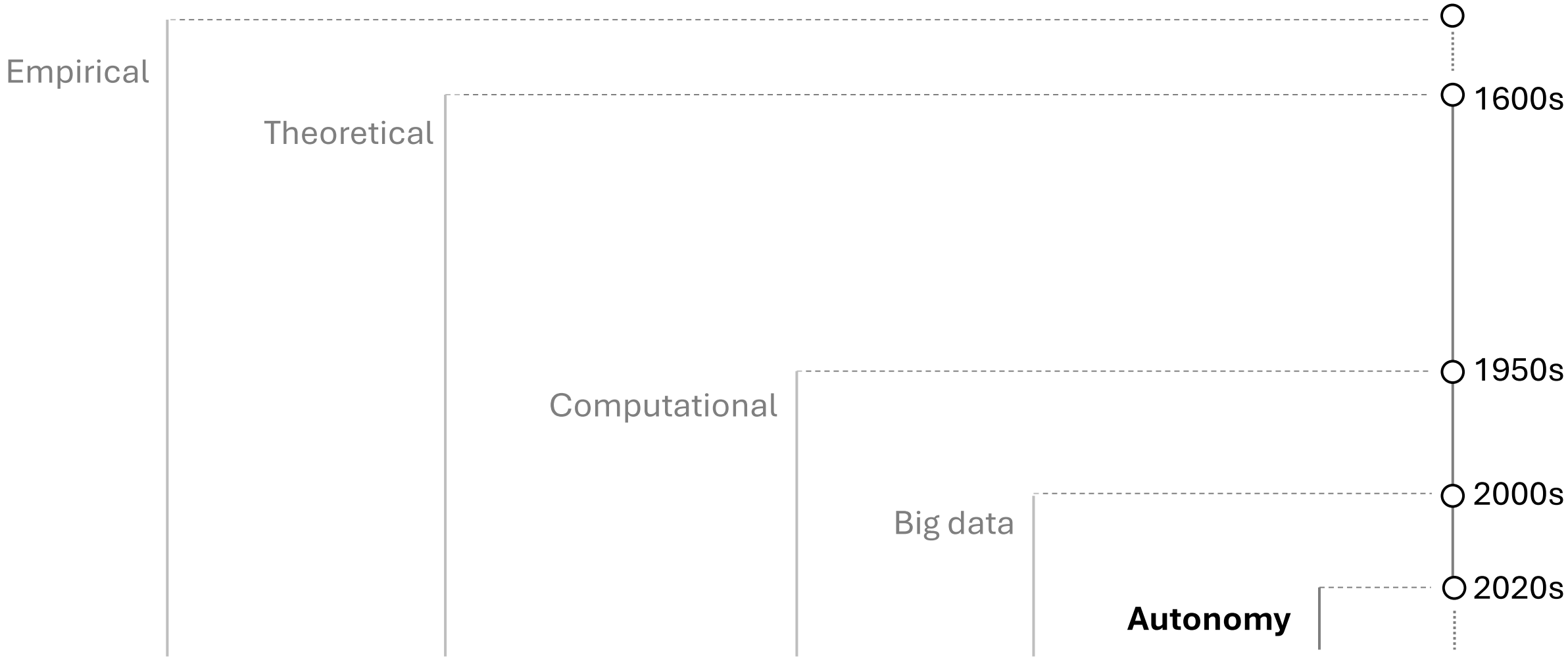
In the 1950s, physics-based computer simulations emerged
(e.g., molecular dynamics and density functional theory)



In the 2000s, models ingested massive sources of digital data
(e.g., the Human Genome Project, AlphaFold)



The future of science is increasingly generative and agentic, tightly coupled with automated experimentation



Combining AI and automation accelerates scientific discovery, improving efficiency, time to results, and reproducibility

AI

Maximize “value” of experiments by

- reducing the number of required experiments
- choosing experiments with most information gain
- predicting properties and quantify uncertainty

Automation

Minimize “burden” of experiments by

- reducing number of tedious, manual steps
- increasing reliability, repeatability, and replicability
- increasing speed and throughput

Combining AI and automation accelerates scientific discovery, improving efficiency, time to results, and reproducibility

AI

Maximize “value” of experiments by

- reducing the number of required experiments
- choosing experiments with most information gain
- predicting properties and quantify uncertainty

Automation

Minimize “burden” of experiments by

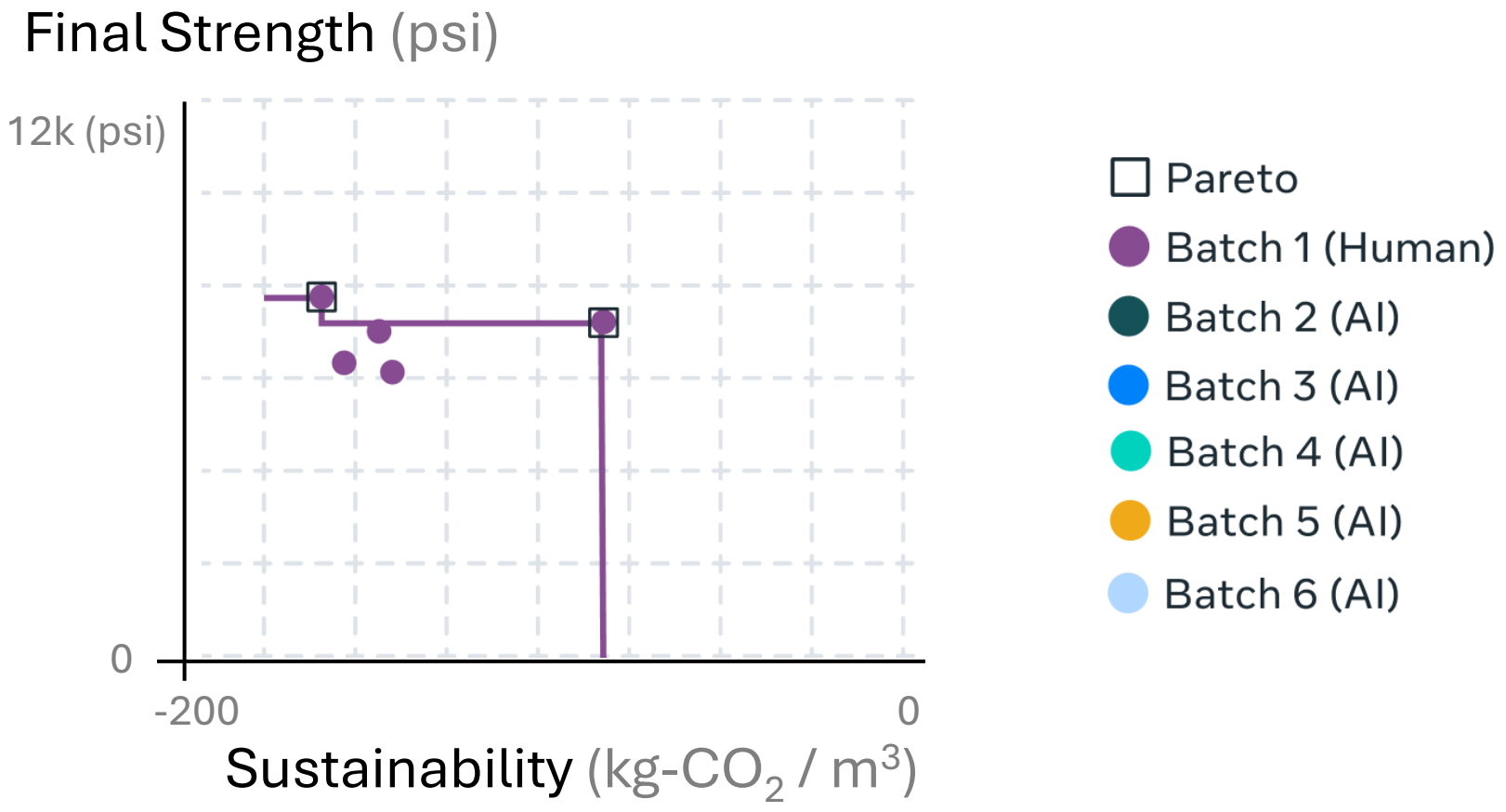
- reducing number of tedious, manual steps
- increasing reliability, repeatability, and replicability
- increasing speed and throughput

Experiments can be incredibly expensive,
and there's always a "bigger fish" (higher fidelity measurements)



Ament et al. *NeurIPS* (2023)

It took Meta, Amrize, and the University of Illinois
6 months to run 6 iterations of experiments (28 days each)



Efforts paid off



“Not all experiments are equally useful. Some experiments provide more information than others.”

Citrine Informatics

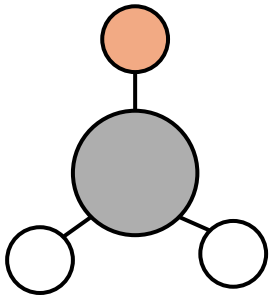

THIS IS WHERE YOU LOST YOUR WALLET?
NO, I LOST IT IN THE PARK. BUT THIS IS WHERE THE LIGHT IS.

Adjusting your search space representation helps provide contextual information to models

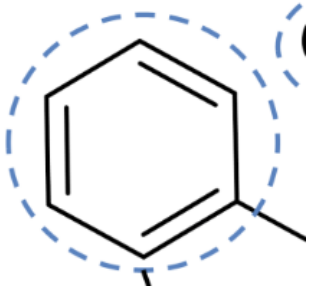
Chemical compounds

C=O

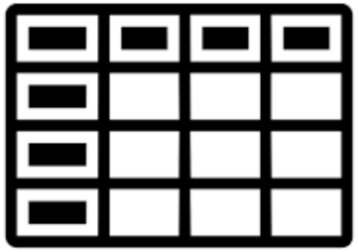
Text



Graph



Fingerprint

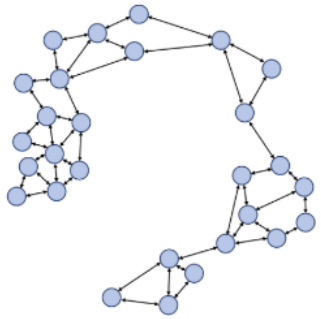


Physical descriptors

Reactions

[C:1]
>>C

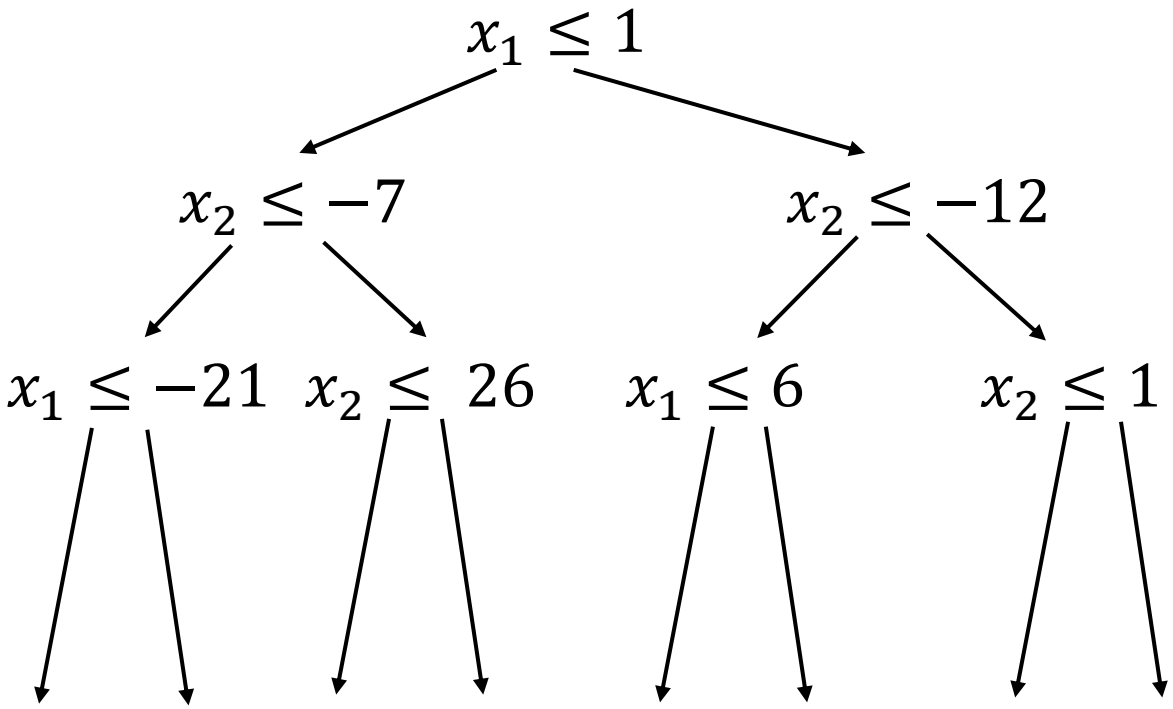
Text



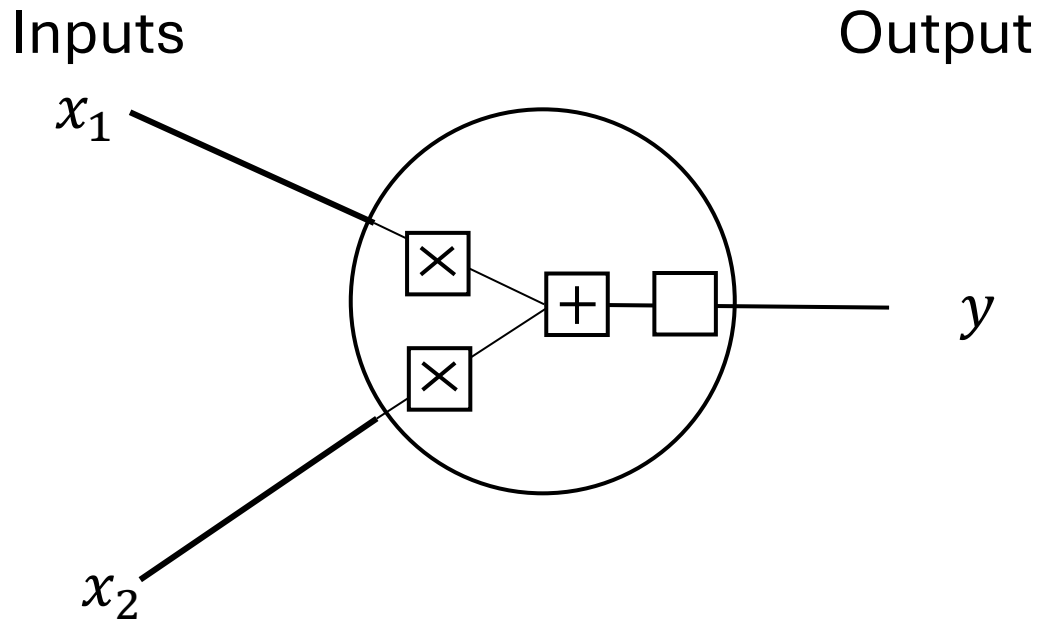
Graph

Classical machine learning models like random forests and neural networks have helped especially with property prediction tasks

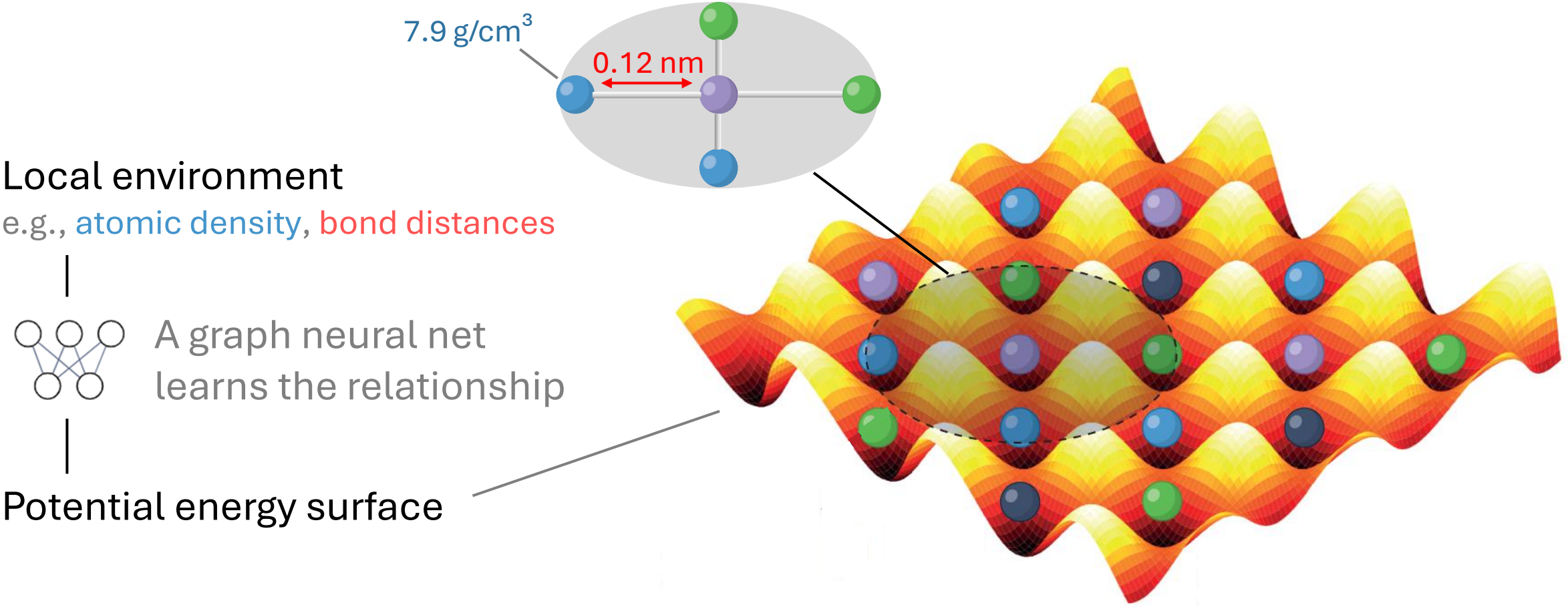
Decision Tree
from a Random Forest (RF)



Neuron
from a neural network (NN)



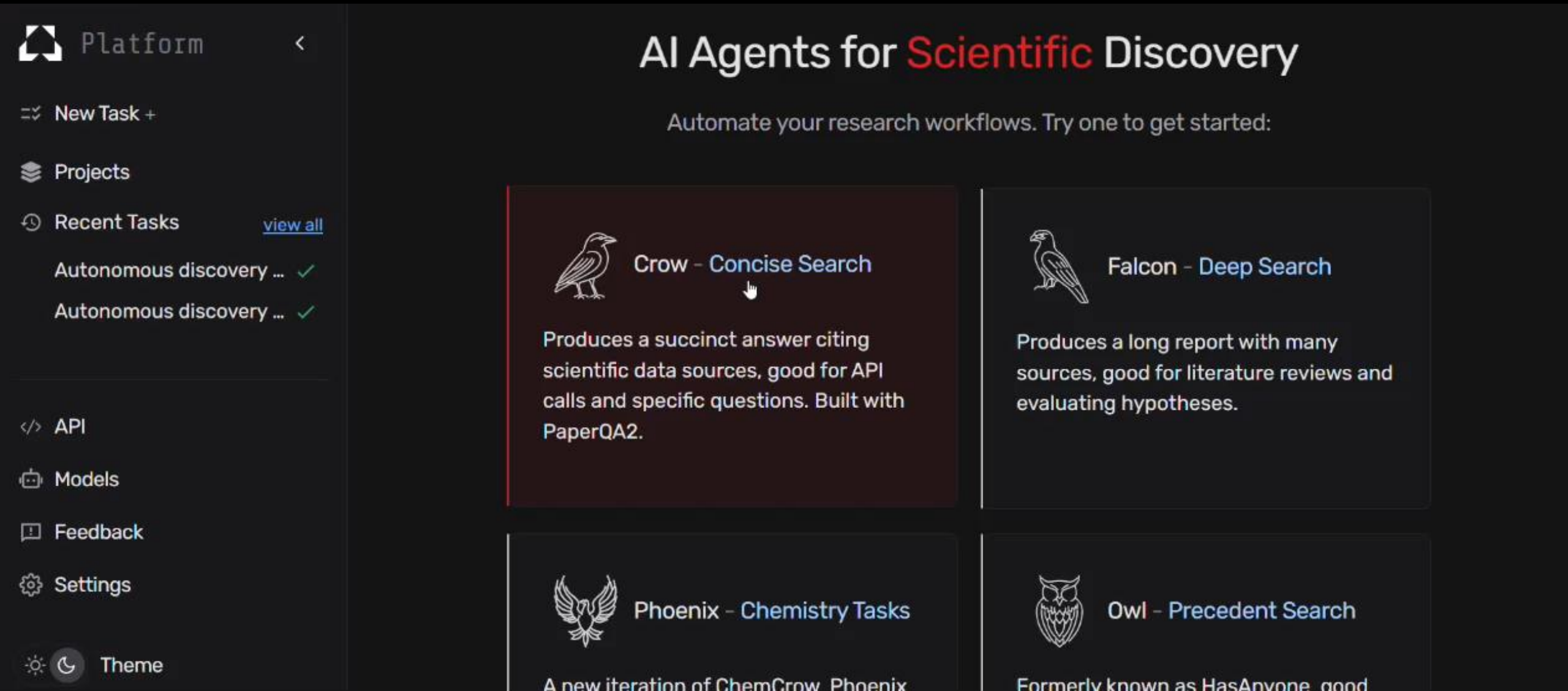
Machine learning models such as graph neural networks (GNNs) are increasingly used to speed up physics-based simulations



Ko et al. *Nat. Comput. Sci.* (2023)

The utility of large language models for materials discovery are being actively explored (literature, predictions, agents)

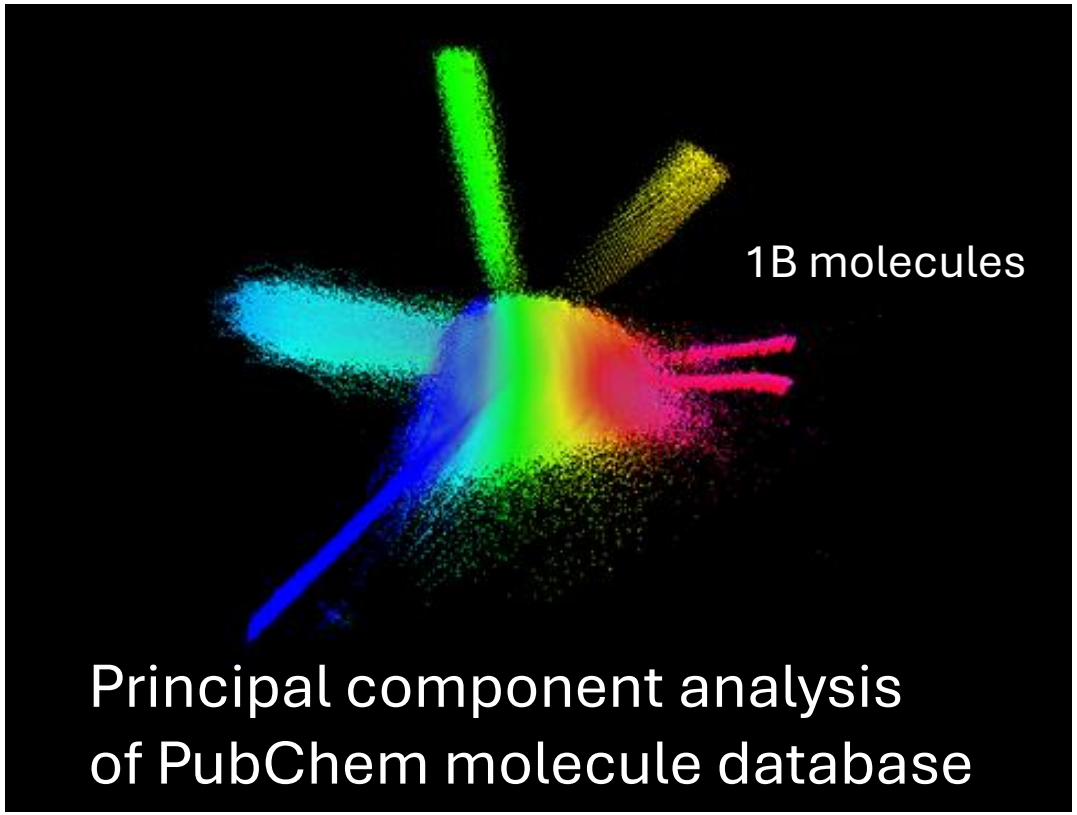
Future House has a suite of AI agents steeped in literature (platform.futurehouse.org)

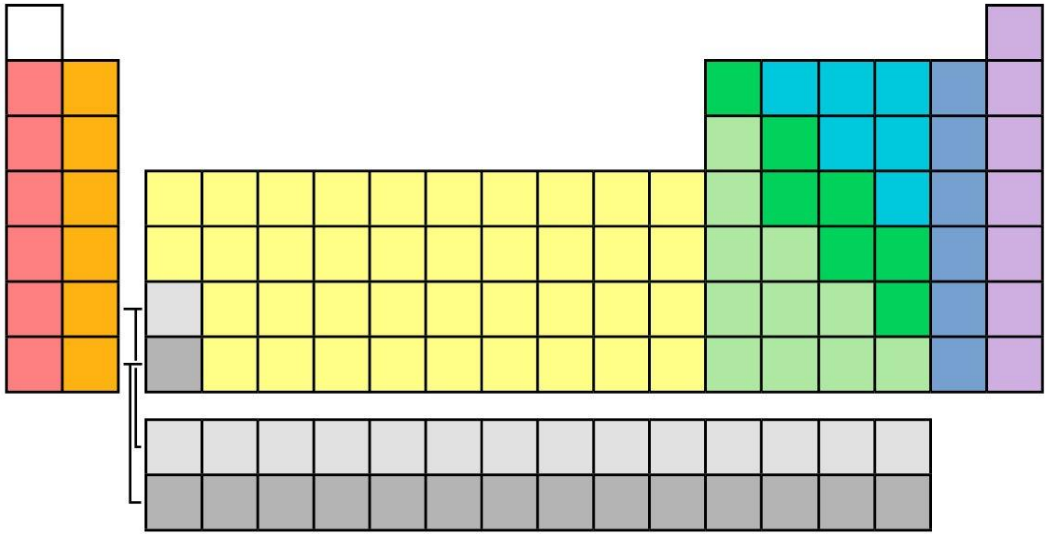


The primary use of generative models in materials discovery is the exploration of "whitespace" (e.g., molecules, crystals, proteins)

The greatest challenge for these models is **synthesizability**

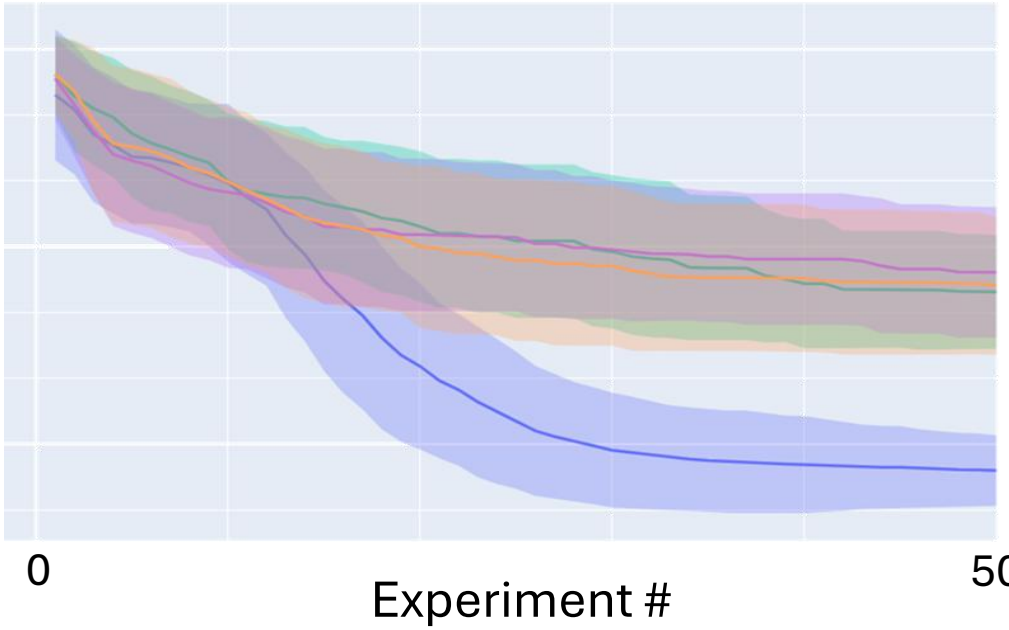
Is the molecule/structure feasible? If so, what's the recipe?





Optimization algorithms (esp. Bayes optimization algorithms) are the heart of self-driving labs (low data regime, many variables)

Task: Find the set of parameters (X) that minimizes a target property (y)



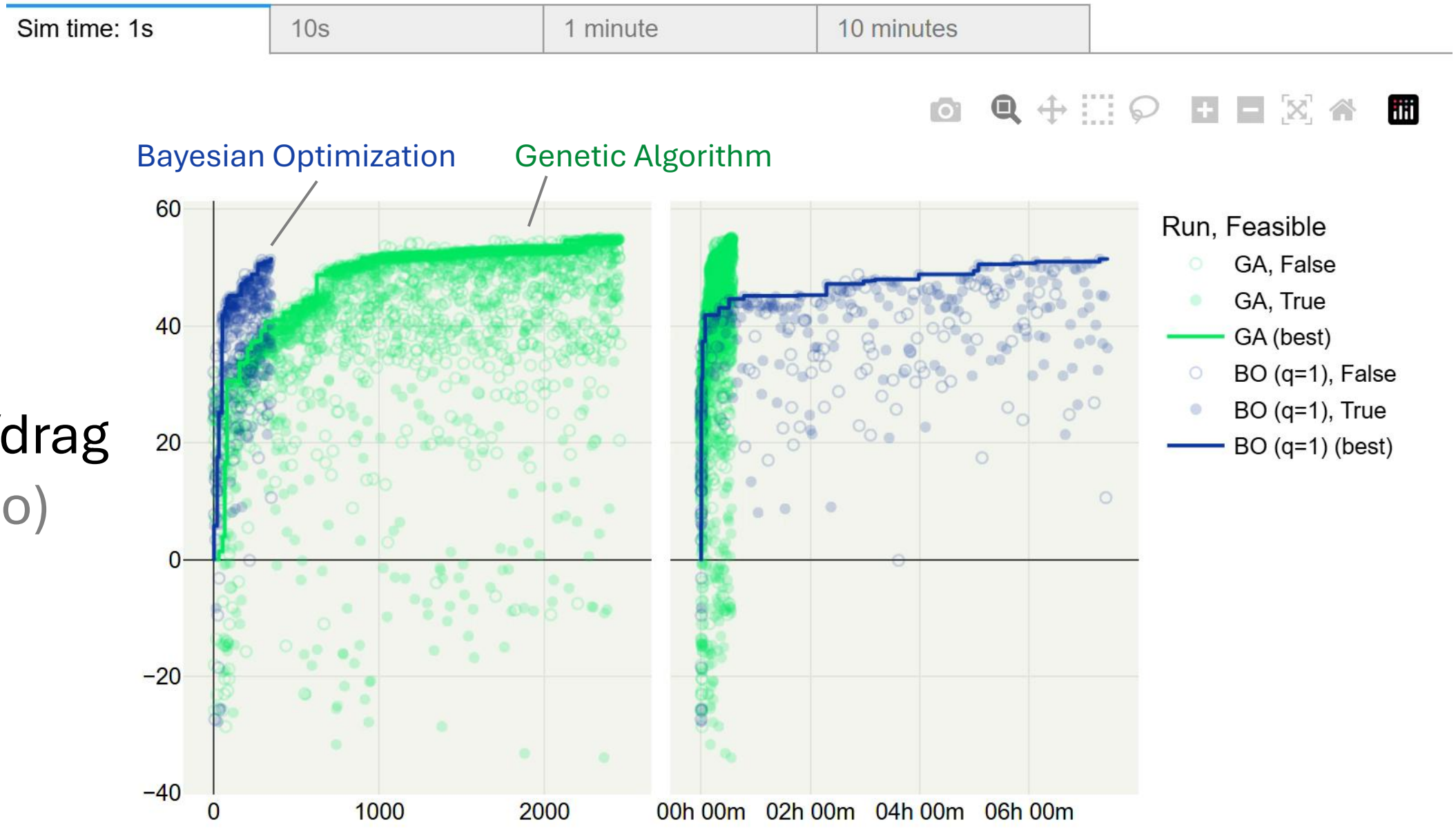
Design of Experiments
(Traditional)

Bayesian Optimization

Task (💊): Find the synthesis pathway (X) that maximizes binding affinity to HIV protease (y)

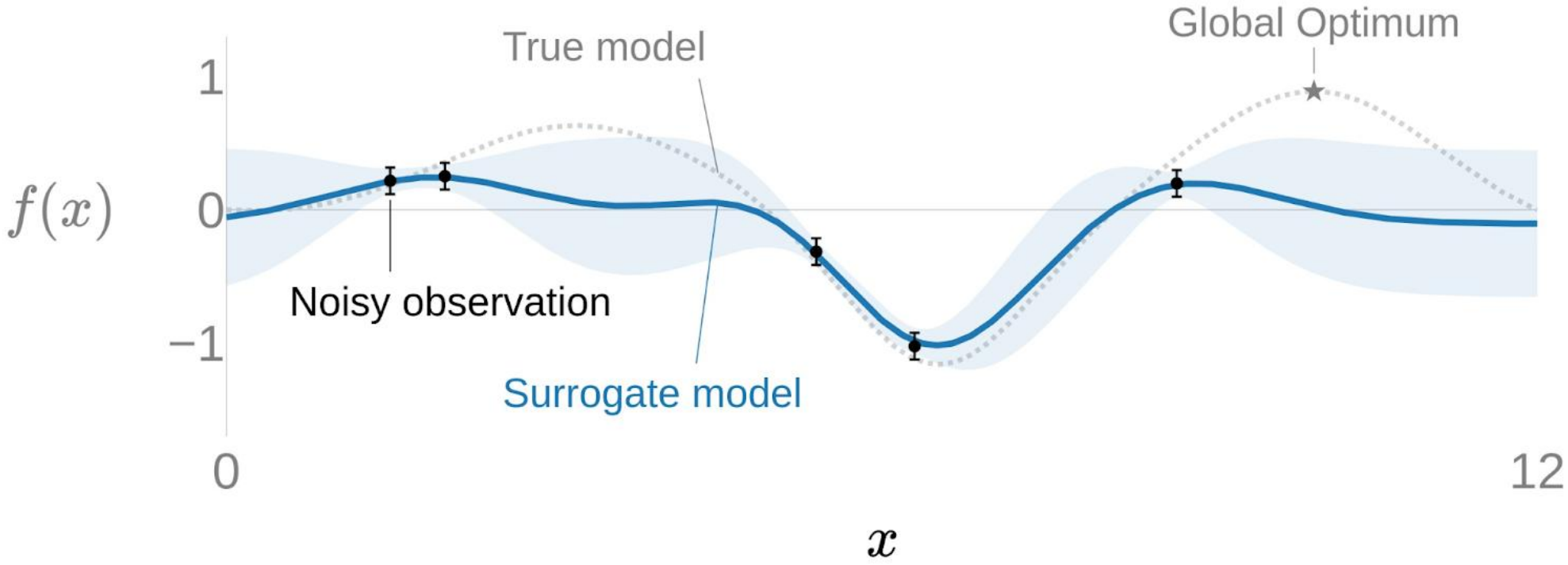
Task (🔩): Find the set of composition/processing parameters (X) that maximizes alloy yield strength (y)

Baird et al. *Matter* (2022)


Sim time: 1s
10s
1 minute
10 minutes
Bayesian Optimization
Genetic Algorithm
Lift/drag
(ratio)
60
40
20
0
−20
−40
0
1000
2000
00h 00m
02h 00m
04h 00m
06h 00m
Run, Feasible
GA, False
GA, True
GA (best)
BO (q=1), False
BO (q=1), True
BO (q=1) (best)
Iteration
Walltime

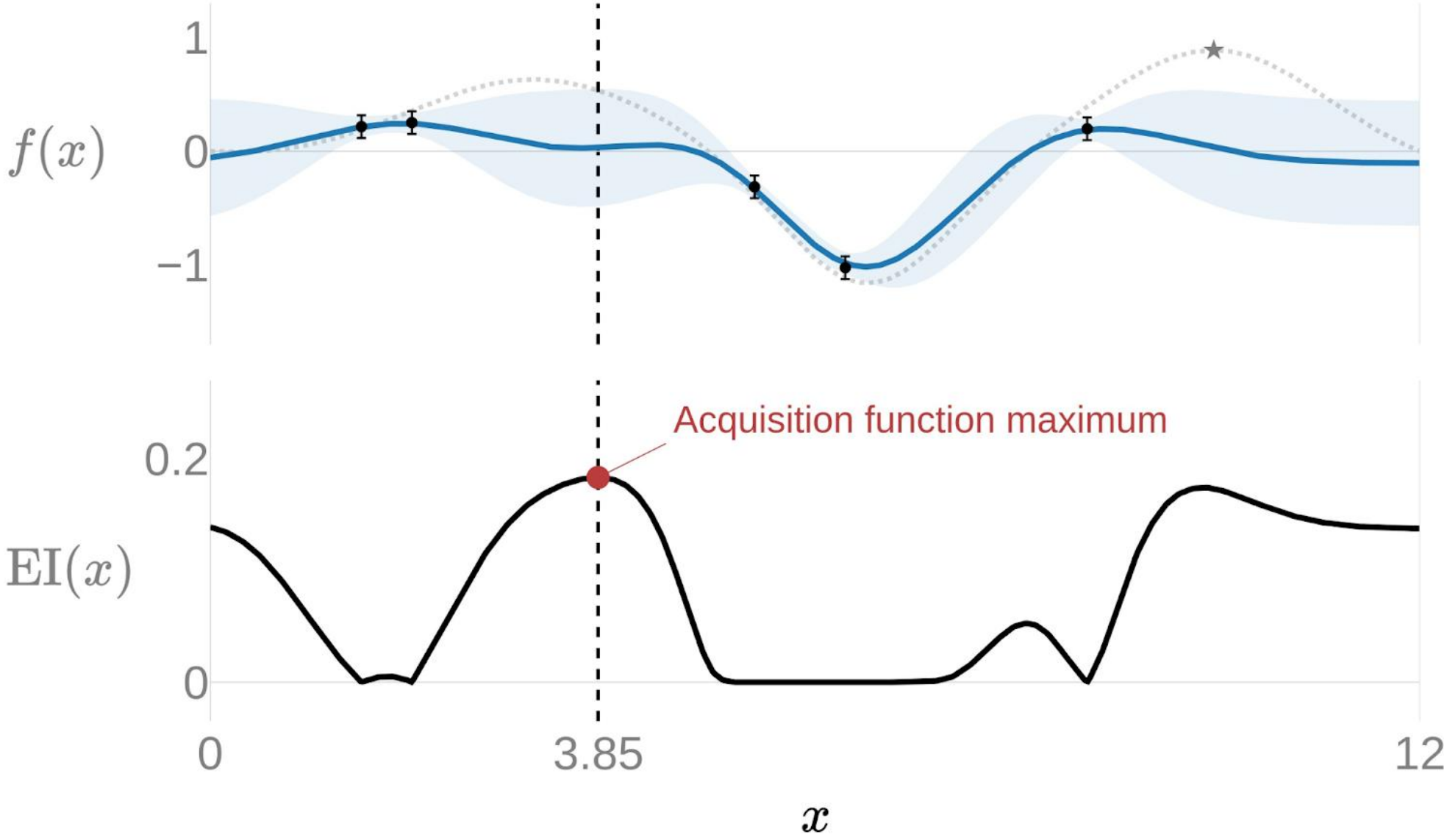
physicsx.ai

The surrogate model predicts properties and quantifies uncertainty



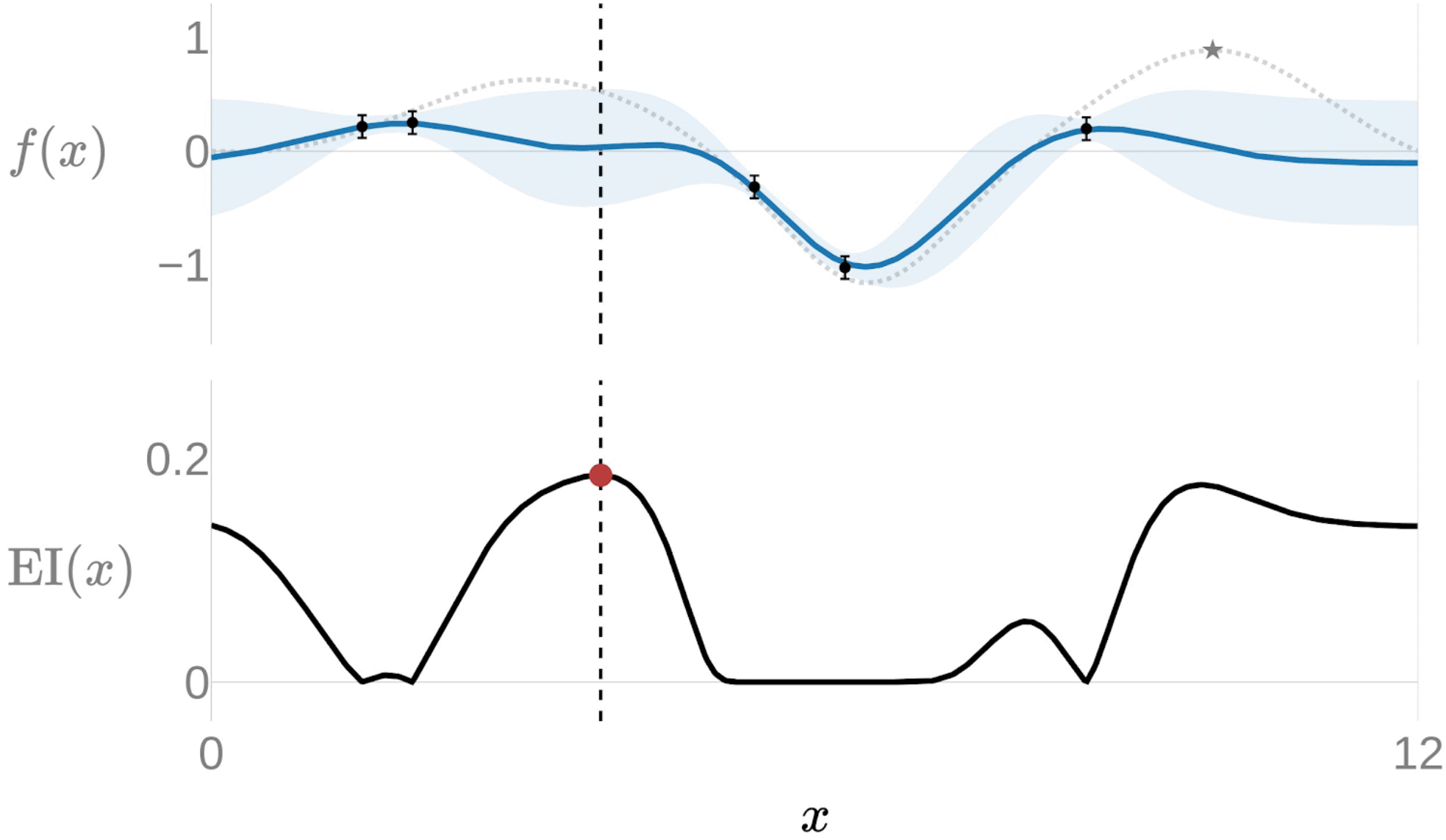
The acquisition function guides how to select the next experiment

(in this case, neither the max predicted nor the max uncertainty)

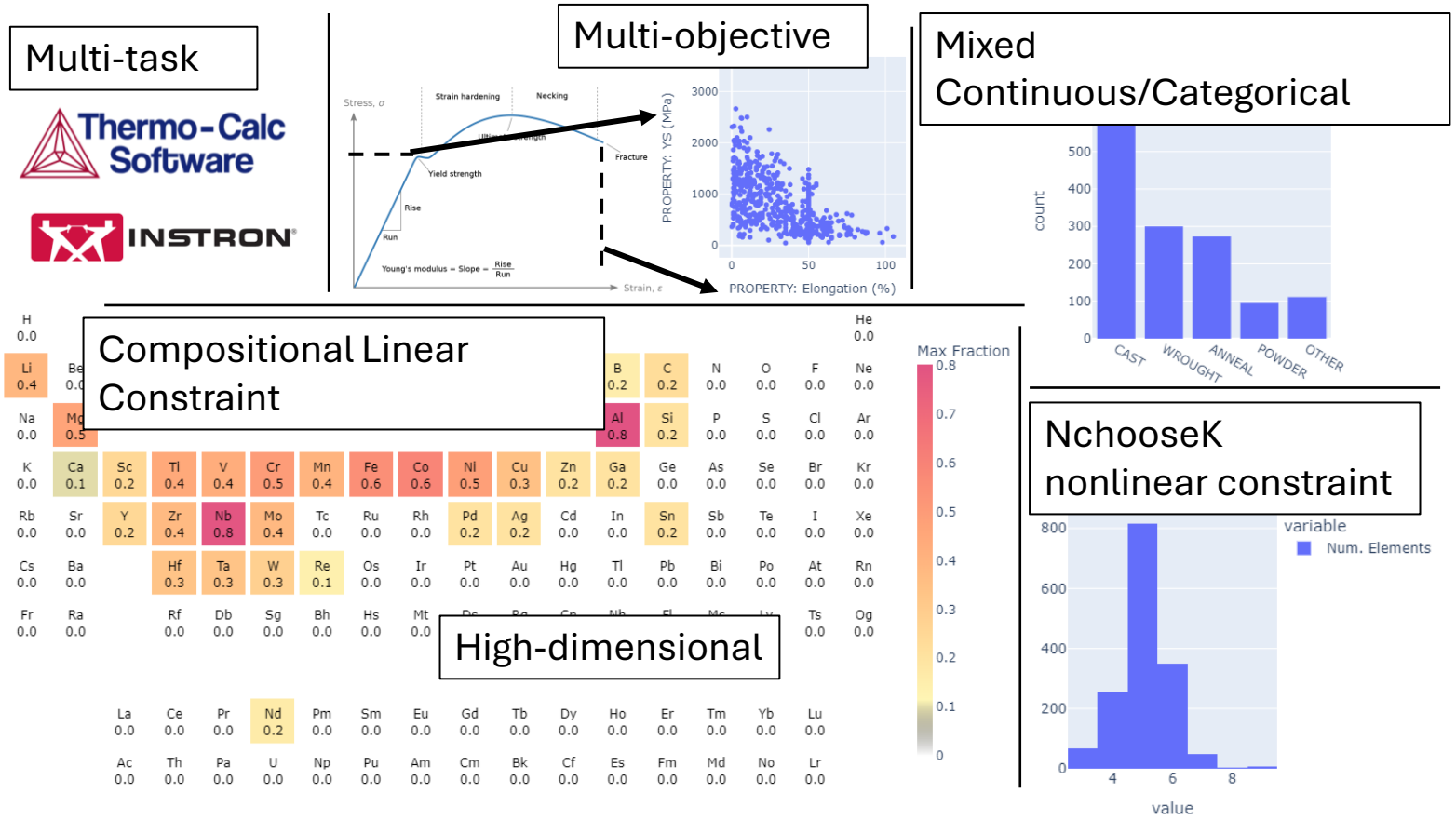


This optimization loop continues iteratively

(1) add data, (2) fit surrogate model (3) optimize acquisition function



Real-world materials discovery tasks require more than “vanilla” optimization



Real-world materials discovery tasks require more than “vanilla” optimization

Multi-task

Multi-objective

Mixed
Continuous/Categorical

Compositional Linear
Constraint

NchooseK
nonlinear constraint

High-dimensional

Real-world materials discovery tasks require more than “vanilla” optimization

Multi-task	Single	Multi
Multi-objective	Single	Multi
Mixed Continuous/Categorical	Continuous	Cont. + Categorical
Compositional Linear Constraint	No Constraint	Linear Constraint
High-dimensional	Low dim.	High dim.
NchooseK nonlinear constraint	No Constraint	Nonlinear Constraint

Real-world materials discovery tasks require more than “vanilla” optimization

	Basic	
Multi-task	Single	Multi
Multi-objective	Single	Multi
Mixed Continuous/Categorical	Continuous	Cont. + Categorical
Compositional Linear Constraint	No Constraint	Linear Constraint
High-dimensional	Low dim.	High dim.
NchooseK nonlinear constraint	No Constraint	Nonlinear Constraint

Real-world materials discovery tasks require more than “vanilla” optimization

	Advanced	
Multi-task	Single	Multi
Multi-objective	Single	Multi
Mixed Continuous/Categorical	Continuous	Cont. + Categorical
Compositional Linear Constraint	No Constraint	Linear Constraint
High-dimensional	Low dim.	High dim.
NchooseK nonlinear constraint	No Constraint	Nonlinear Constraint

Honegumi lets you quickly create examples

Basic

Single obj.	Multi obj.
Single task	Multi task
Continuous	Cont. + Cat.
No Constraint	Linear
Low dim.	High dim.
No Constraint	Nonlinear

Advanced

Single obj.	Multi obj.
Single task	Multi task
Continuous	Cont. + Cat.
No Constraint	Linear
Low dim.	High dim.
No Constraint	Nonlinear

```python
from ax.service.ax_client import AxClient
from ax.utils.measurement.synthetic_functions import branin


ax_client = AxClient()
ax_client.create_experiment(
    parameters=[
        {"name": "x1", "type": "range", "bounds": [-5.0, 10.0]},
        {"name": "x2", "type": "range", "bounds": [0.0, 10.0]},
    ],
    objective_name="branin",
    minimize=True,
)

for _ in range(15):
    parameters, trial_index = ax_client.get_next_trial()
    results = branin(parameters["x1"], parameters["x2"])
    ax_client.complete_trial(trial_index=trial_index, raw_data=results)


best_parameters, metrics = ax_client.get_best_parameters()
```

21 lines

3x

62 lines

Pairs well with LLMs

https://honegumi.readthedocs.io/

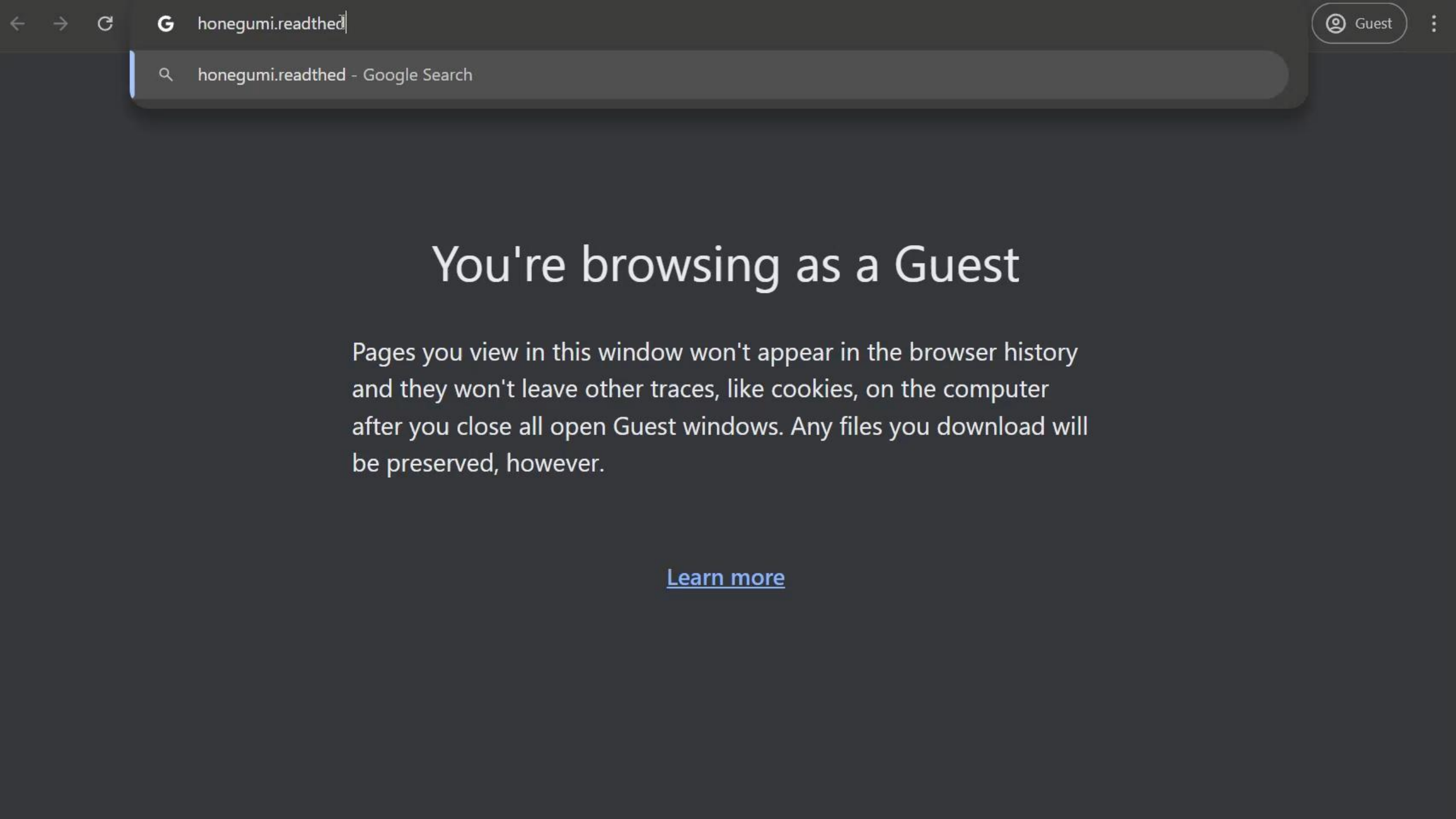
Honegumi: An Interface for Accelerating the Adoption of Bayesian Optimization in the Experimental Sciences

Sterling G. Baird[a,b], Andrew R. Falkowski[a], Taylor D. Sparks[a,b]







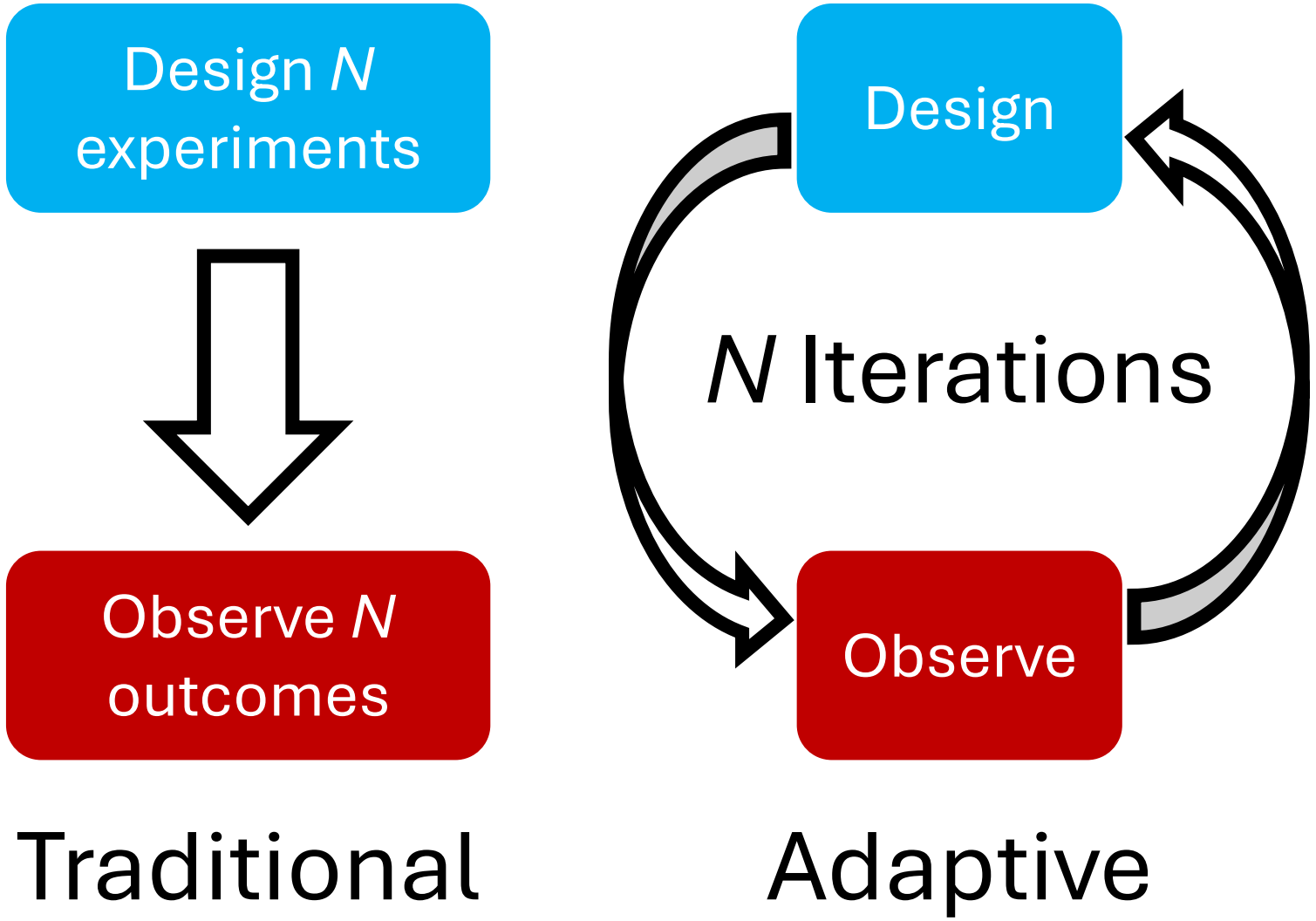

honegumi.readthed
honegumi.readthed - Google Search
Guest
You're browsing as a Guest
Pages you view in this window won't appear in the browser history and they won't leave other traces, like cookies, on the computer after you close all open Guest windows. Any files you download will be preserved, however.
Learn more

The future is increasingly looking
autonomous, generative, agentic, and human-collaborative

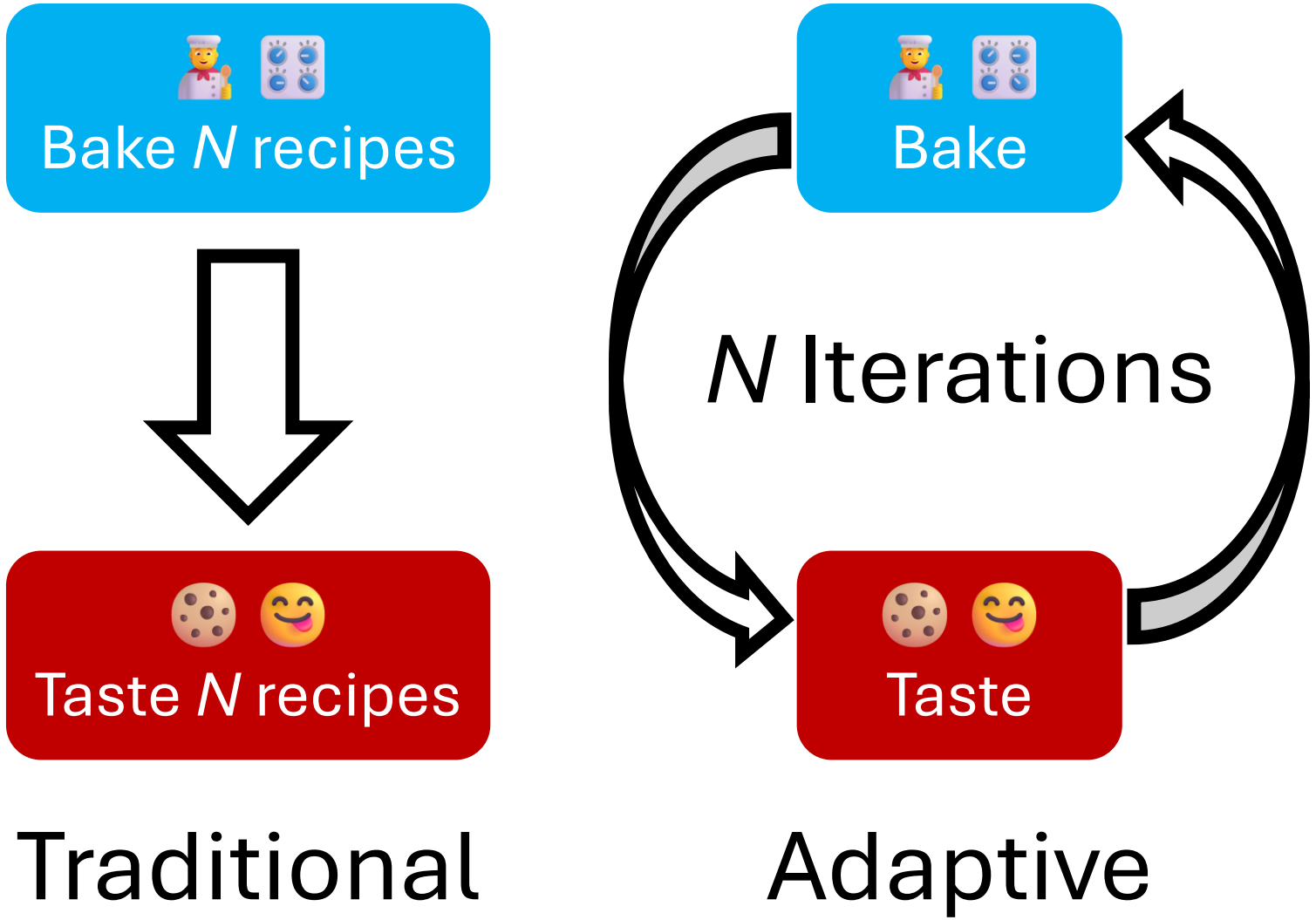
What is your research problem?

[Discuss in groups of 2-3]

Adaptive experimentation is a more efficient version of design of experiments

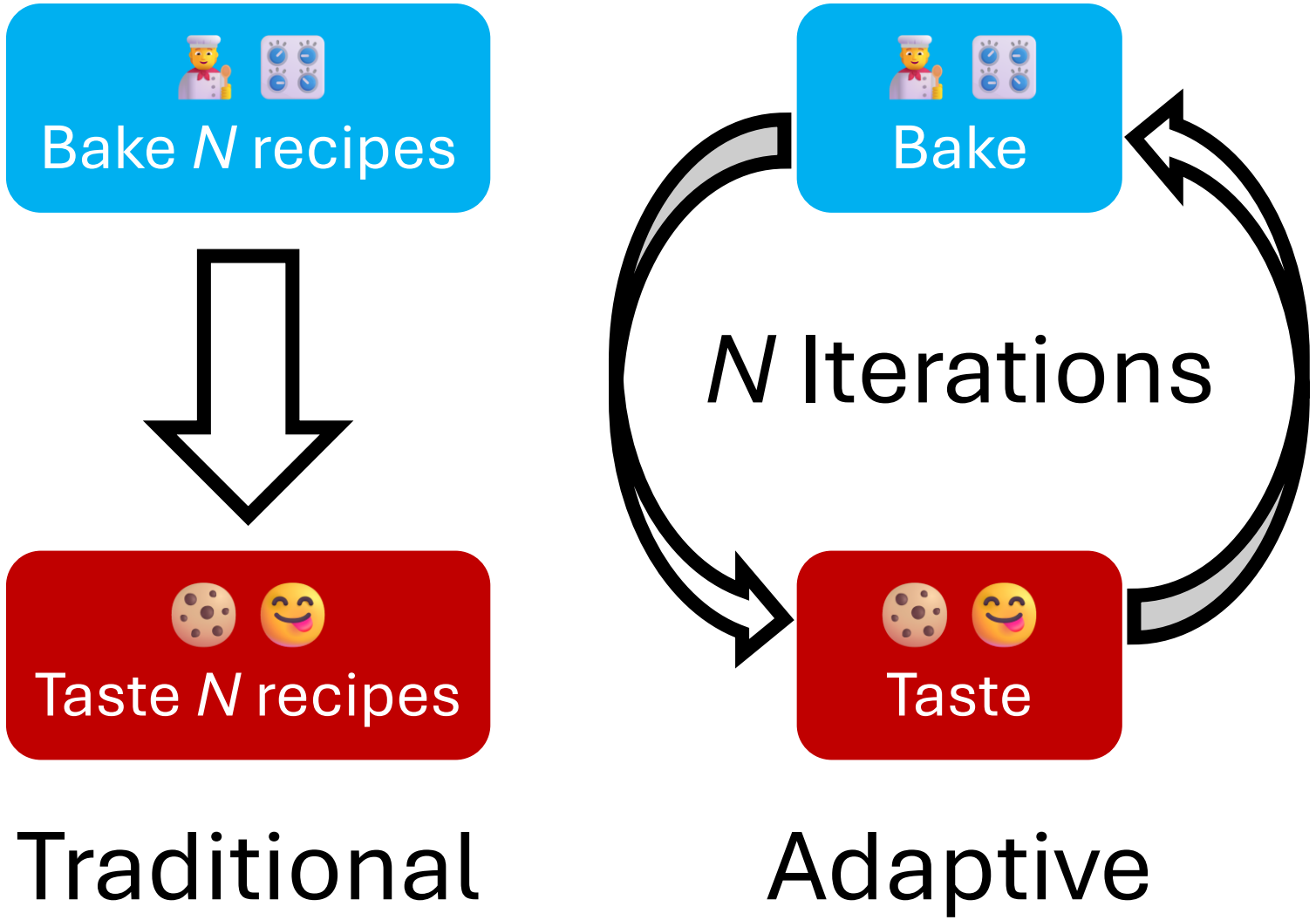


Adaptive experimentation is a more efficient version of design of experiments

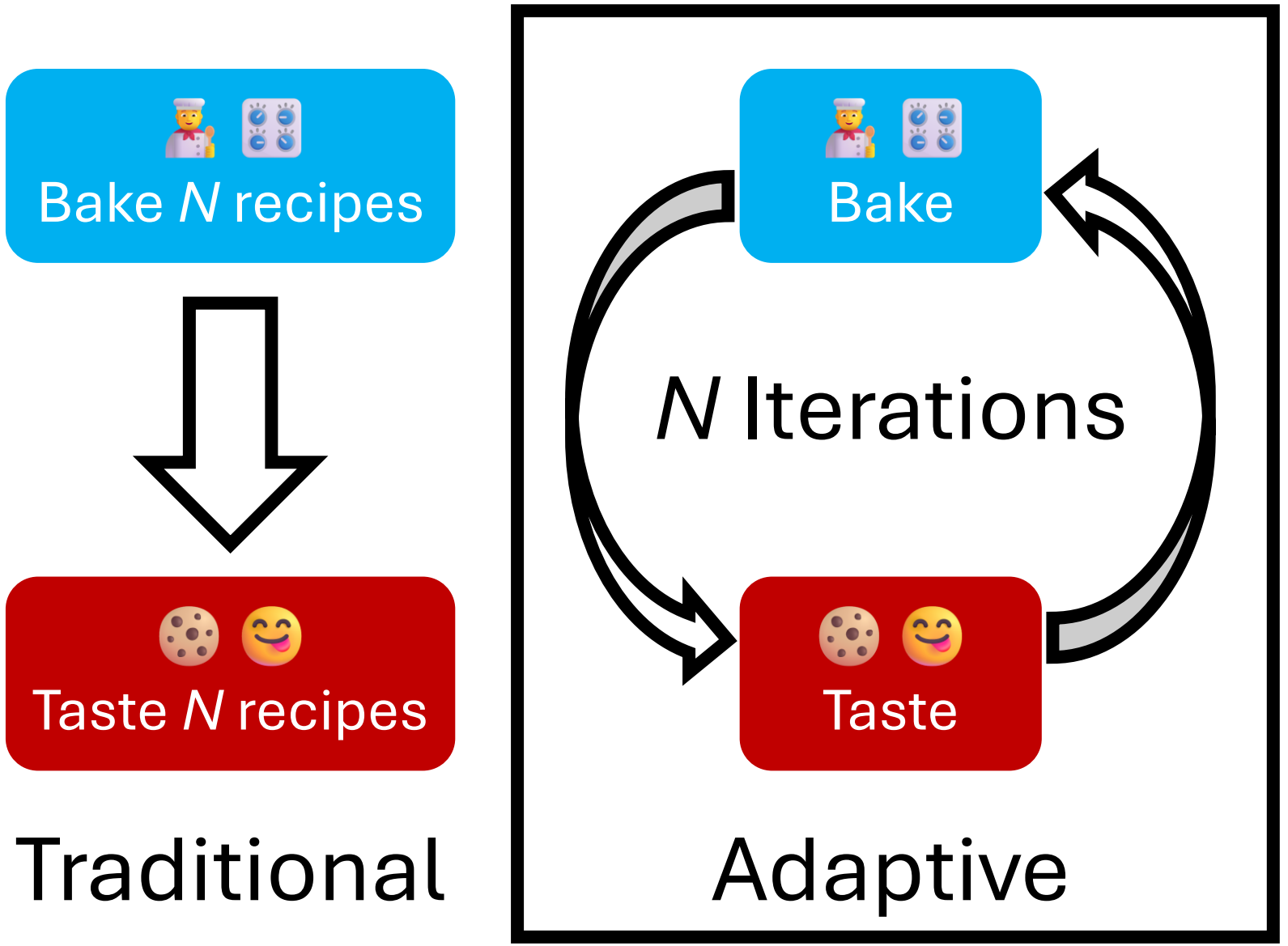




How did Marie find the optimal cookie?



How did Marie find the optimal cookie?



What are Marie’s **objectives**?

What **parameter** did she change?

What are the **constraints**?

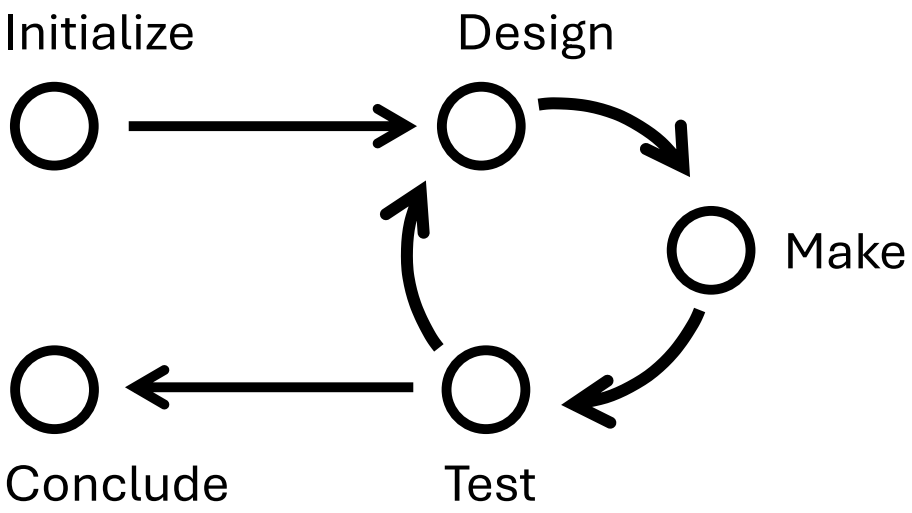
Parameters+Constraints=**Search Space**

A materials optimization task is defined by its **search space and objectives**

What is your materials discovery task?

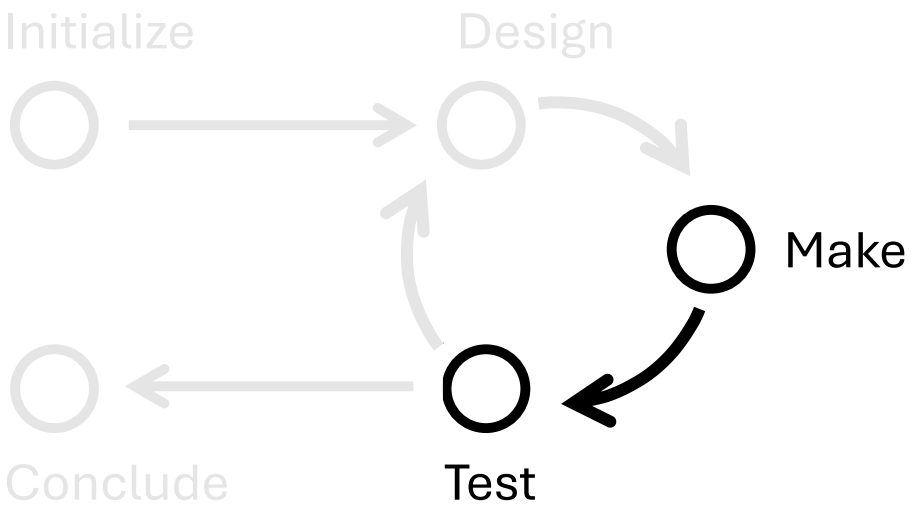
[Discuss in groups of 2-3]

So, you want to build a self-driving lab?



1. AI
   to maximize value
2. Automation
   to minimize burden
3. Integration
   to scale quickly

So, you want to build a self-driving lab?



1. AI
   to maximize value
2. Automation
   to minimize burden
3. Integration
   to scale quickly



Automation is used in organic small molecule discovery

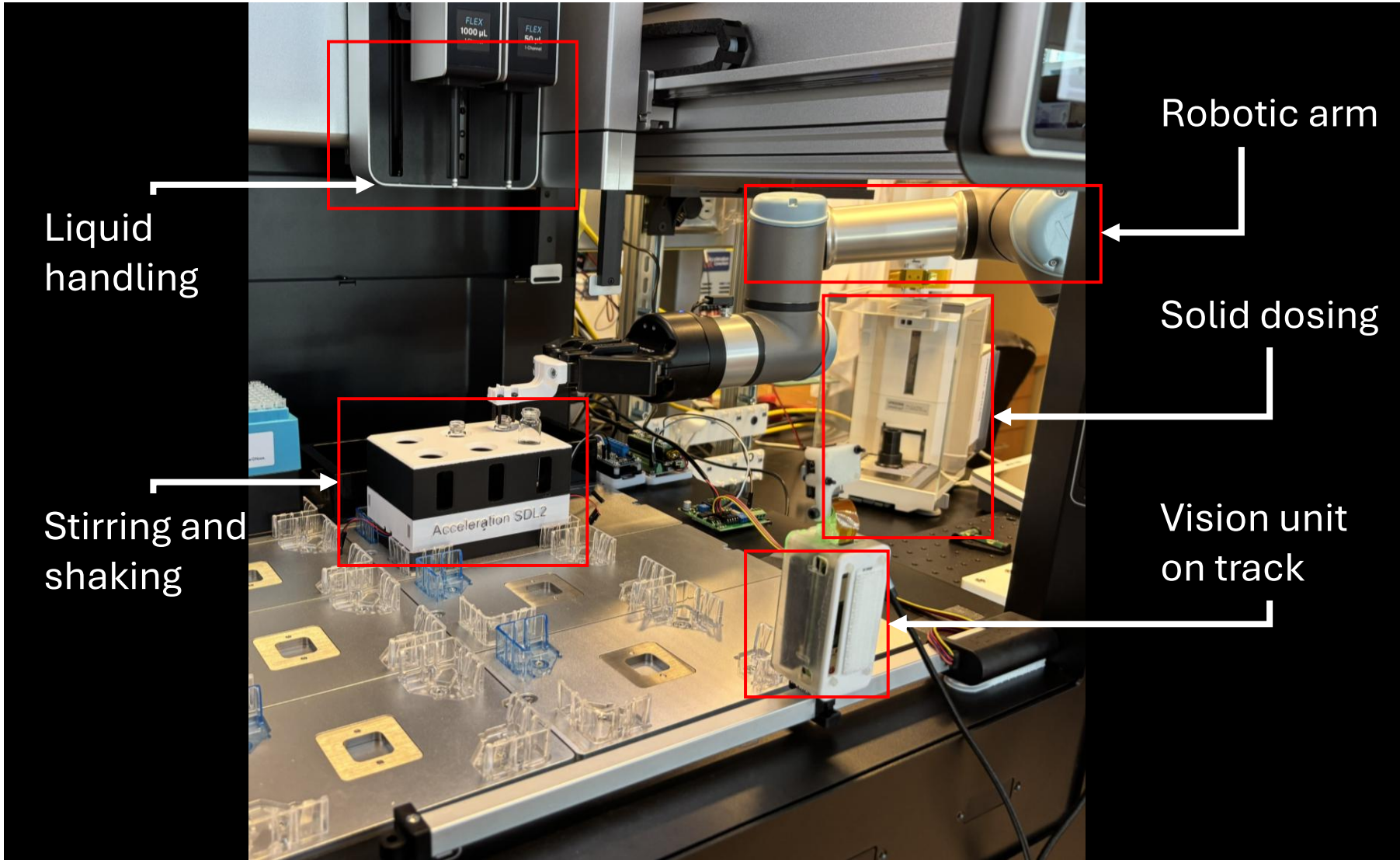
Sample preparation

↓

Reaction

↓

Work-up



Organic Lab @ AC

Automation is used in organic small molecule discovery

Sample preparation

↓

Reaction

↓

Work-up



Synthesis
(high-throughput)

Heater/shaker

Organic Lab @ AC

Automation is used in organic small molecule discovery

Sample preparation

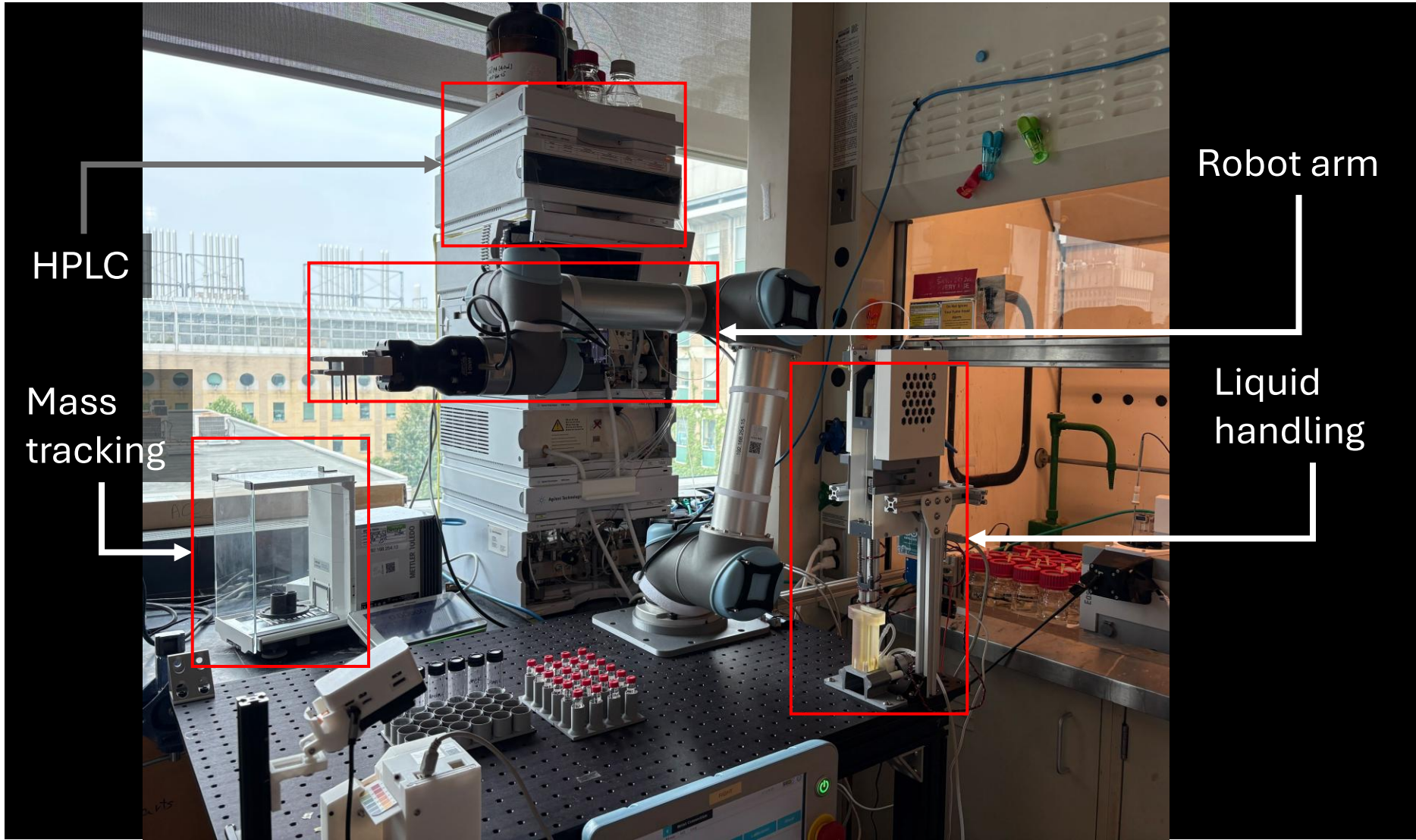
↓

Reaction

↓

Work-up

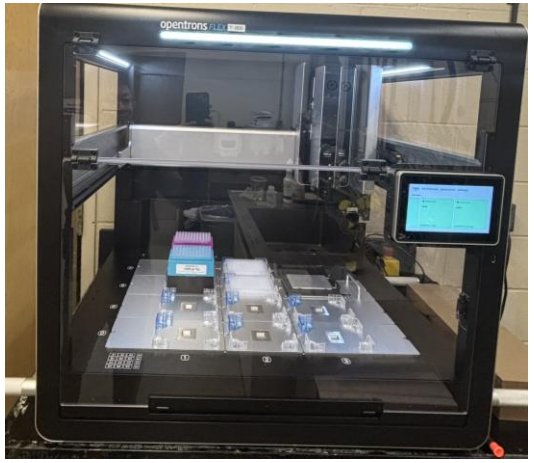
Organic Lab @ AC



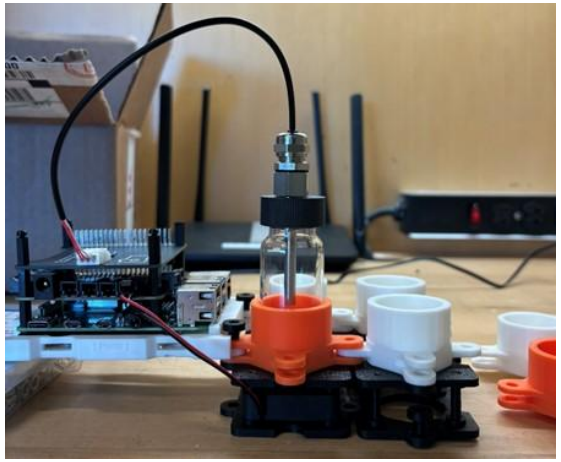
A likely challenge will be finding middle-paths between buy vs. build



All-in-One



Device Integration



Fully Custom

A likely challenge will be finding middle-paths between …

Research outputs	vs	Service/Infrastructure

A likely challenge will be finding middle-paths between …

Research outputs	vs	Service/Infrastructure
Time to deliverable	vs	Documentation

A likely challenge will be finding middle-paths between …

Research outputs	vs	Service/Infrastructure
Time to deliverable	vs	Documentation
Tailored solutions	vs	Universal conventions

A likely challenge will be finding middle-paths between …

Research outputs	vs	Service/Infrastructure
Time to deliverable	vs	Documentation
Tailored solutions	vs	Universal conventions
Autonomy level	vs	Human intervention

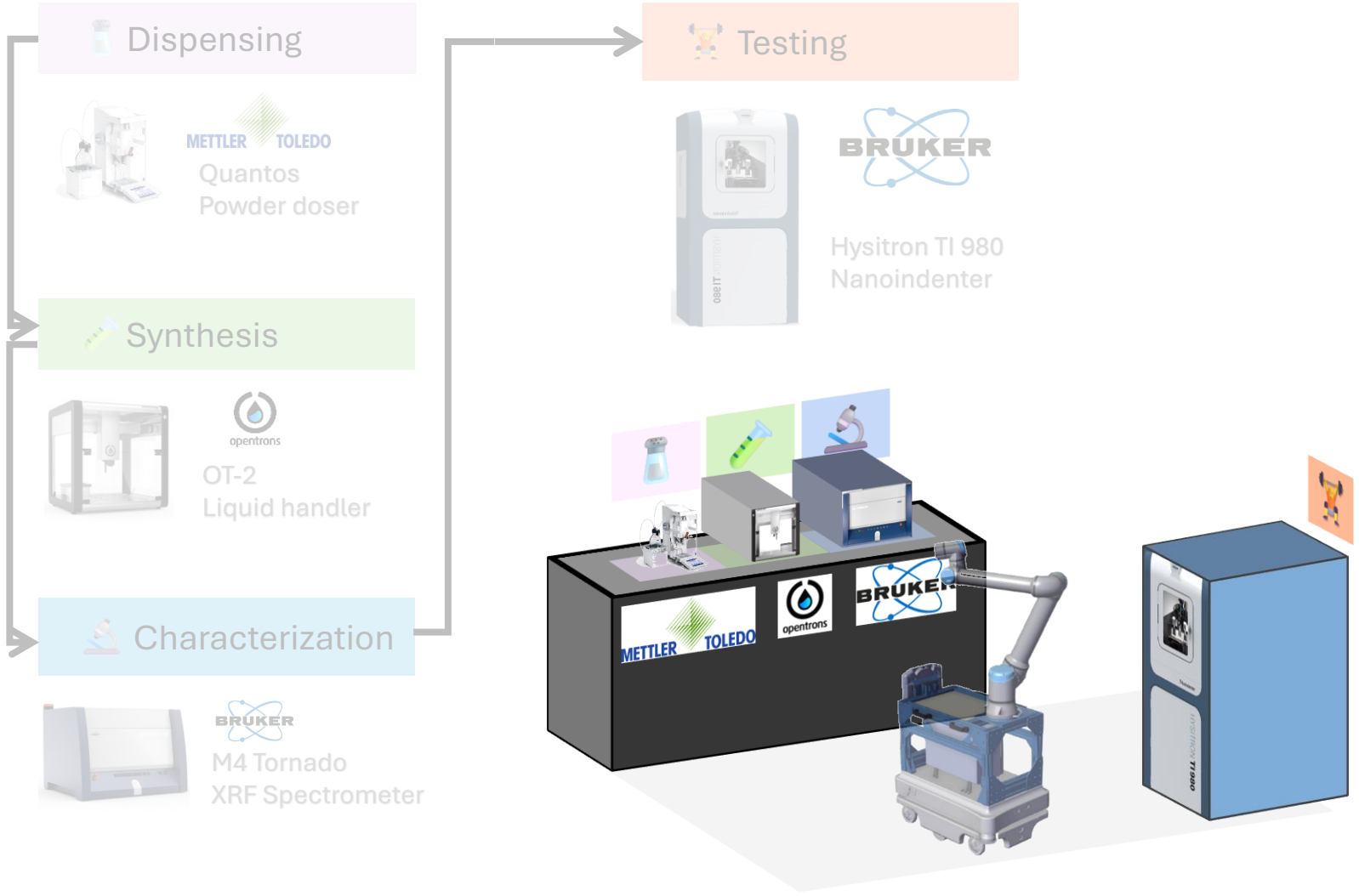
A likely challenge will be finding middle-paths between …

Research outputs	vs	Service/Infrastructure
Time to deliverable	vs	Documentation
Tailored solutions	vs	Universal conventions
Autonomy level	vs	Human intervention
Equipment Cost	vs	Research Fidelity

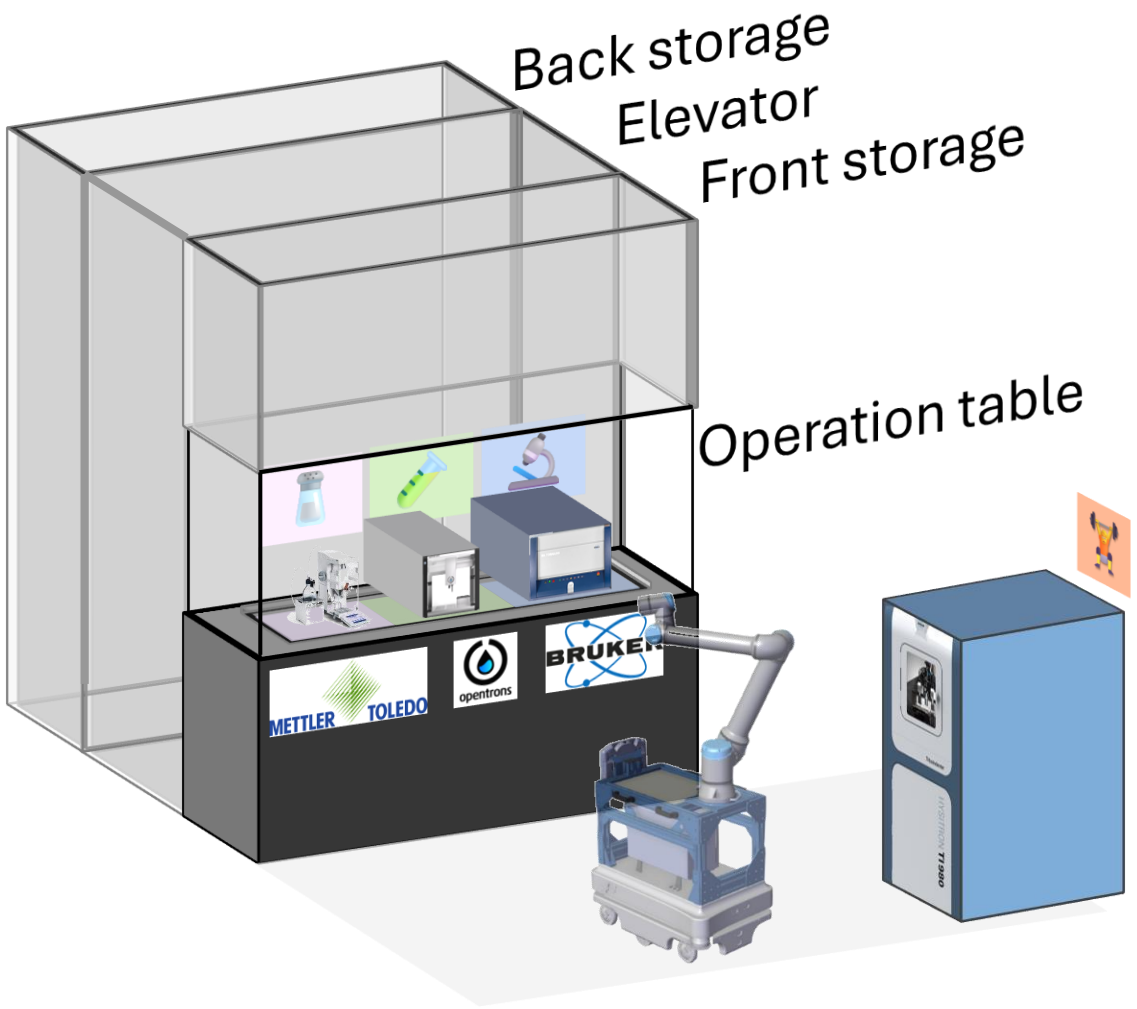
A likely challenge will be finding middle-paths between ...

Research outputs	vs	Service/Infrastructure
Time to deliverable	vs	Documentation
Tailored solutions	vs	Universal conventions
Autonomy level	vs	Human intervention
Equipment Cost	vs	Research Fidelity
Open-source dev	vs	Closed-source dev

In a vertical lab, equipment is brought to the operator



In a vertical lab, equipment is brought to the operator



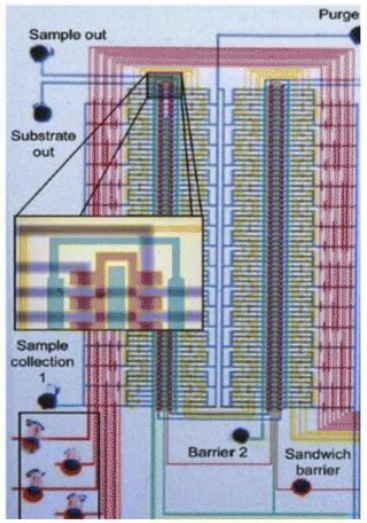
What is the hardest step to automate in your workflow?

[Discuss in groups of 2-3]

Hardware can be designed for speed or flexibility

Increased speed and reliability

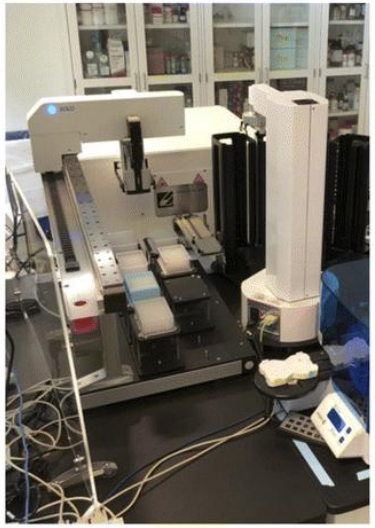
Increased flexibility



integrated



fixed



flexible



configurable

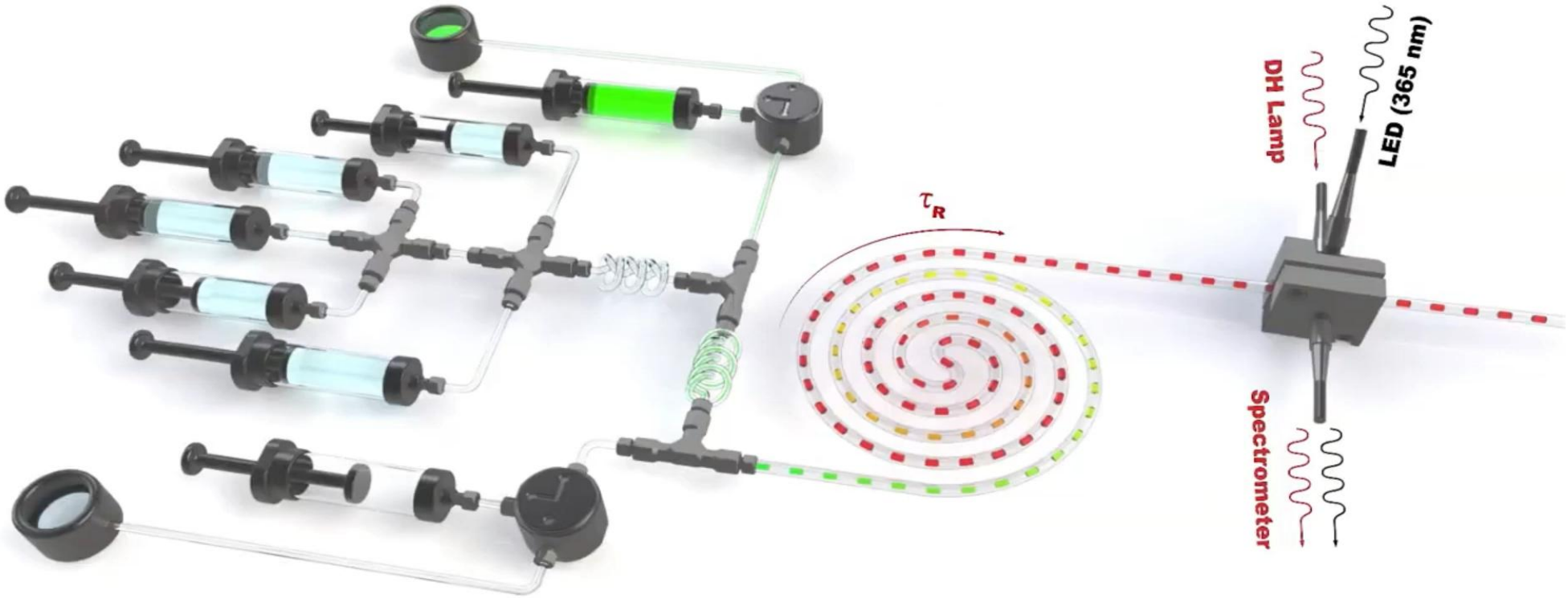


mobile



manual

Epps et al. shows an integrated setup for nanoparticle synthesis



Epps et al. *Advanced Materials* (2020)

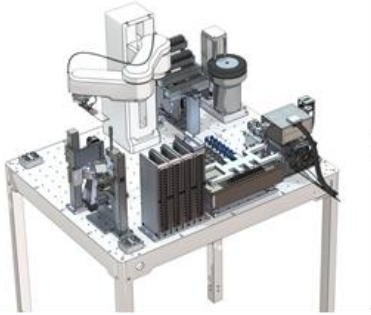
Lunt et al. shows a flexible setup for automated solid-state synthesis and powder X-ray diffraction



Lunt et al. *Chemical Science* (2024)

MacLeod et al. shows an evolving, hybrid setup with standalone modules connected by robot arms

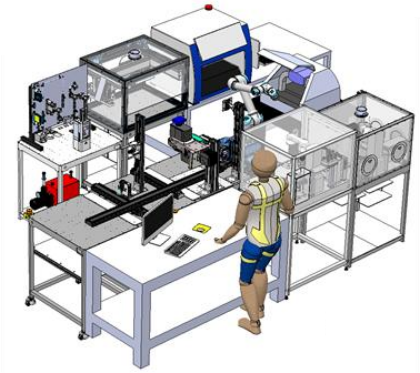
2018



Ada

solar cells

2021



Ada-catalyst

electrocatalysts

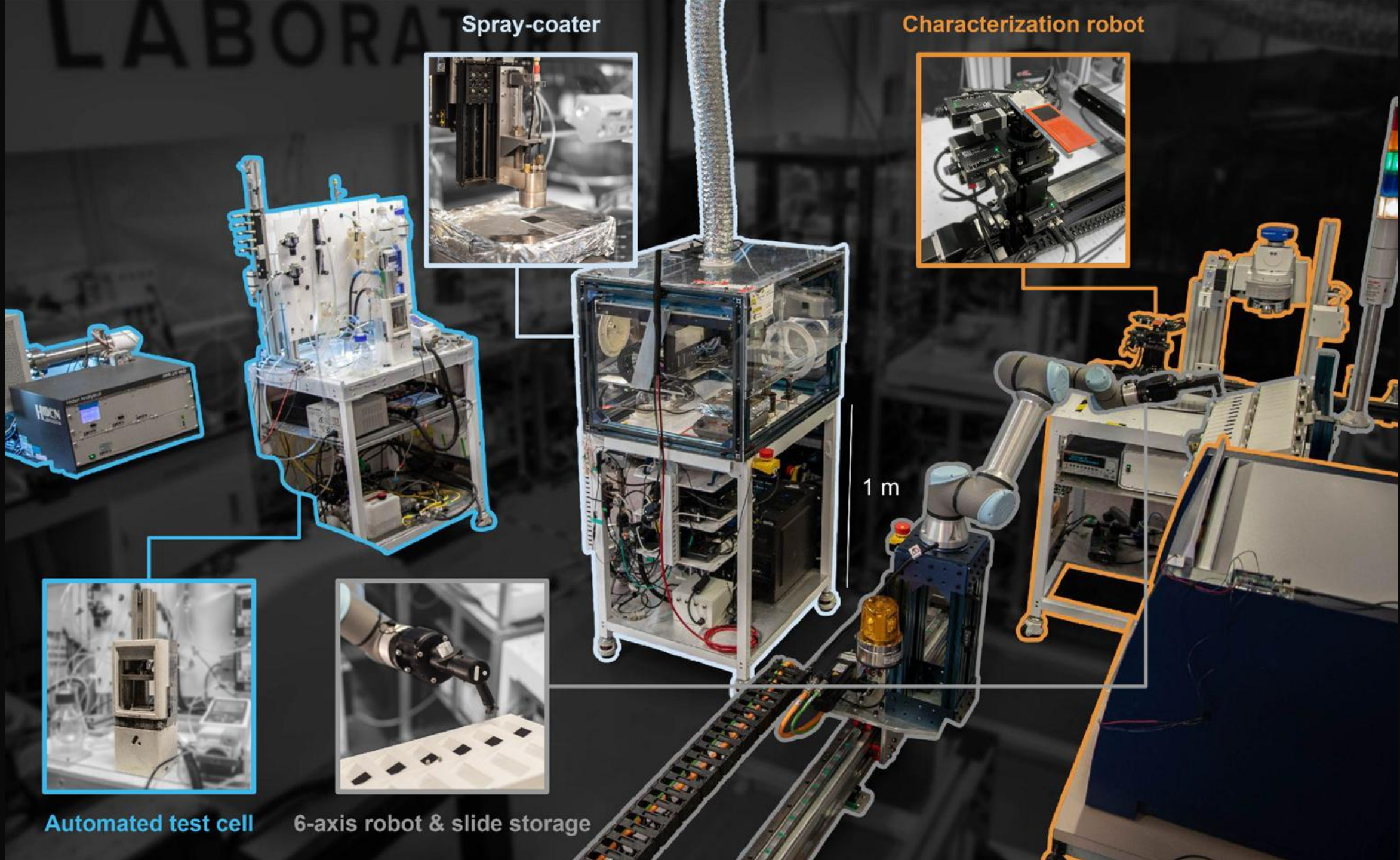
2024

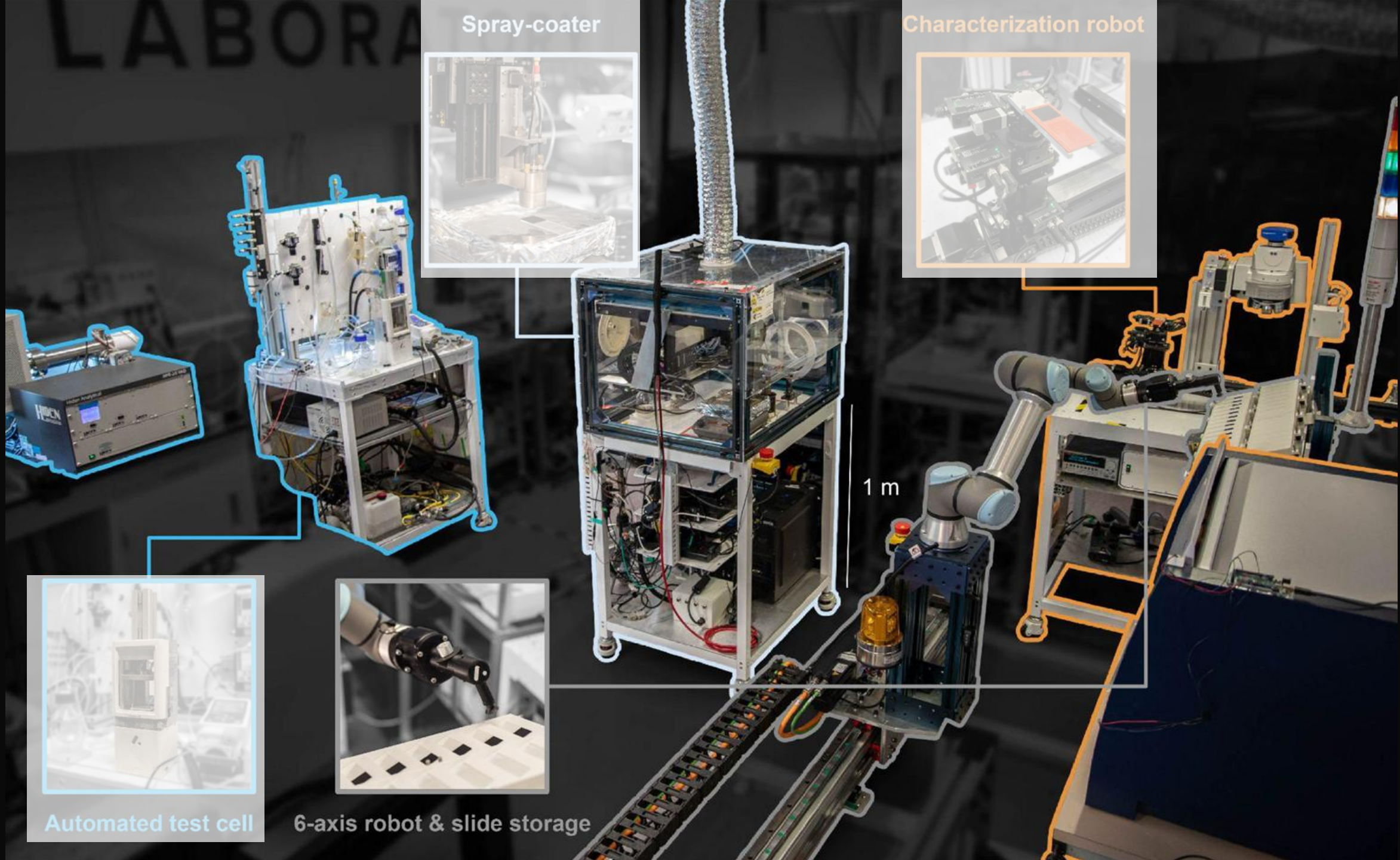


Ada-carbon

gas diffusion electrodes

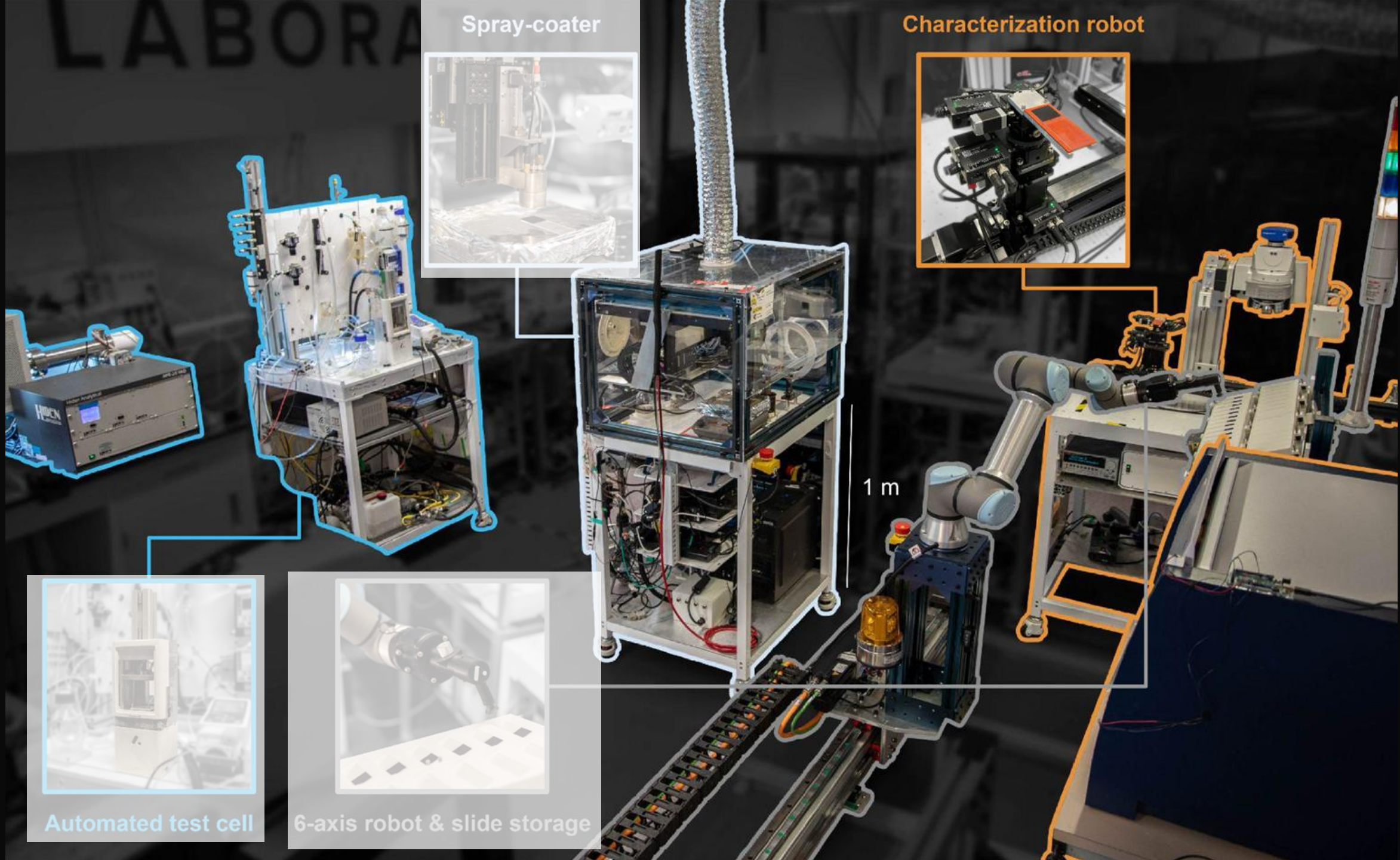
MacLeod et al. *Sci Adv* (2020)
MacLeod et al. *Nat Commun* (2022)
Rupnow et al. *Cell Rep Phys Sci* (2023)


Spray-coater
Characterization robot
1 m
Automated test cell
6-axis robot & slide storage

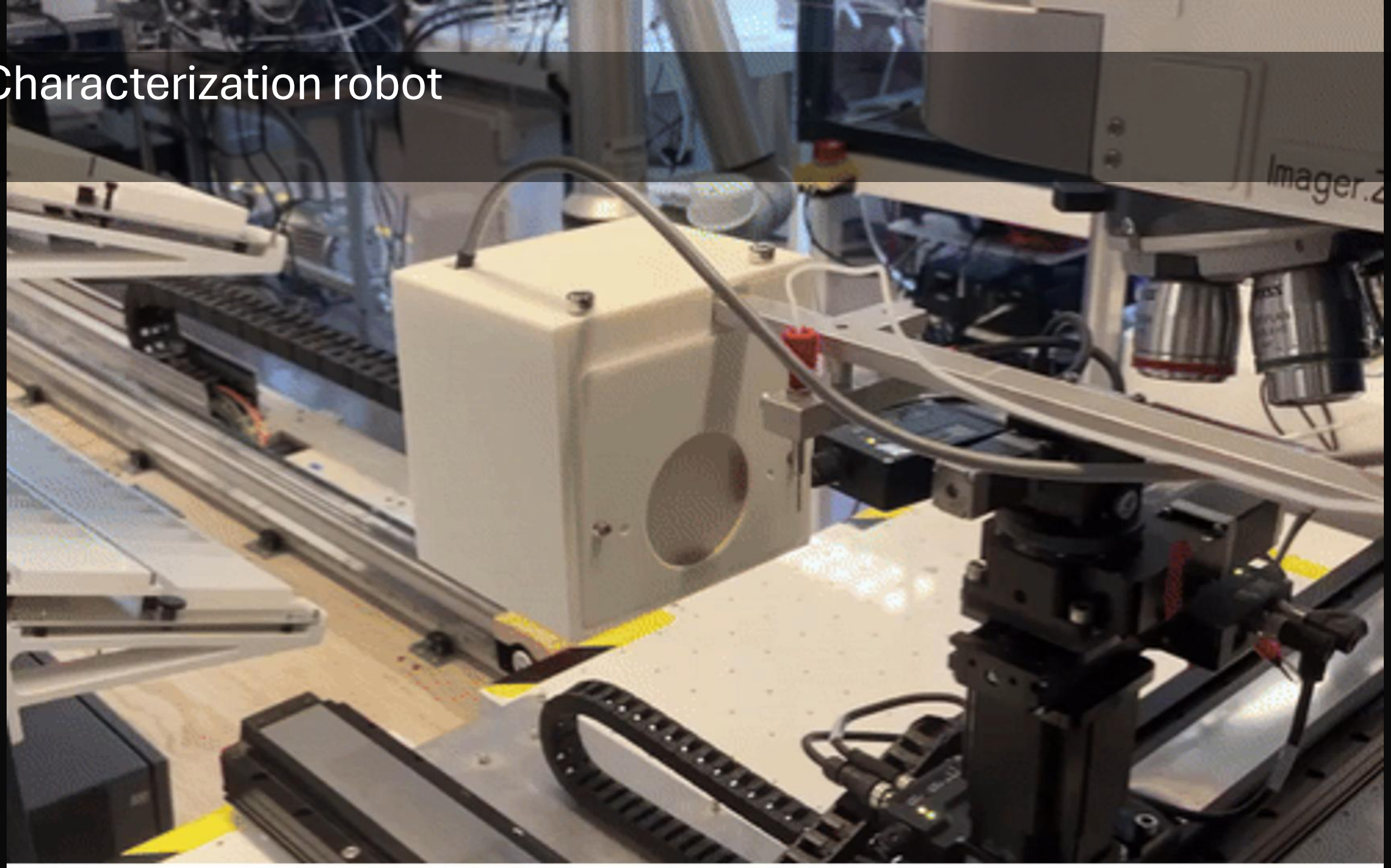

Spray-coater
Characterization robot
1 m
Automated test cell
6-axis robot & slide storage

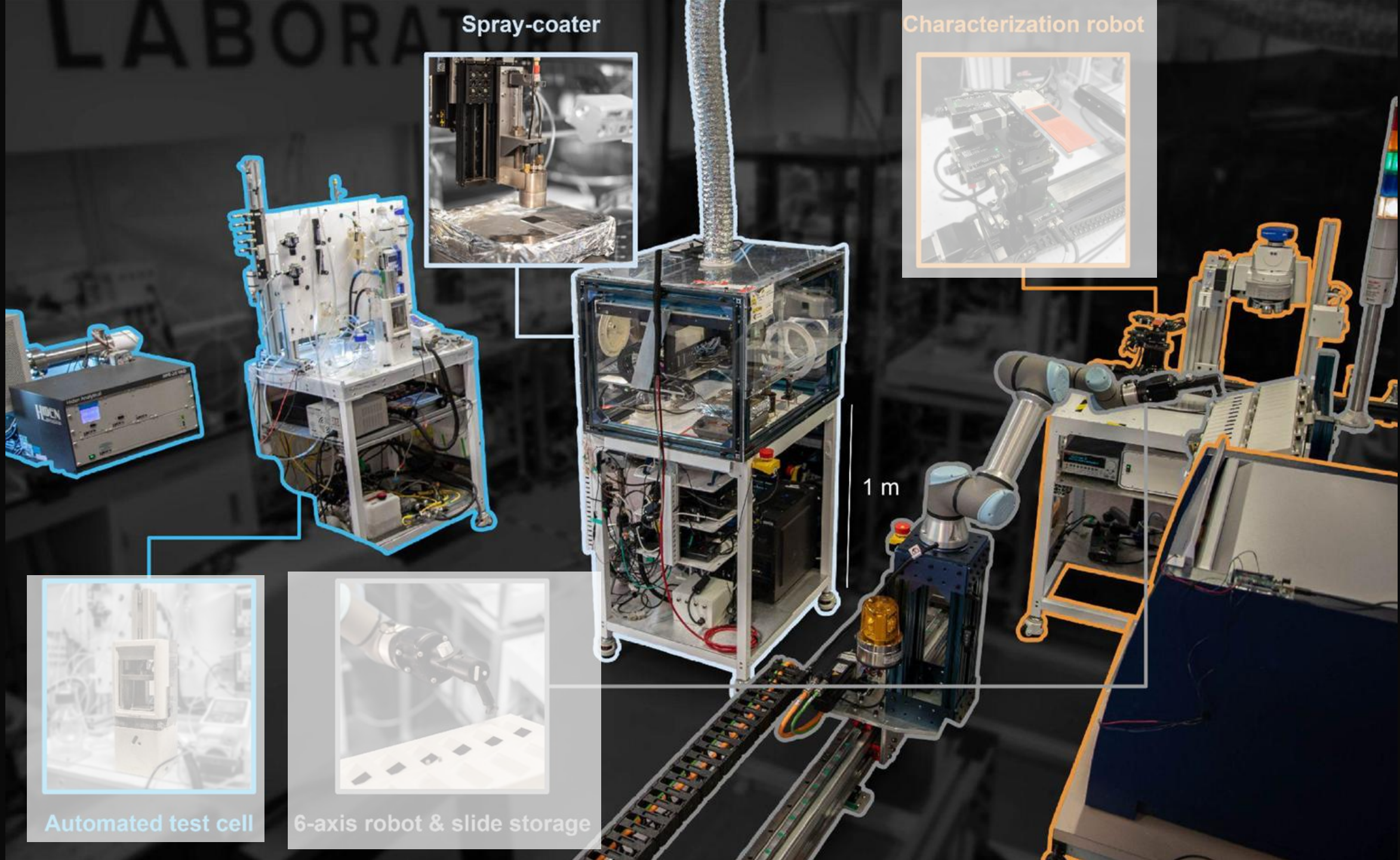
6-axis robot & slide storage



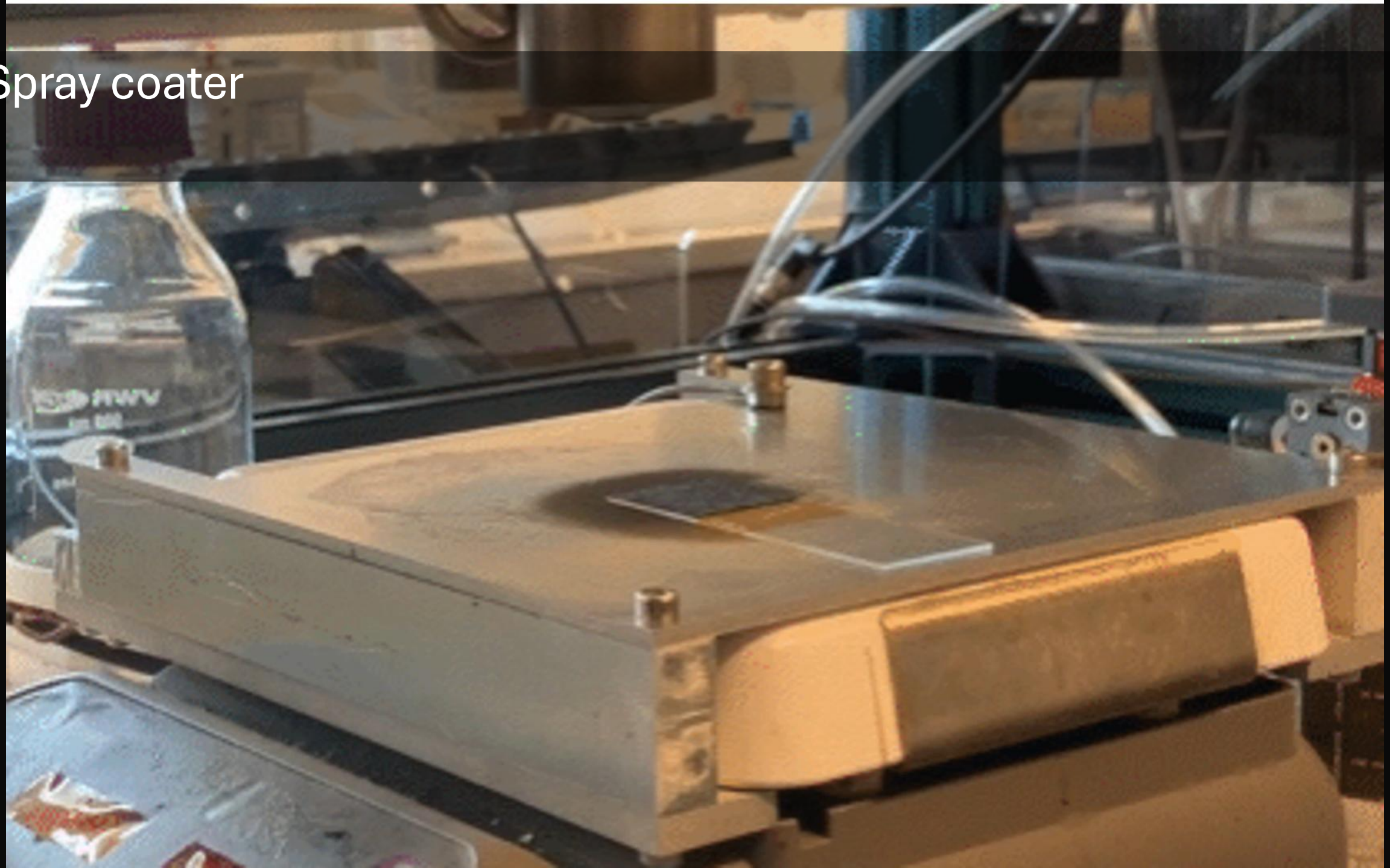

Spray-coater
Characterization robot
1 m
Automated test cell
6-axis robot & slide storage

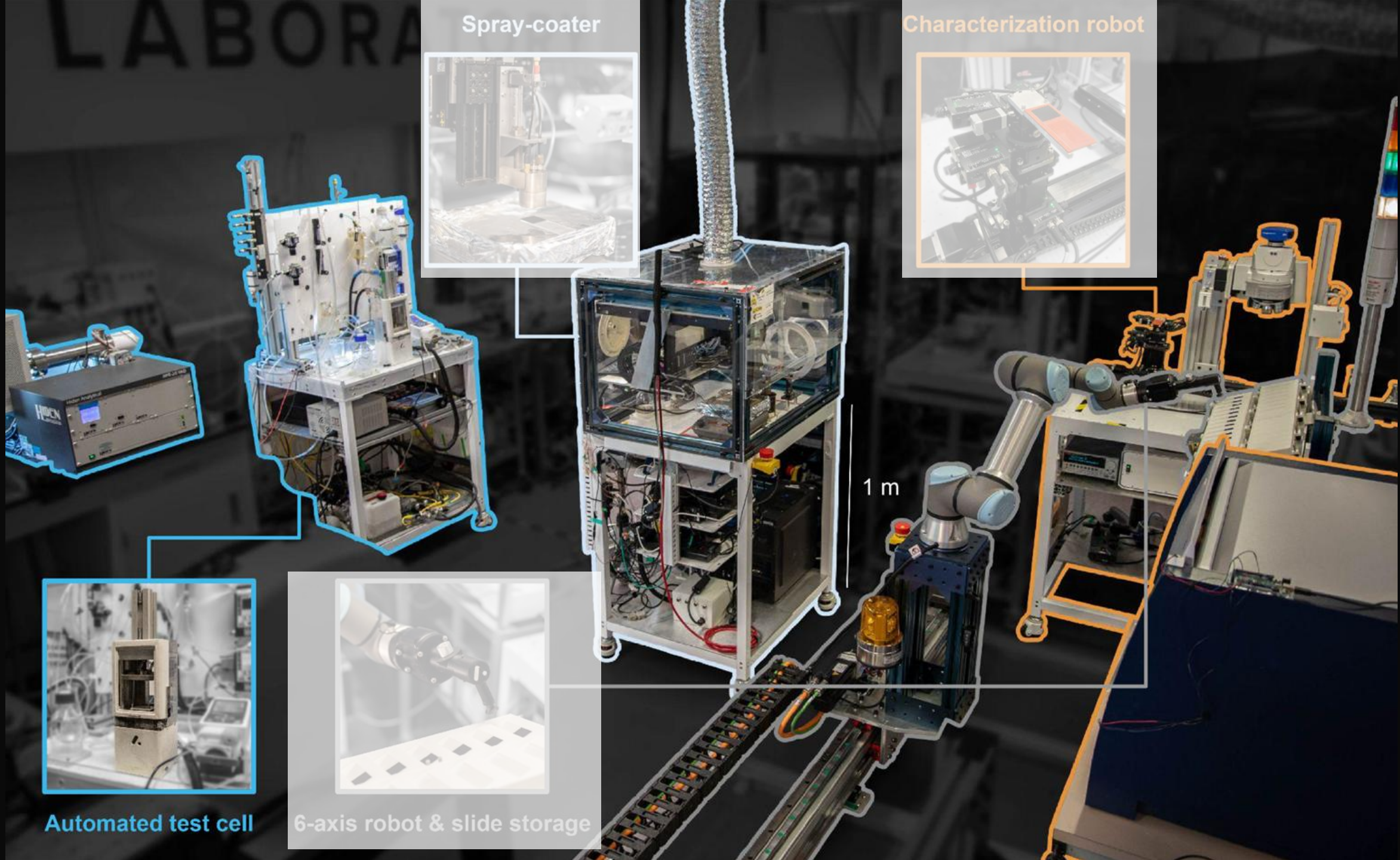
Characterization robot




Spray-coater
Characterization robot
1 m
Automated test cell
6-axis robot & slide storage

Spray coater

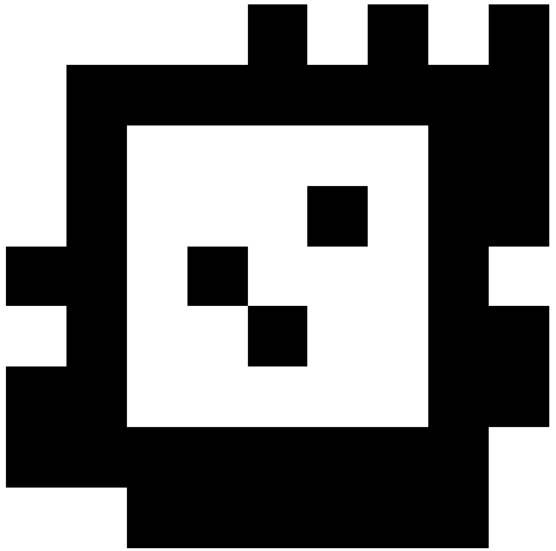



Spray-coater
Characterization robot
1 m
Automated test cell
6-axis robot & slide storage

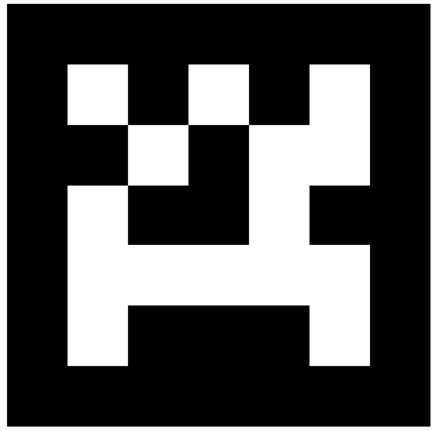
Automated test cell



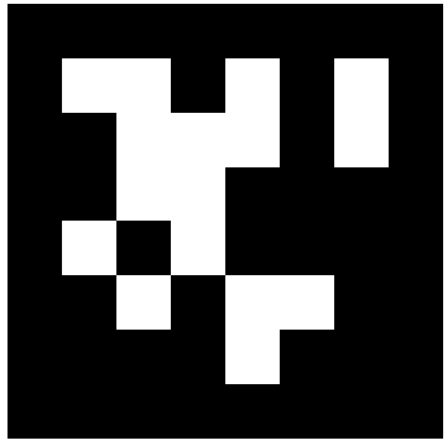
AprilTags are a high-contrast visual marker that help enable flexible and modular SDLs



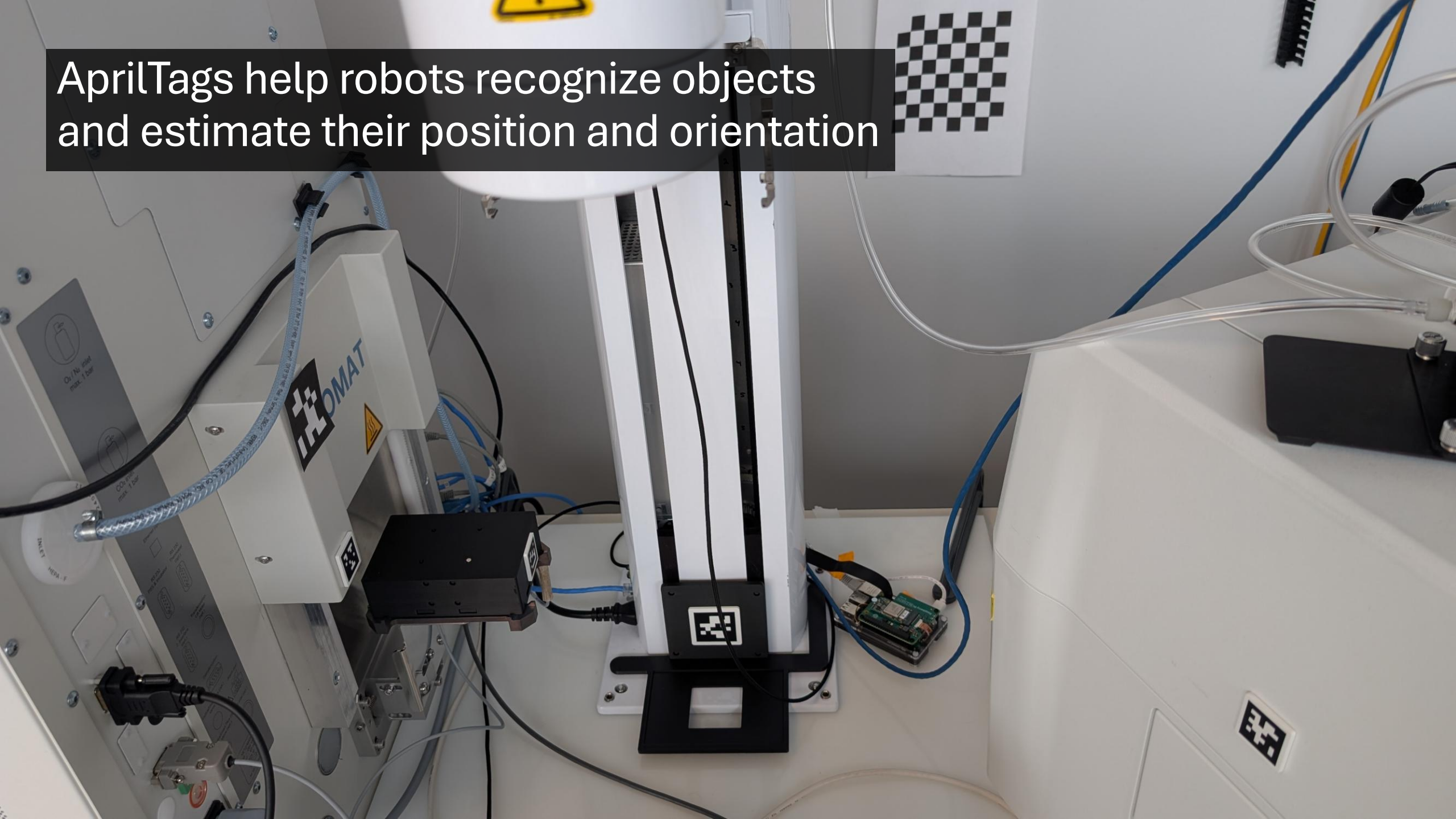
tag41_12_00110



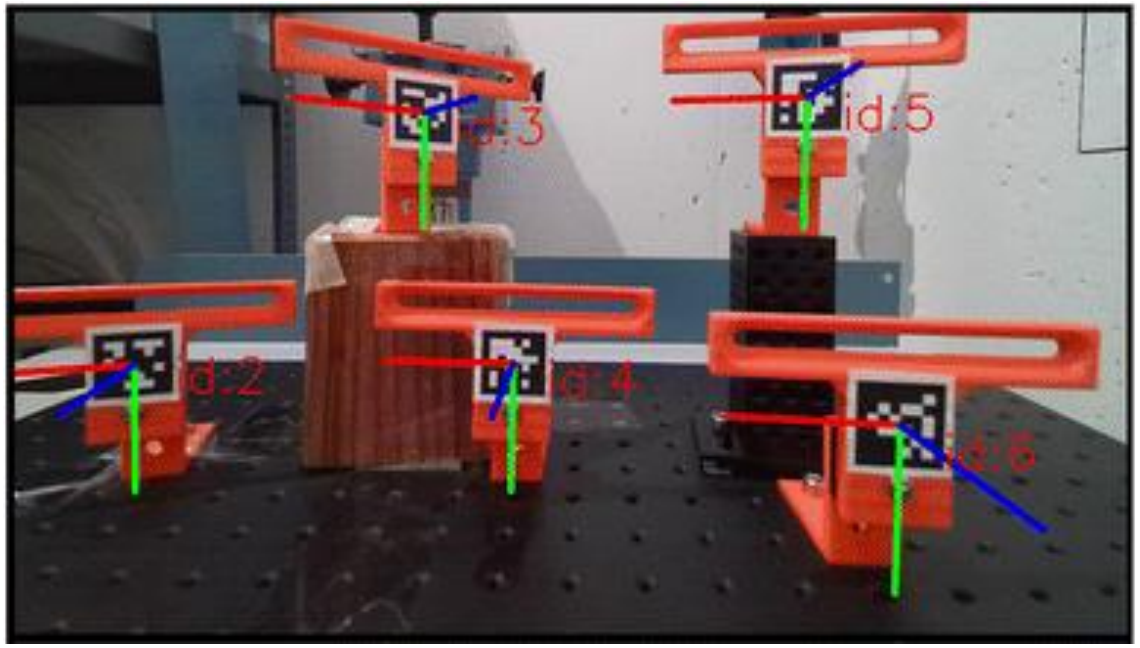
tag25_09_00000

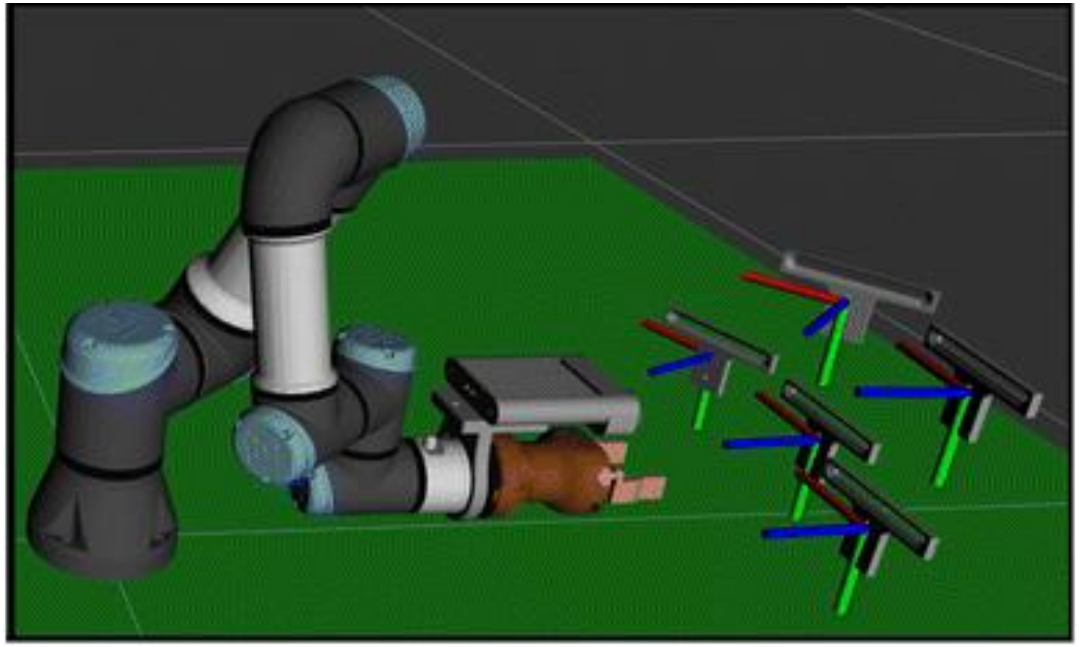


tag36_11_00000


AprilTags help robots recognize objects
and estimate their position and orientation

At Argonne National Laboratory, AprilTags are used to identify and place X-ray experiment samples precisely





Fernando et al., Digital Discovery (2025)

AprilTag tutorials are available at
ac-training-lab.readthedocs.io

Digital Discovery

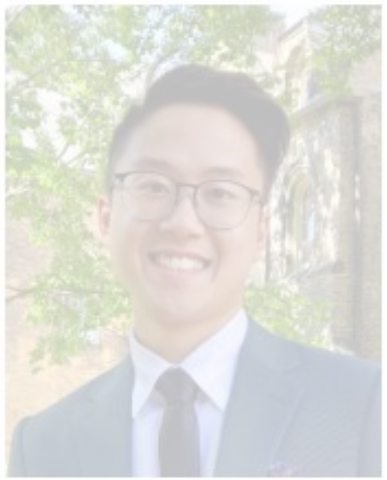
ROYAL SOCIETY OF CHEMISTRY

Check for updates

Cite this: *Digital Discovery*, 2024, 3, 842

Review of low-cost self-driving laboratories in chemistry and materials science: the "frugal twin" concept

Stanley Lo, †*[a] Sterling G. Baird, †*[bc] Joshua Schrier, [d] Ben Blaiszik, [eg] Nessa Carson, [f] Ian Foster, [eg] Andrés Aguilar-Granda, [h] Sergei V. Kalinin, [io] Benji Maruyama,[j] Maria Politi, [k] Helen Tran, [abl] Taylor D. Sparks †*[cm] and Alán Aspuru-Guzik †*[abln]



Stanley Lo

Stanley Lo obtained his BSc degree in Chemistry at the University of Toronto in 2022. Prior to starting graduate studies, he spent 4 months at Kebotix Inc. developing graph neural network architecture for molecular property prediction. Currently, he is a doctoral student under the joint supervision of Alán Aspuru-Guzik and Helen Tran, focusing on the automation of polymer post-functionalization for depolymerization.

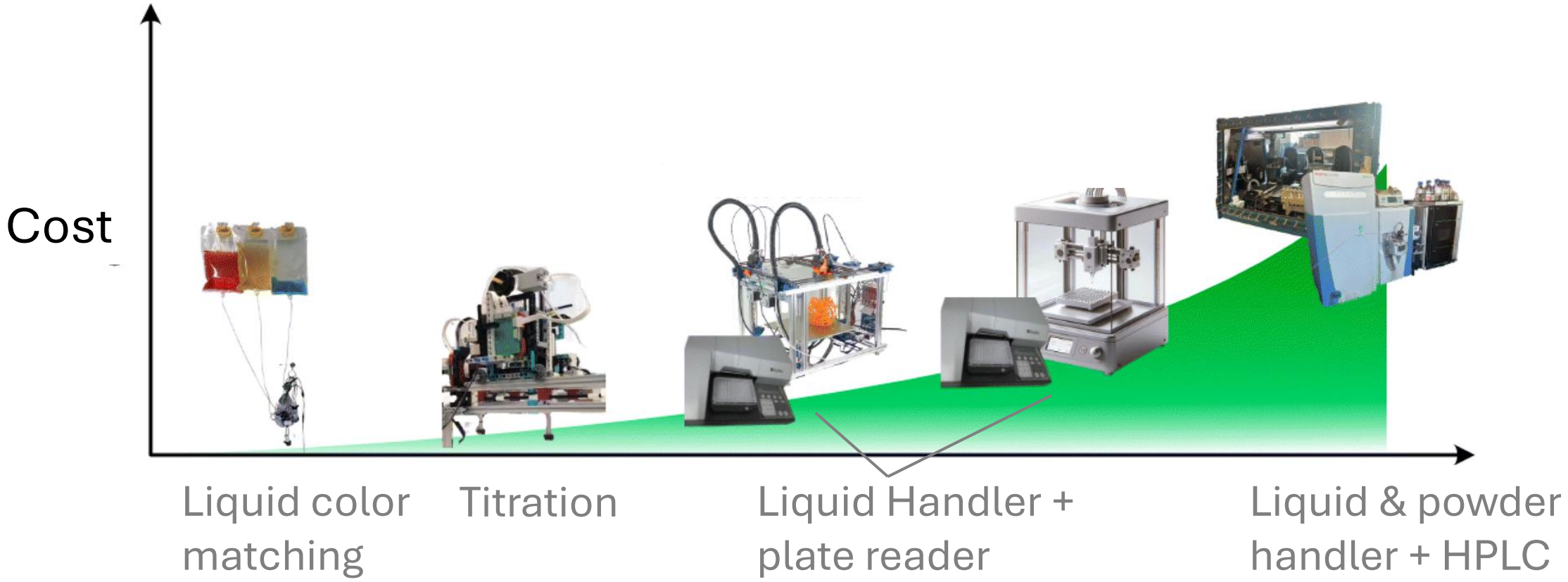


Sterling G. Baird

Sterling G. Baird is the Director of Training and Programs at the Acceleration Consortium within the University of Toronto. He received his BSc in Applied Physics and his MSc in Mechanical Engineering at Brigham Young University. He completed his PhD in Materials Science & Engineering at the University of Utah. Sterling is accelerating materials discovery through advanced Bayesian optimization, self-driving laboratories, and educational platforms.

A frugal twin SDL is a low-cost counterpart to a high-cost SDL

for education, prototyping, and multi-fidelity research



Lo*; **Baird***; et al. *Digital Discovery* (2024)

A Ph.D. student designed and built an electrocatalyst SDL at the AC in three months for under $5000, then replicated it after the stay



Nis Fisker-Bødker

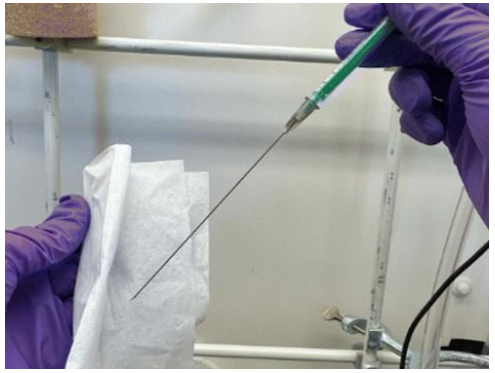
Other contributors:
Daniel Persaud
Jae Hattrick-Simpers

Reactor cleaning

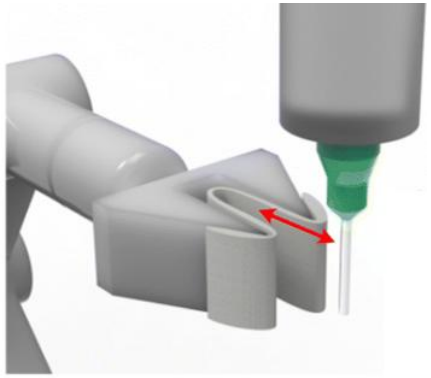
Fisker-Bødker, *DD* (2025

We can use existing hardware in less intuitive ways
e.g., when cleaning a syringe or mixing liquids

Human-in-the-loop



Human-inspired



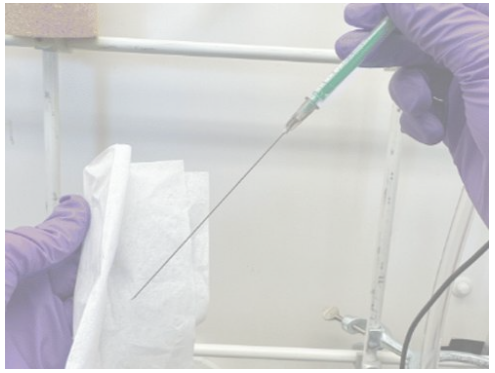
Hardware-centric

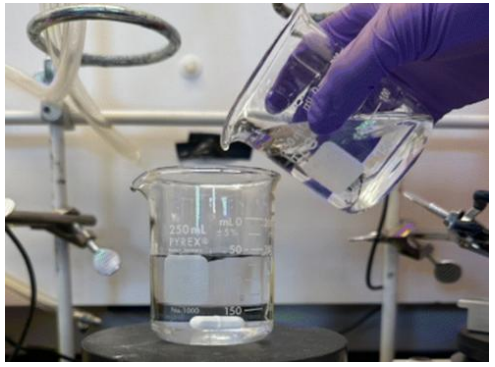


Lo*; **Baird***; et al. *Digital Discovery* (2024)

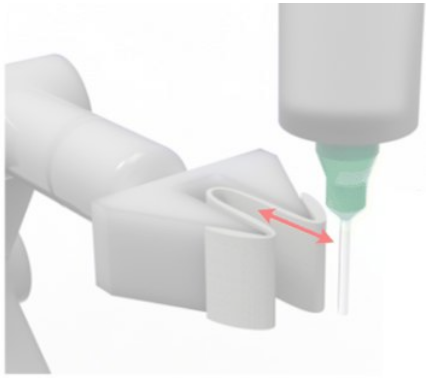
We can use existing hardware in less intuitive ways
e.g., when cleaning a syringe or mixing liquids

Human-in-the-loop



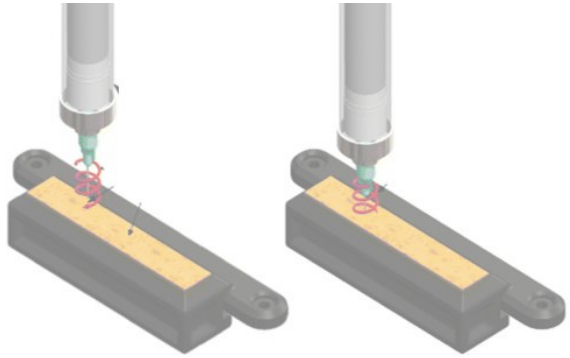


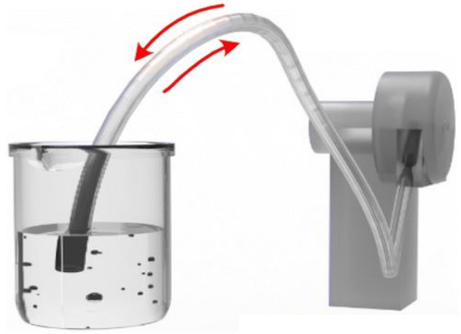
Human-inspired





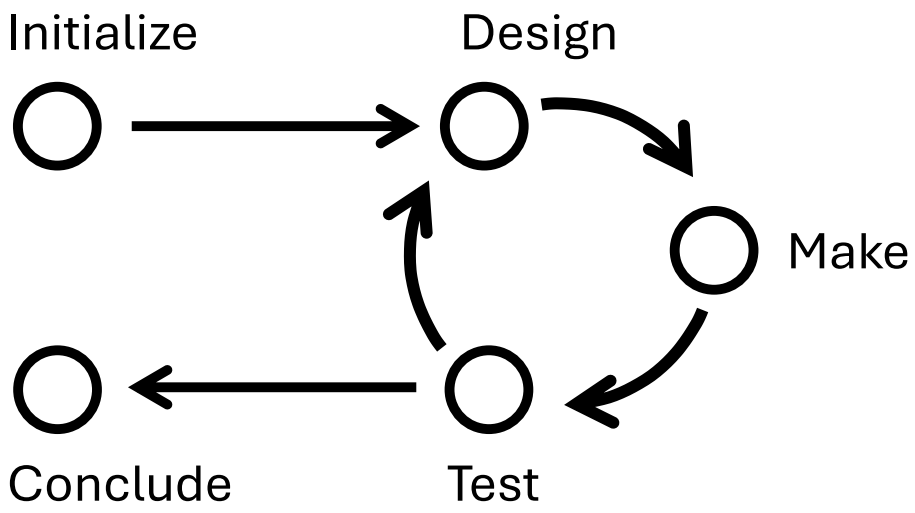
Hardware-centric





Lo*; **Baird***; et al. *Digital Discovery* (2024)

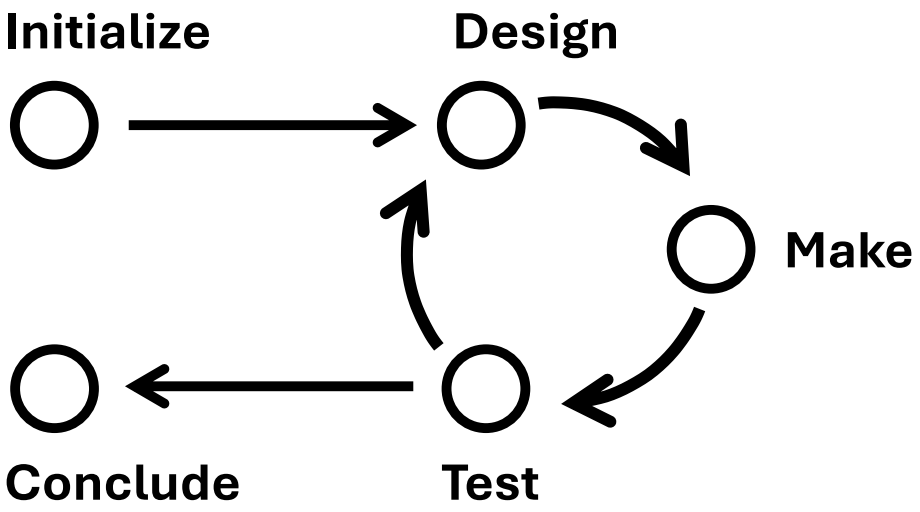
So, you want to build a self-driving lab?



1. AI
   to maximize value
2. Automation
   to minimize burden
3. Integration
   to scale quickly



So, you want to build a self-driving lab?

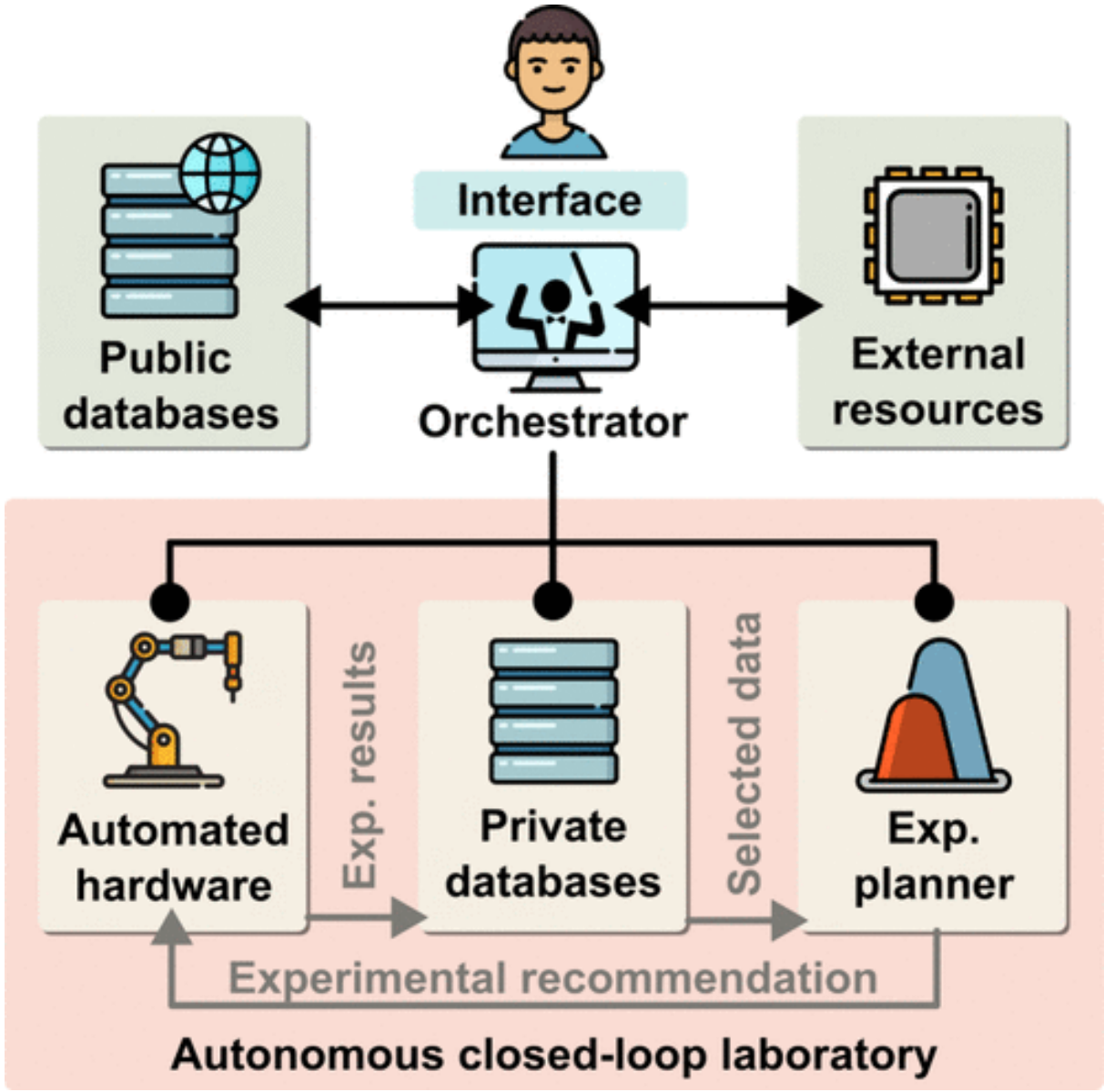


1. AI
   to maximize value
2. Automation
   to minimize burden
3. Integration
   to scale quickly



Orchestrators coordinate the design-make-test cycle

workflow planning
instrument control
data analysis



Tom et al. *Chem. Rev.* (2024)

This color matching SDL integrates two devices: a liquid handler and a wireless color sensor tool

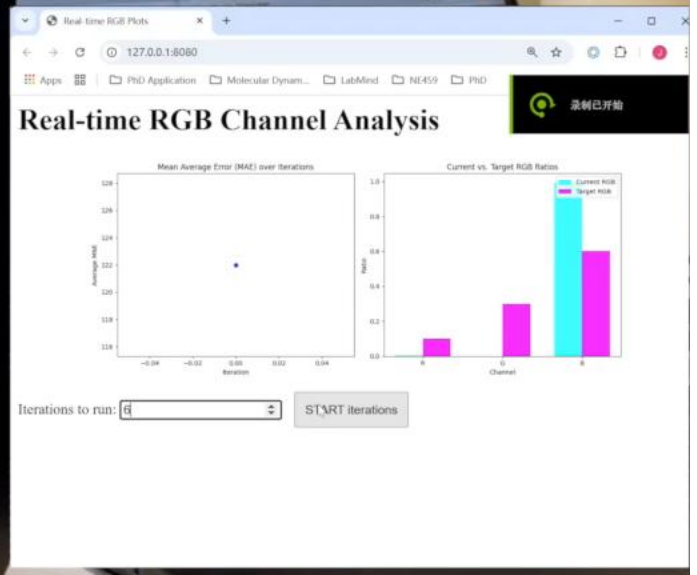
Wireless charging base for 8-channel spectrophotometer



Yanghuang Liu

RYB Colors





This color matching self-driving lab is set up for remote access

https://huggingface.co/spaces/AccelerationConsortium/OT-2-LCM

https://huggingface.co/spaces/AccelerationConsortium/OT-2-LCM

Guest

You're browsing as a Guest

Pages you view in this window won't appear in the browser history and they won't leave other traces, like cookies, on the computer after you close all open Guest windows. Any files you download will be preserved, however.

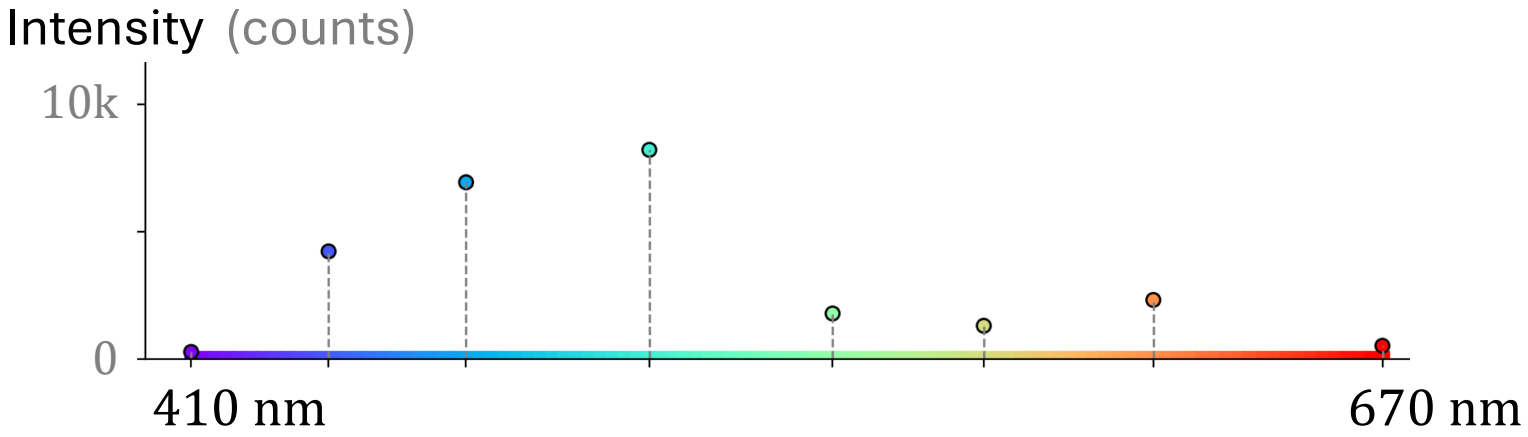
Learn more

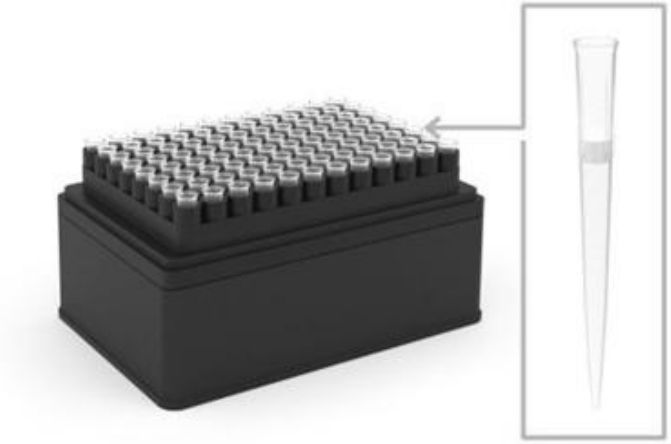
What information do you need to capture in this color-mixing experiment?

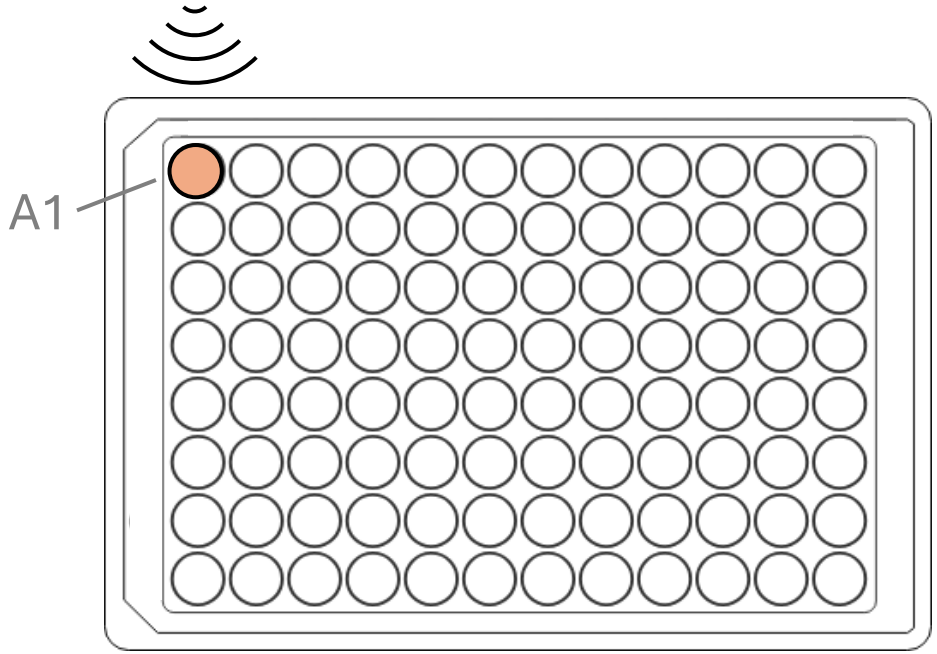


What information do you need to capture in this workflow?

How would you represent this info to a computer? (group Q, 2-3)









How would you represent this info to a computer?

[Discuss in groups of 2-3]

Three common approaches are
document, relational, and graph databases

Document-based databases are simple to set up and intuitive (Python dictionaries) but scale poorly for large, complex systems



```
{
  "_id": "5cf0029caff5056591b0ce7d",
  "firstname": "Jane",
  "lastname": "Wu",
  "address": {
    "street": "1 Circle Rd",
    "city": "Los Angeles",
    "state": "CA",
    "zip": "90404"
  }
  "hobbies": ["surfing", "coding"]
}
```



Relational databases are more structured and efficient, but it's difficult to "get the schema right" and later, evolve a schema

Students Table

Student	ID*
John Smith	084
Jane Bloggs	100
John Smith	182
Mark Antony	219

Participants Table

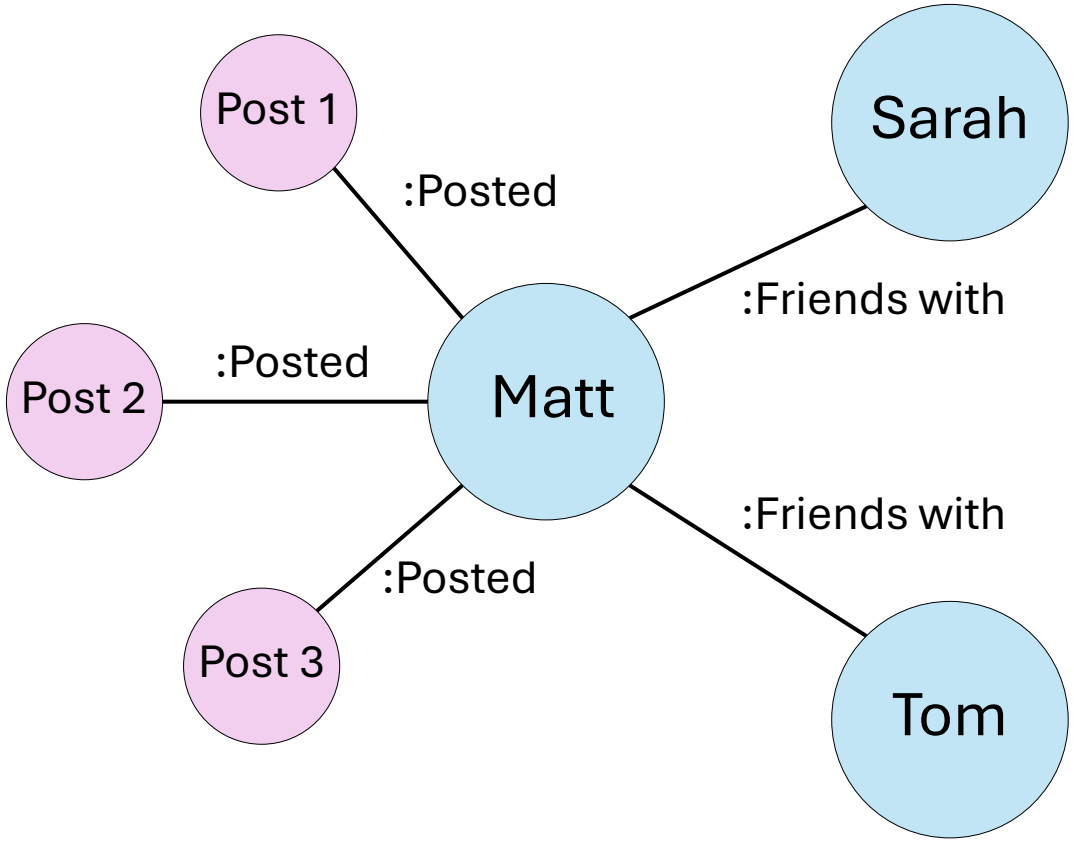
ID*	Activity*
084	Tennis
084	Swimming
100	Squash

Activities Table

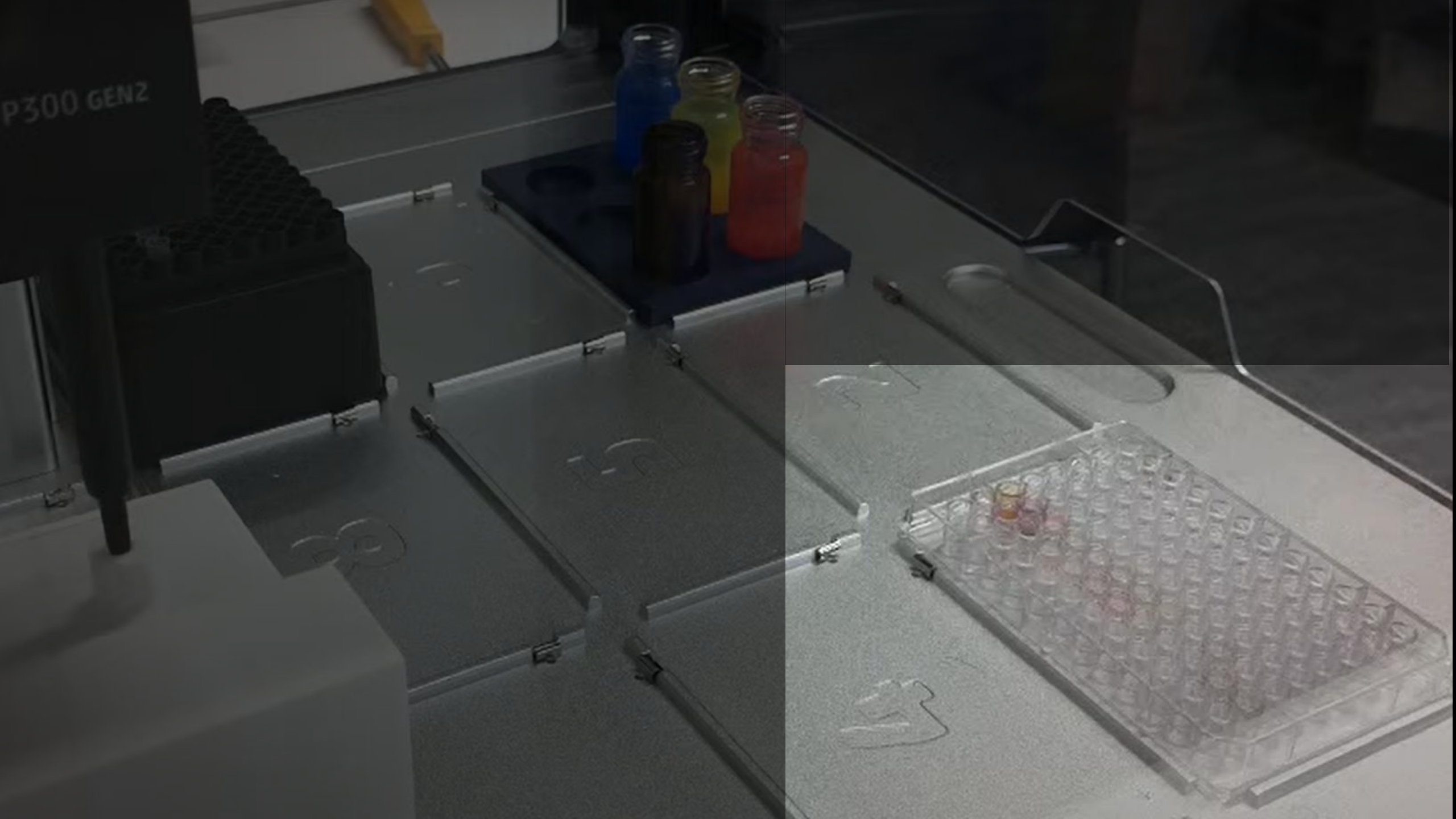
Activity*	Cost
Squash	$40
Swimming	$15
Tennis	$36



Graph databases are the most complex to design and learn but excel at modeling complex relations (e.g., social networks)



neo4j


P300 GEN2



Excuse me human, the wellplate slots are all used.
Please replace the wellplate with a fresh one.

— **Prefect Workflow Orchestrator**

default
Search workspace ⌘ k
Home
Runs
Flows
Deployments
Work pools
Automations
Events
Settings
Help

Deployments / wells-maintenance

Run

SUCCESS RATE
52.63%



AVERAGE LATENESS
5s



AVERAGE DURATION
30s



Runs | Upcoming | Parameters | Configuration | Description

22 Flow runs

Search by run name

All except scheduled

Newest to oldest

request-wells-maintenance > dashing-antelope
Completed 2025/04/14 04:57:44 PM 1 Parameter 20s 1 Task run
Deployment wells-maintenance Work Pool ot2-pool Work Queue default

request-wells-maintenance > stoic-octopus
Completed 2025/04/14 04:49:06 PM 1 Parameter 14s 1 Task run
Deployment wells-maintenance Work Pool ot2-pool Work Queue default

request-wells-maintenance > malachite-tench

Flow request-wells-maintenance
Work Pool ot2-pool
Work Queue default

Schedules
+ Schedule

Triggers
+ Add

Status
Ready

Created
2025/04/14 01:47:28 PM

Created By
omega6-betazed

Last Updated
2025/04/14 02:08:37 PM

Updated By
omega6-betazed

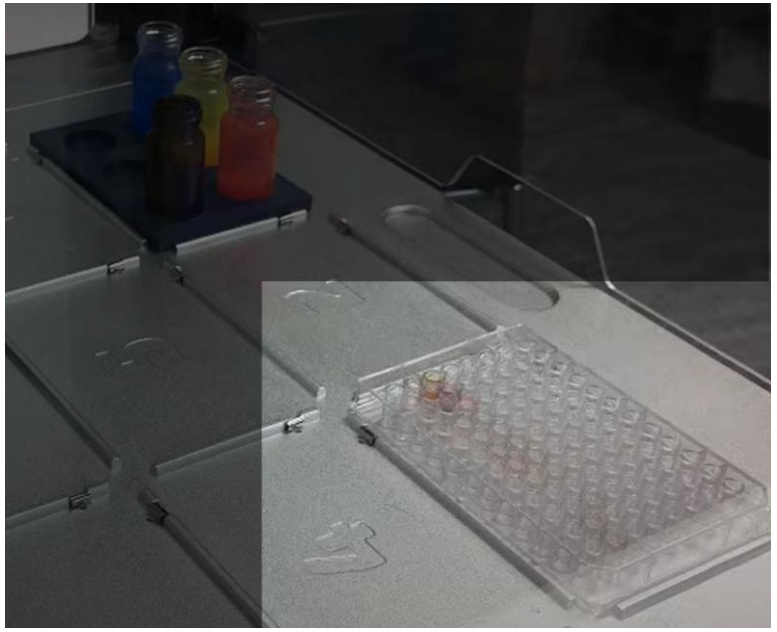
Entrypoint
maintenance_flow.py:request_wells_maintenance

Path
None

Concurrency Limit
None

There are three primary types of human intervention

Maintenance by *The Technician*



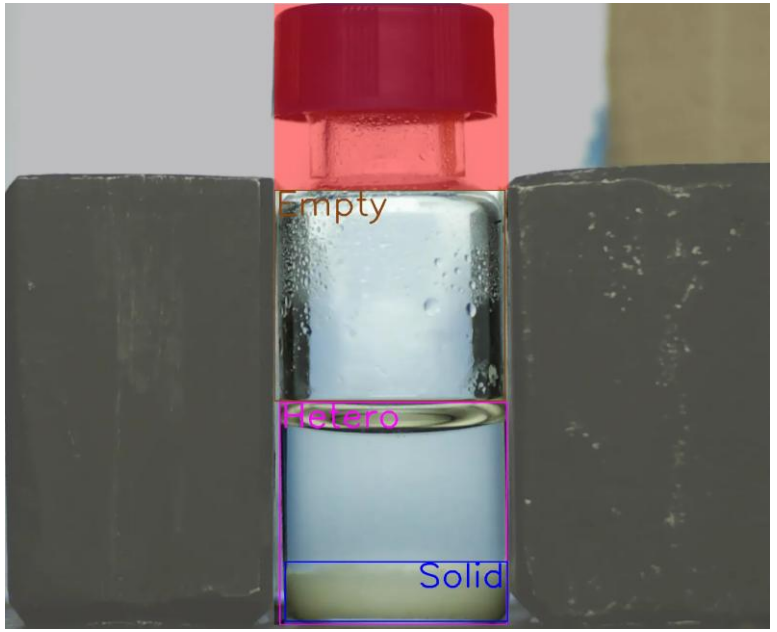
All wells are used.
Please swap the plate.

Error-handling by *The Engineer*



Sample not detected.
Please investigate.

Decision-making by *The Scientist*



Should I continue the experiment or stop it?

Don’t underestimate the pain of manual equipment and graphical user interfaces (GUIs)

APIs

Q: Does it have an application programming interface (API)? (**Python + open-source**)

Please provide an in-person or virtual demo of the API in use

Sample transfer

Q: Do samples have unobstructed access from top and/or side?

Please provide a recording of the sample loading and unloading process

The community faces significant challenges due to an absence of standard choices in tools



A universal convention supplies all of maintainability, clarity, consistency, and a foundation for good programming habits too.

What it doesn’t do is insist that you follow it against your will.

That’s Python!

Tim Peters
Zen of Python

As SDL technologies become more mature and widespread, the role of the researcher may shift toward translating the results from autonomous experimentation into scientific understanding

Tom et al. *Chem Rev* (2024)





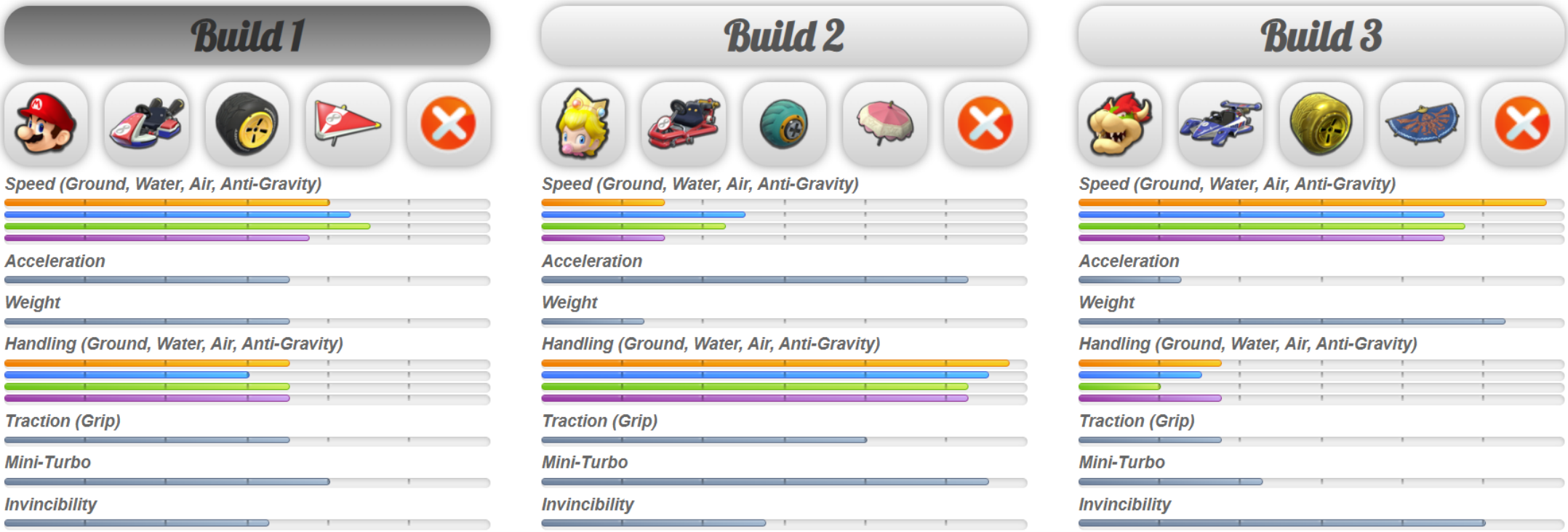
B

Mario

A OK

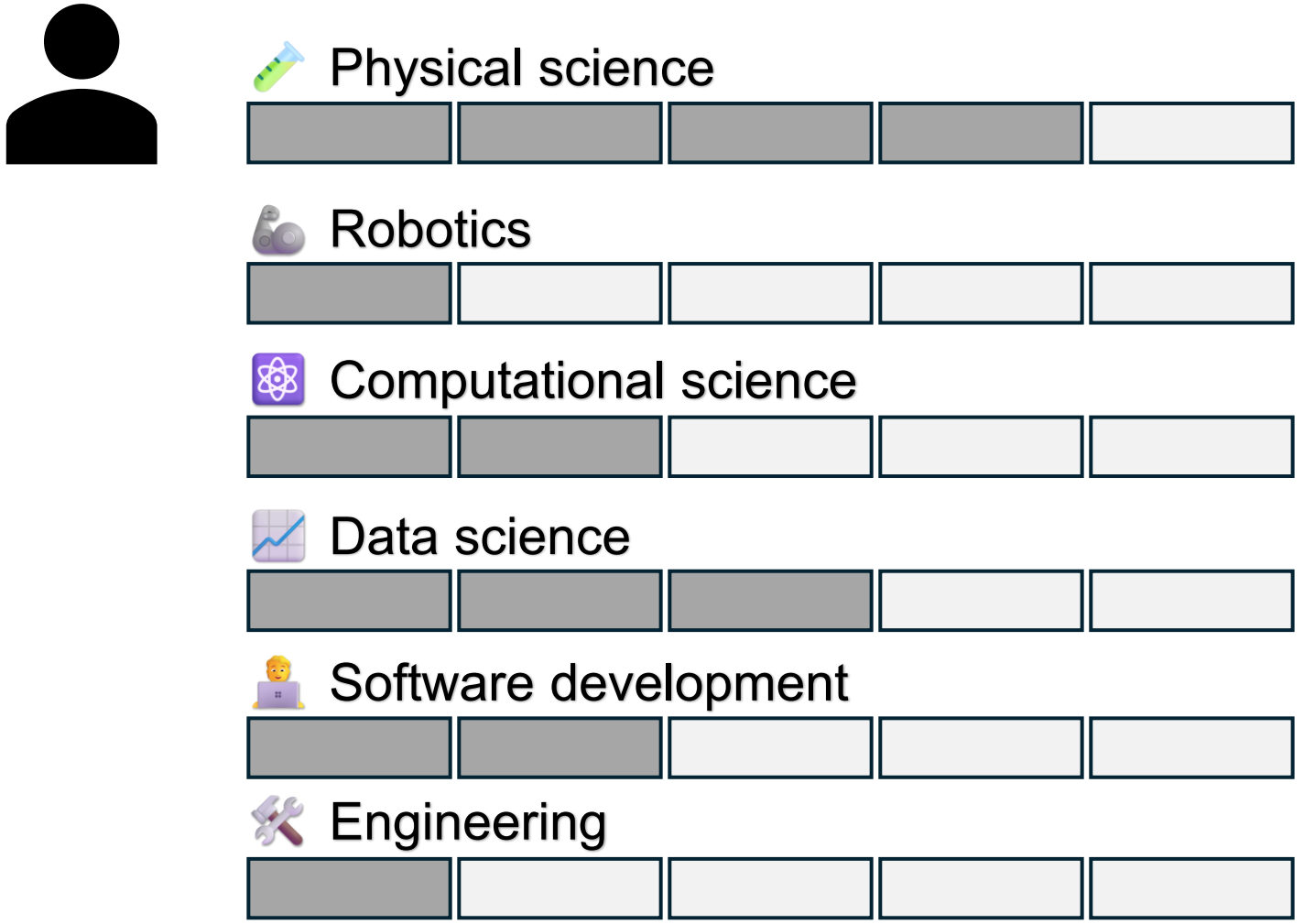

Speed
Acceleration
Weight
Handling
Grip
B
Flame Rider
-/+
A OK

Would you hire Mario, Baby Peach, or Bowser?



https://mk8dxbuilder.com/

Highly multidisciplinary skillsets are required to build self-driving labs



Someone/something needs to “glue” cross-disciplinary skillsets



こんにちわ

Aryl Halide

Bonjour

Pose Estimate





Hallo

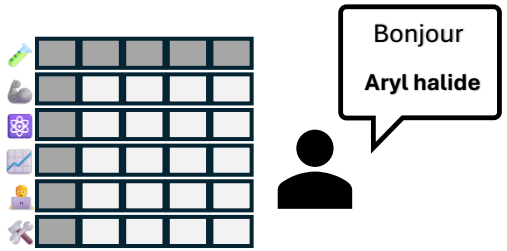
Projector
Augmented Wave

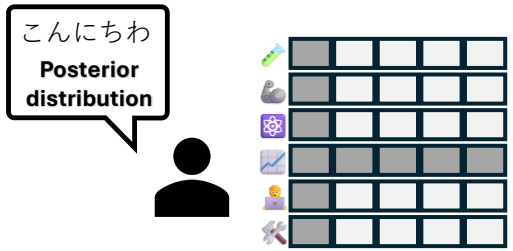
안녕하세요

Posterior
distribution



Effective cross-disciplinary communication requires shared vocabulary and conceptual understanding





DEFINITION

An aromatic compound where one or more hydrogen atoms in the aromatic ring are replaced by halogen atoms

A conditional probability that updates the prior probability with information from the likelihood via Bayes' rule

FORMULA

$c1c(X)ccc(X)c1$

$C_6H_{6-n}X_n$

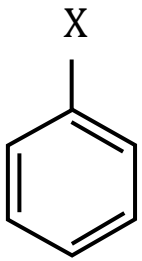
observation location

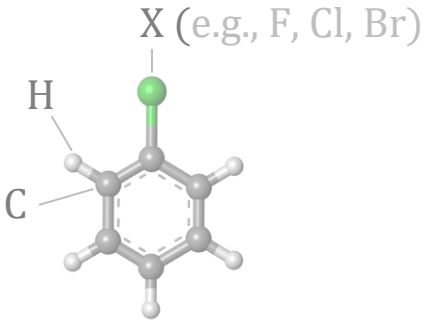
objective function $f(x)$

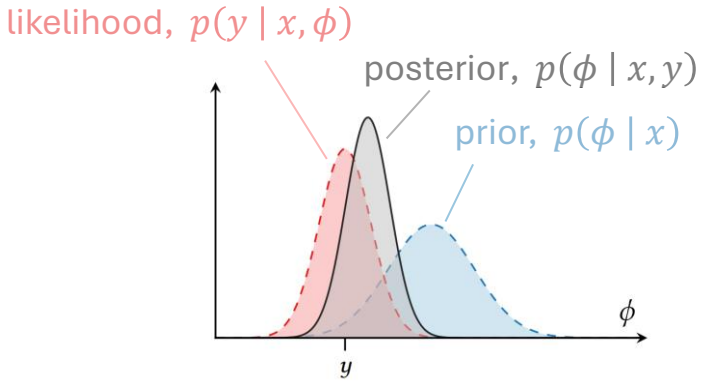
measured value

$$p(\phi \mid x, y) = \frac{p(x, y \mid \phi)}{p(x, y)} p(\phi)$$

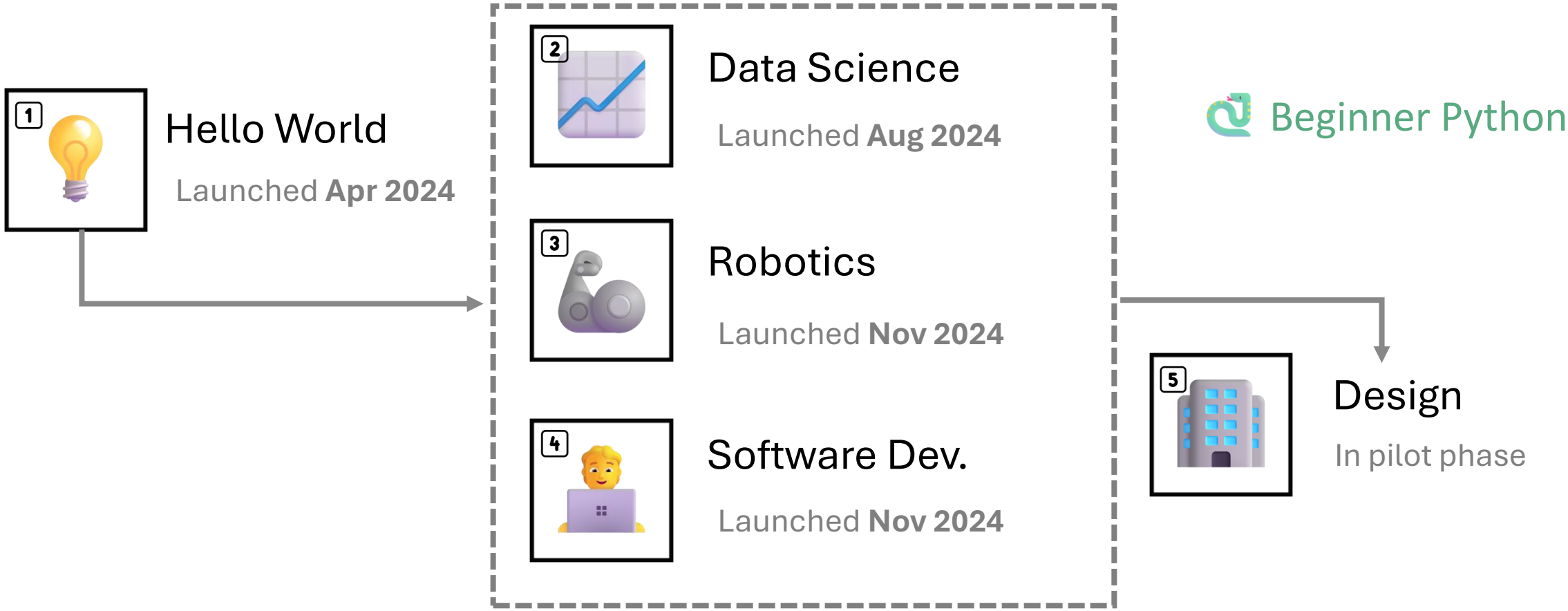
VISUALIZATION





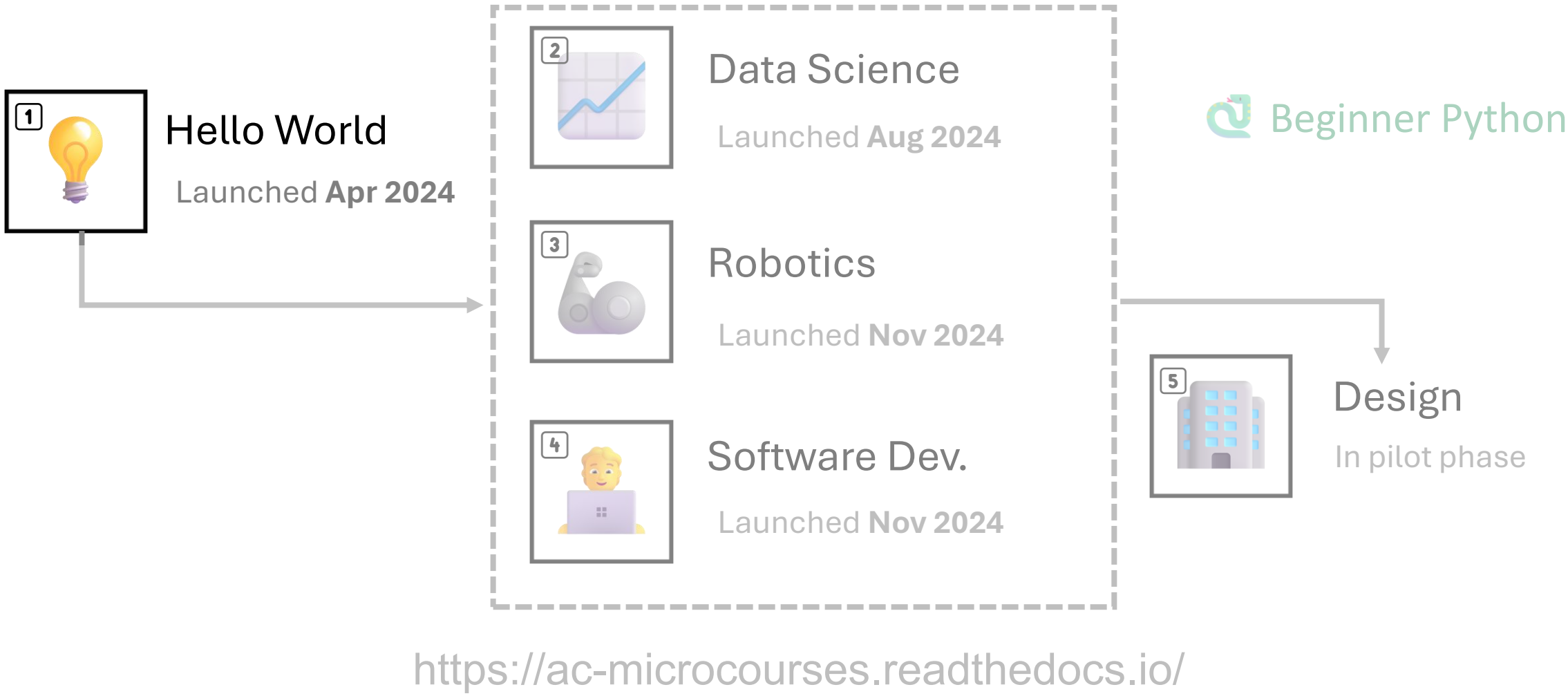


AC Microcourses help provide the “glue” for autonomous labs



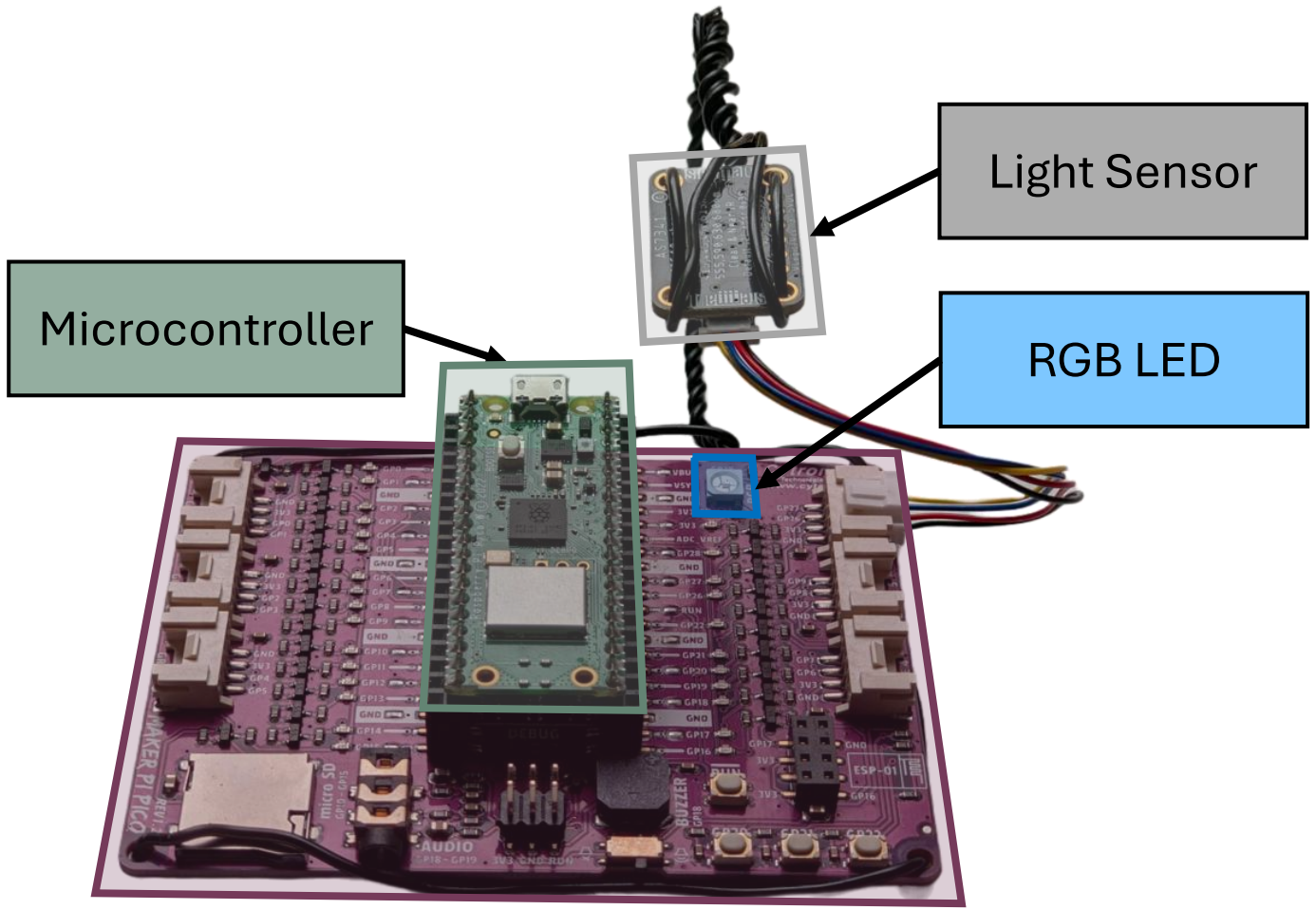
https://ac-microcourses.readthedocs.io/

AC Microcourses help provide the “glue” for autonomous labs

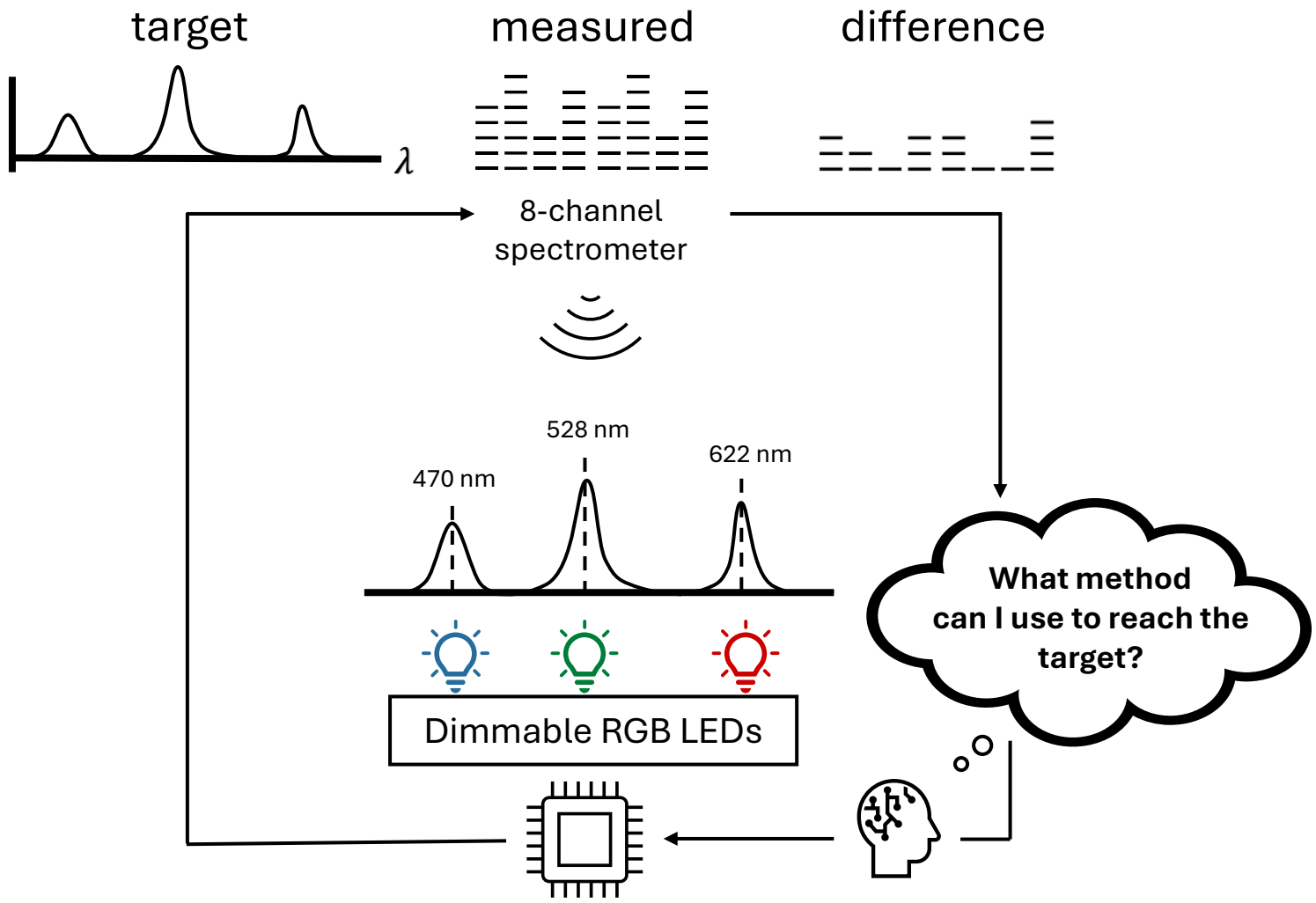


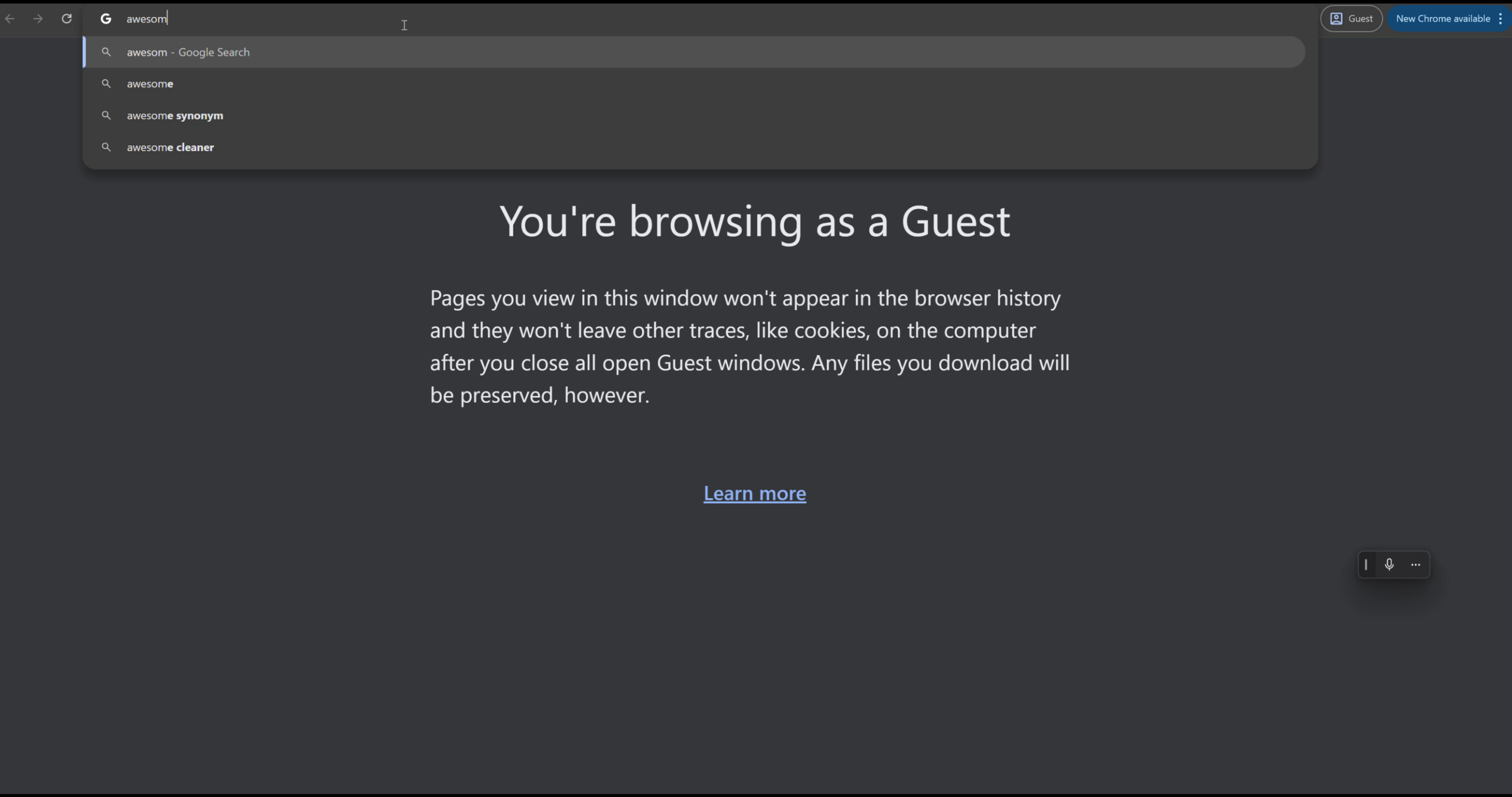
https://ac-microcourses.readthedocs.io/

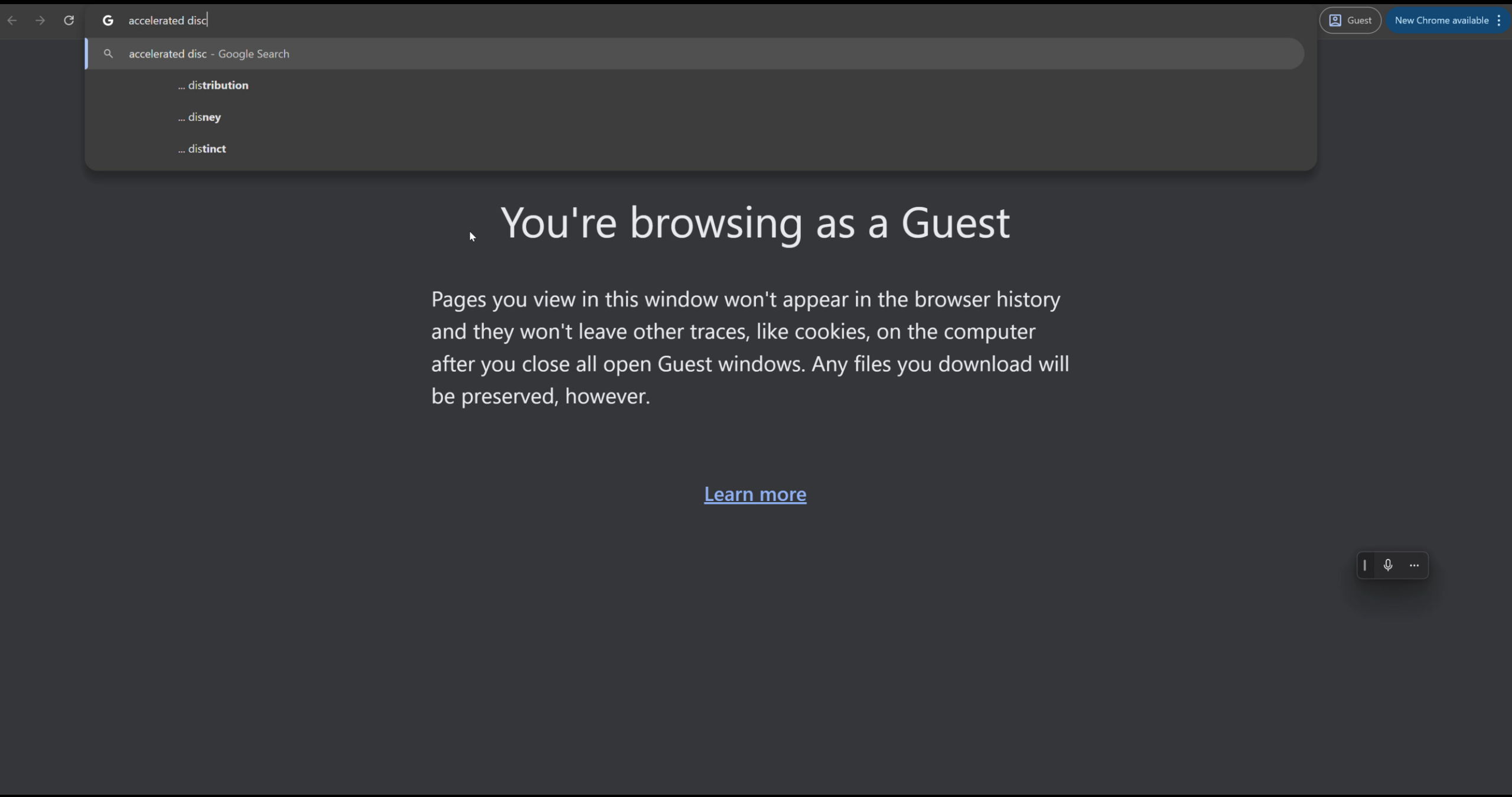
Design and build a self-driving lab demo from scratch



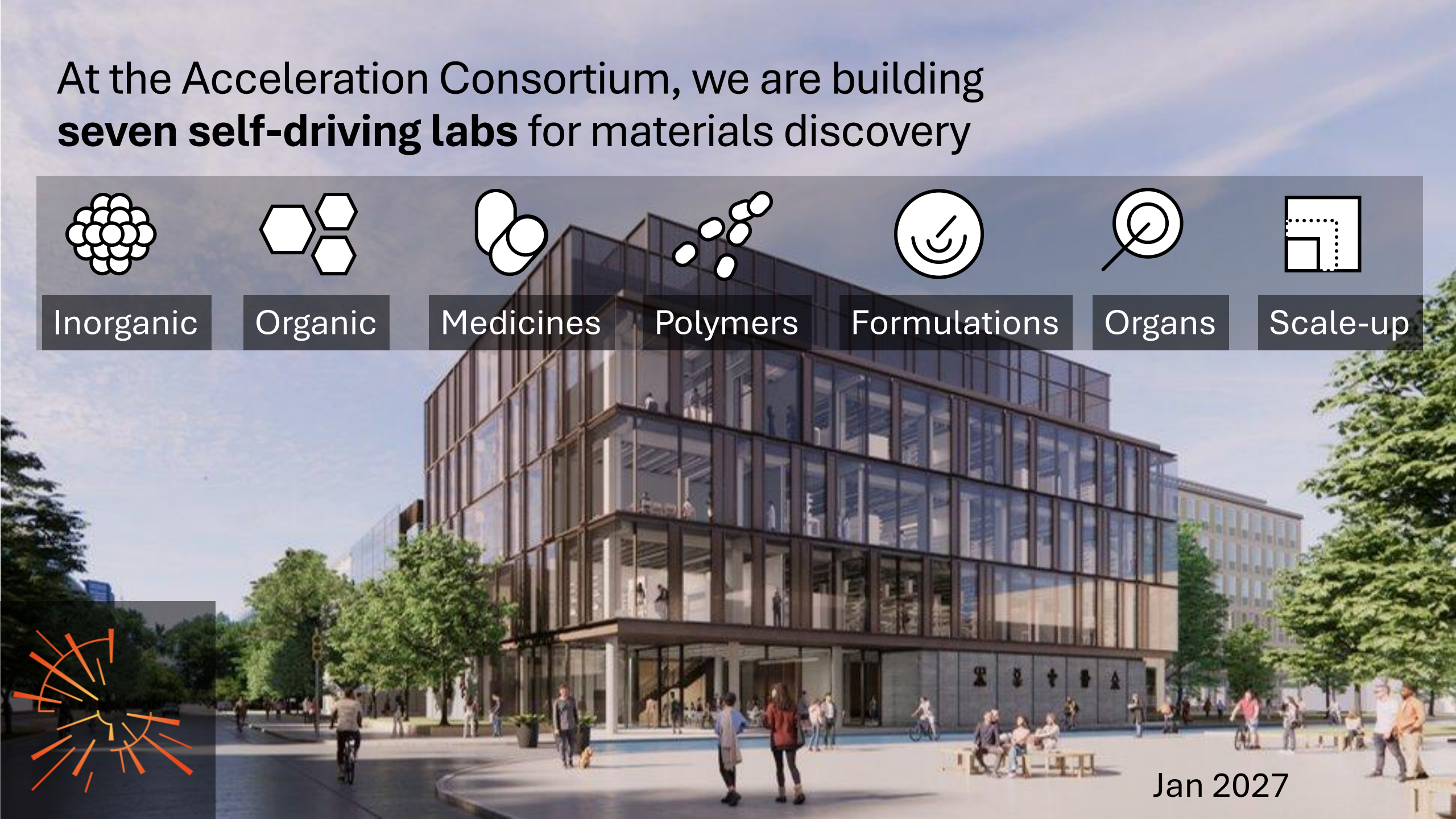
Design and build a self-driving lab demo from scratch




awesom
awesom - Google Search
awesome
awesome synonym
awesome cleaner
Guest
New Chrome available
You're browsing as a Guest
Pages you view in this window won't appear in the browser history and they won't leave other traces, like cookies, on the computer after you close all open Guest windows. Any files you download will be preserved, however.
Learn more


accelerated disc
accelerated disc - Google Search
... distribution
... disney
... distinct
Guest
New Chrome available
You're browsing as a Guest
Pages you view in this window won't appear in the browser history and they won't leave other traces, like cookies, on the computer after you close all open Guest windows. Any files you download will be preserved, however.
Learn more

At the Acceleration Consortium, we are building **seven self-driving labs** for materials discovery





None of us should be like the fisherman who thinks he has been fishing all day when in reality he has spent most of his time getting to and from the water, eating lunch, and fussing with his equipment.

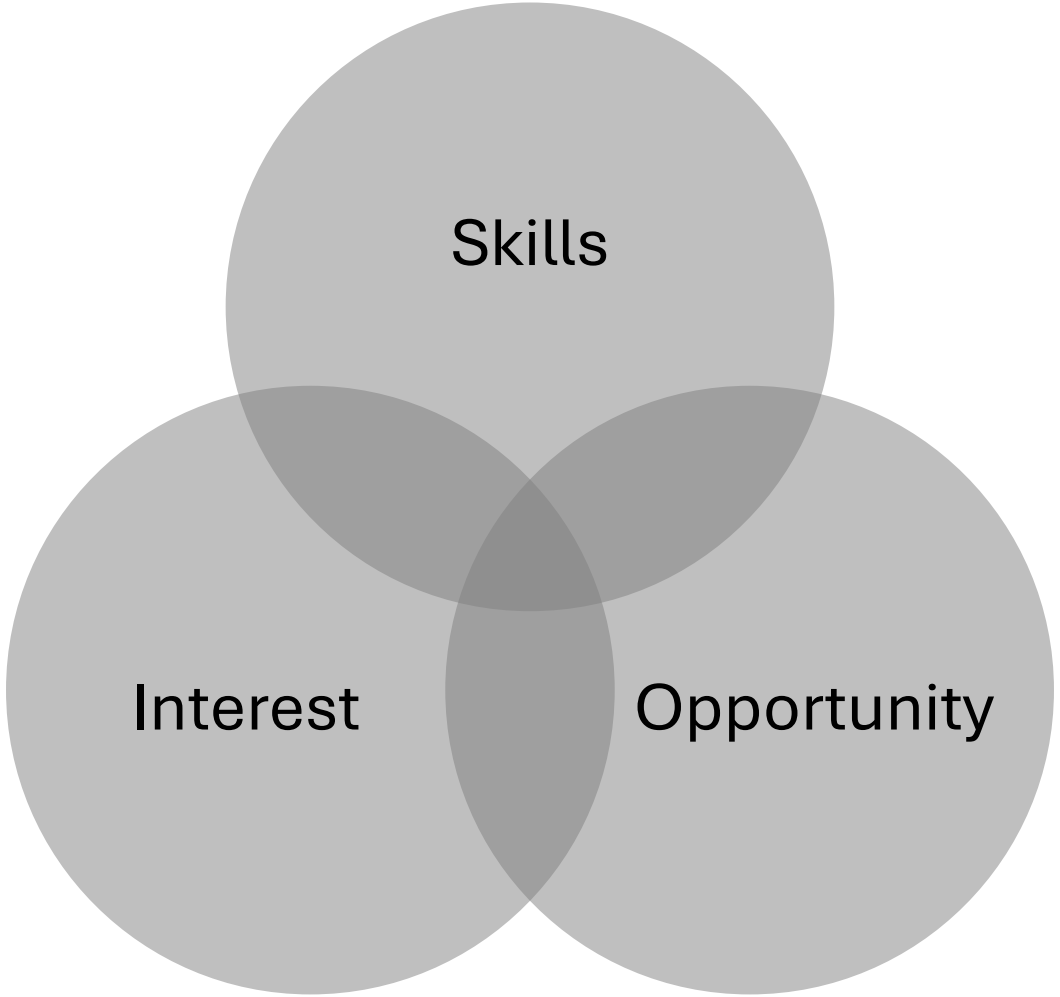
Fishing success is related to how long you have your line in the water, not to how long you are away from the apartment. Some fishermen are away from home for twelve hours and have their line in the water for ten hours. Other fishermen are away from home for twelve hours and have their line in the water for only two hours.

This last type may wonder why they do not have the same success as others.

- Dallin H. Oaks

Don’t let building a self-driving lab stop you from doing real science

Your next steps are an intersection of your skills, interests, and the opportunities of your situation



What are your next steps?

[Discuss in groups of 2-3]



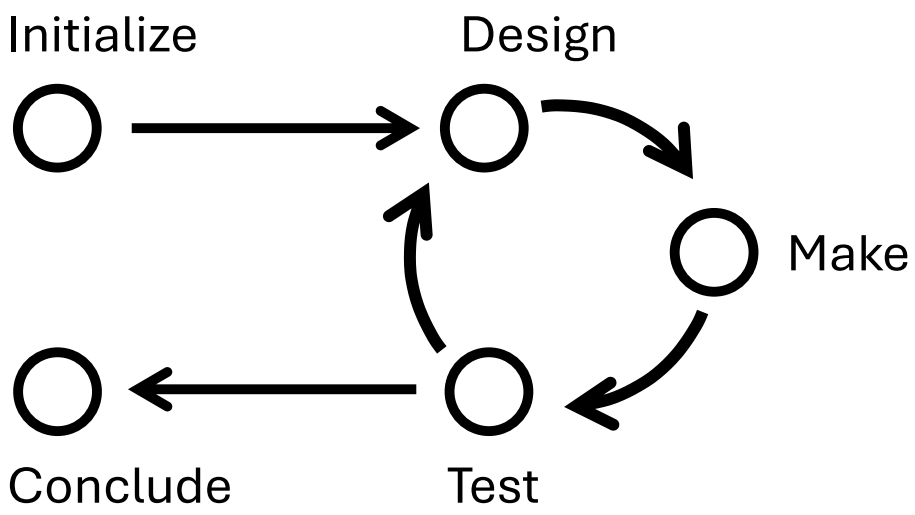
Wishing you success
with your discoveries

https://linktr.ee/acceleration_consortium

Acceleration Consortium

sterling.baird@utoronto.ca

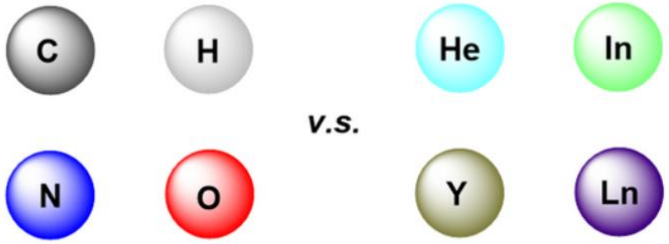
So, you want to build a self-driving lab?



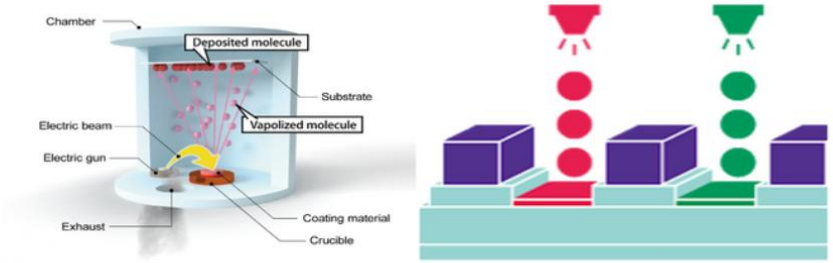
1. AI
   to maximize value
2. Automation
   to minimize burden
3. Integration
   to scale quickly



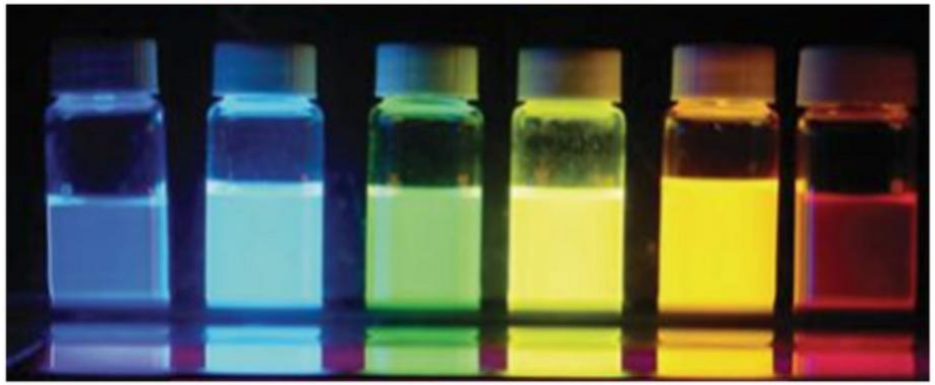
Investigating new molecular gain materials for organic solid-state lasers is important



Sustainability



Low-cost fabrication

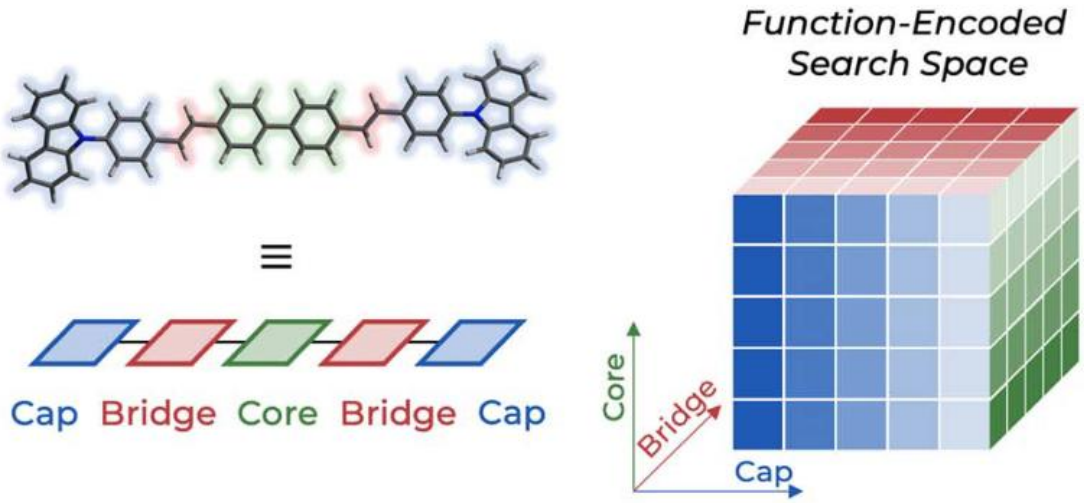


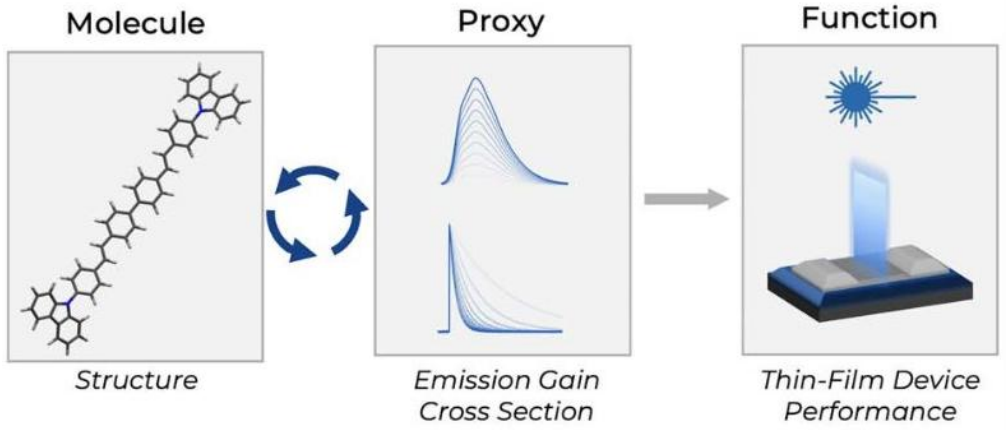
Tunability



Versatile substrates

Initialize by searching through discrete and enumerated space

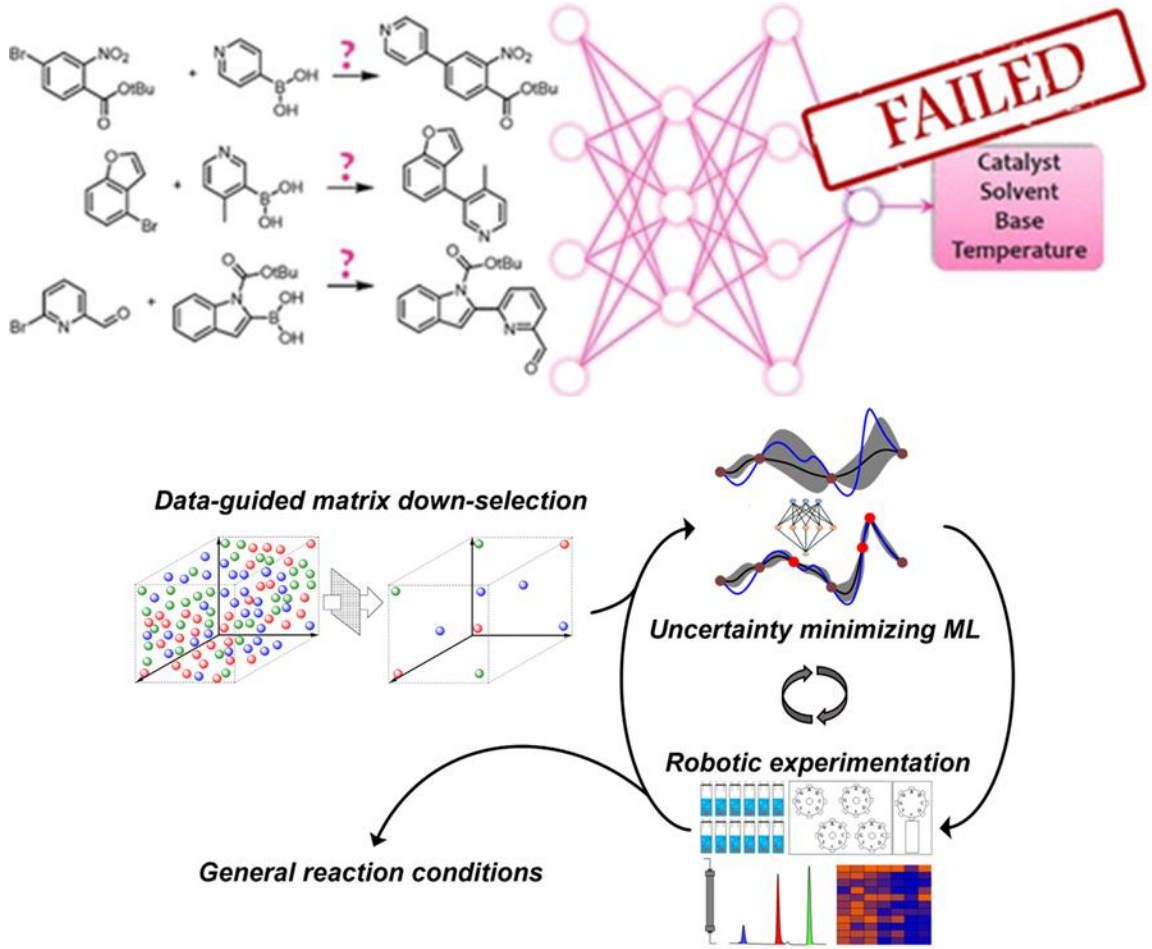




Initialize: define the search space

- Discover new compounds by operating on the **chemical space of molecules.**
- AI-planner searches through **discrete and enumerated** space based on Cap/Bridge/Core fragments.
- Maximize the **emission gain cross section**, which serves as a proxy.

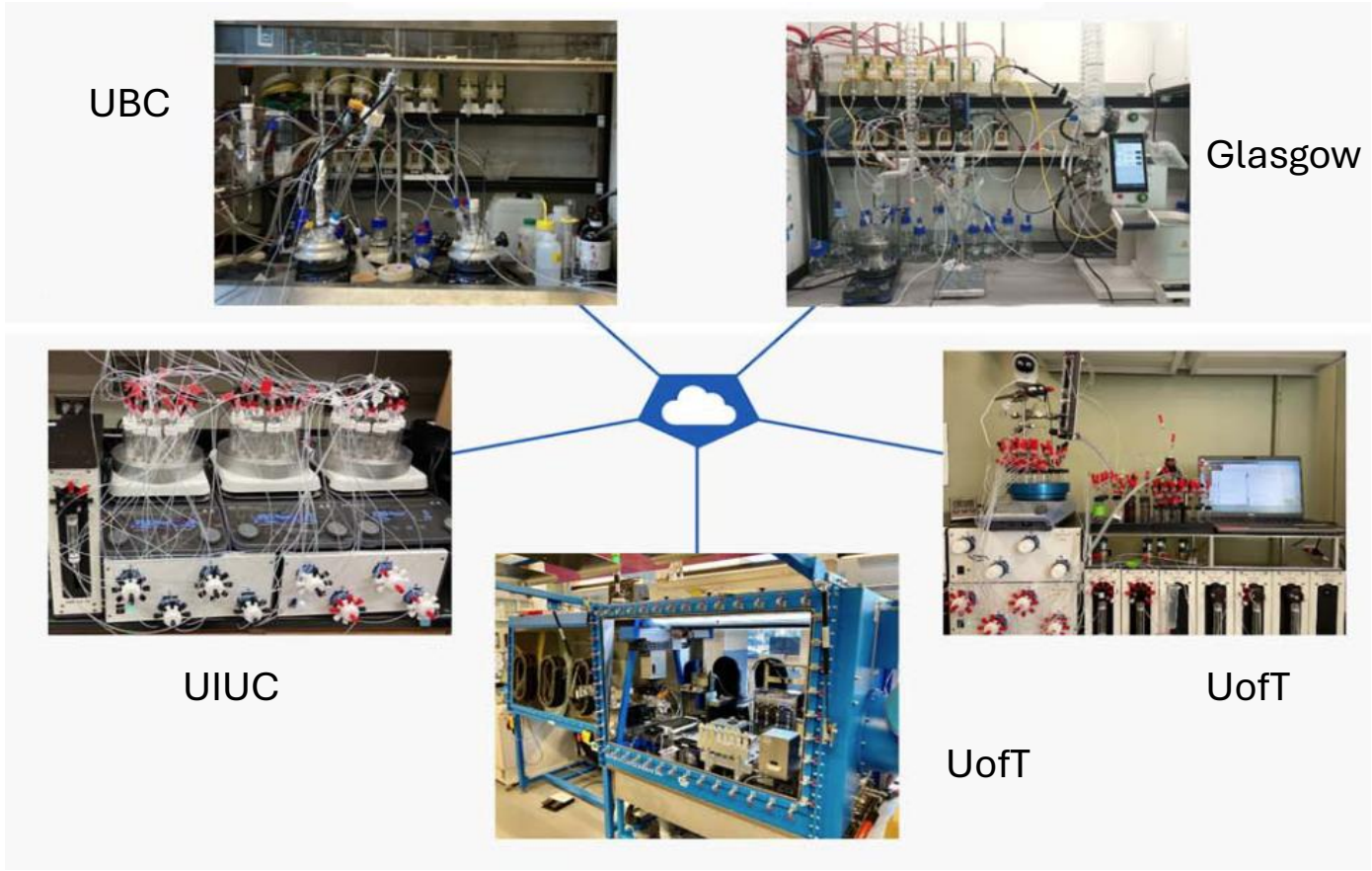
Design by using Suzuki-Miyaura coupling reactions to link fragments together



Design: design the experiment

- Suzuki-Miyaura coupling reactions to link fragments together.
- Well established method for coupling fragments, but poor yields
- Determining the best general reaction conditions required **separate SDL optimization**
- 21-46% improvement

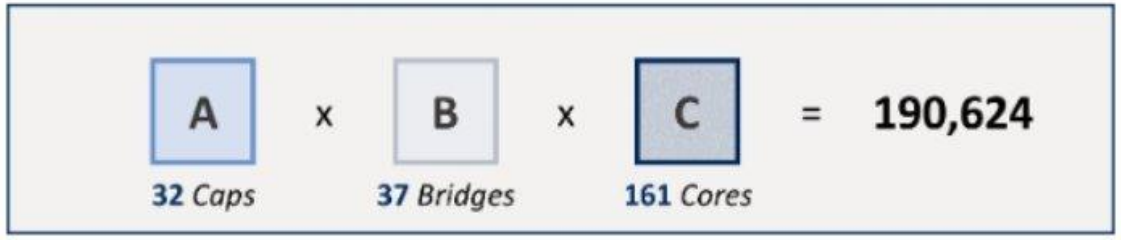
Make by automating synthesis

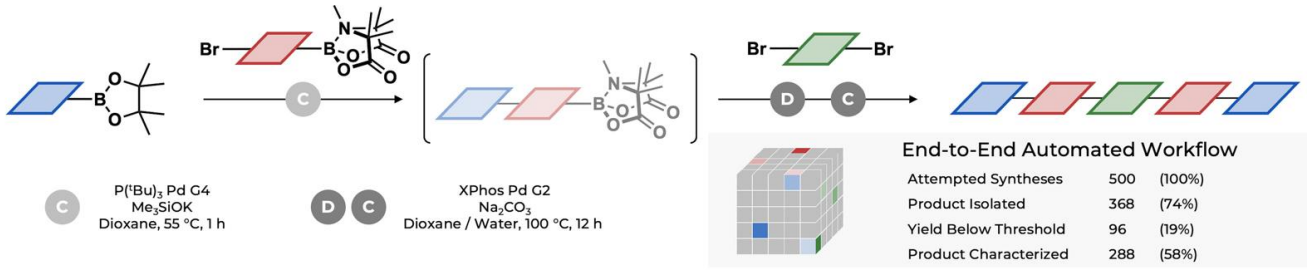


Make: automate synthesis

- Synthesize LEGO-like **building blocks** *(UBC and UGlasgow)*.
- Two-step, one-pot Suzuki–Miyaura couplings allows for **batch reactors** connected by flow-systems *(UIUC and UofT)*.

Make by automating synthesis



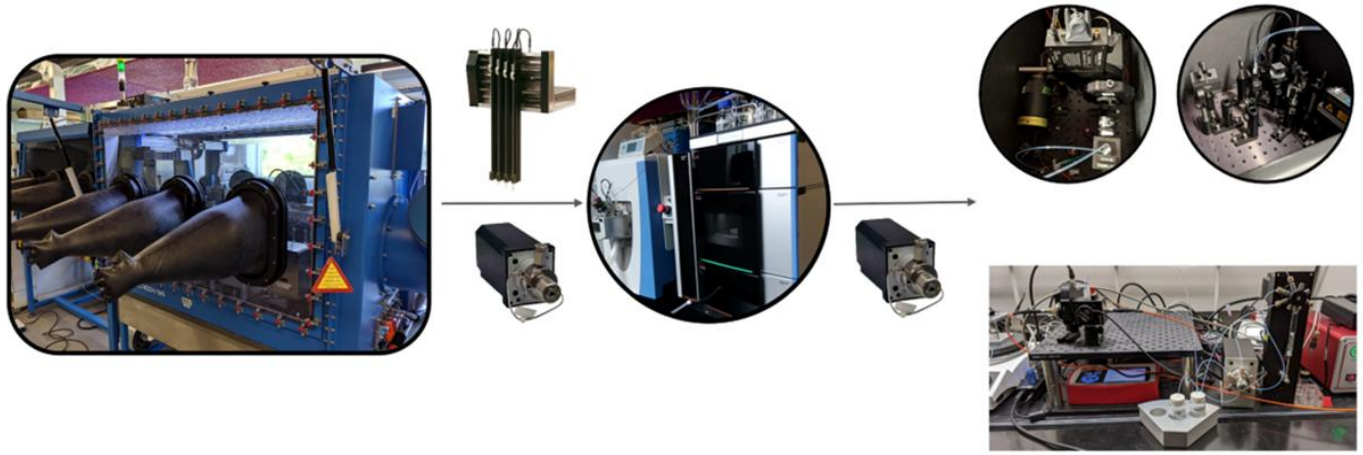


- Product was crashed out with hexane/diethyl ether and dried
- Purified just enough to detect HPLC-MS

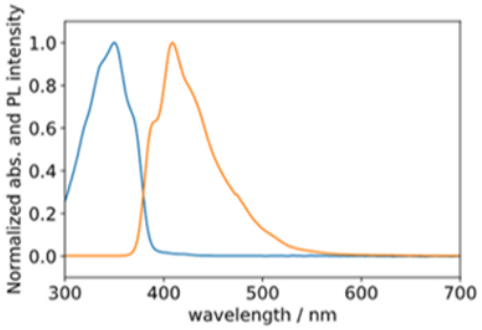
Make: one-pot reaction

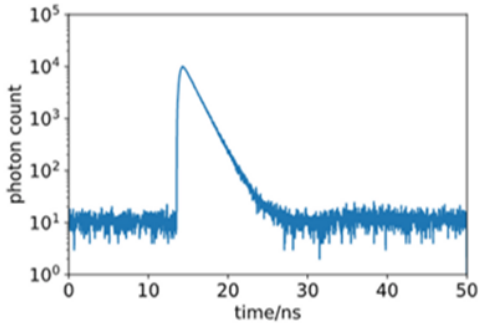
- Synthesize LEGO-like **building blocks** *(UBC and UGlasgow)*.
- Two-step, one-pot Suzuki–Miyaura couplings allows for **batch reactors** connected by flow-systems *(UIUC and UofT)*.

Test by automating characterization and analysis



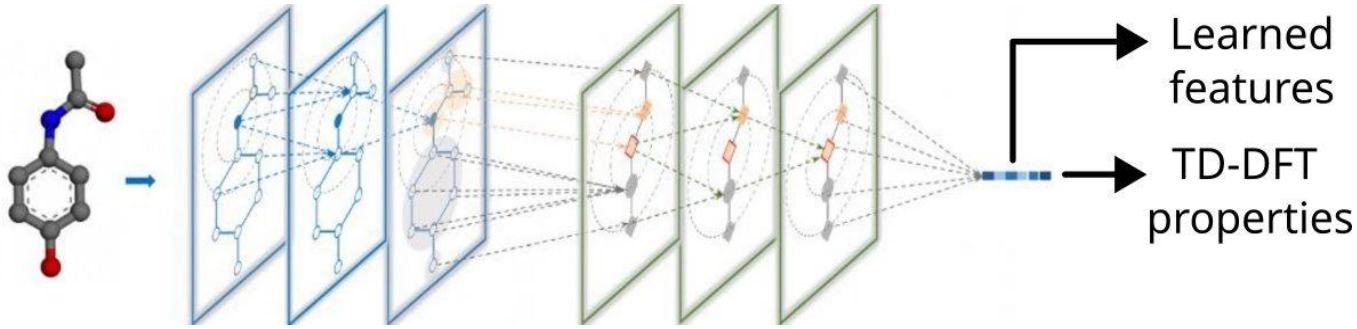
Test: Characterize and test

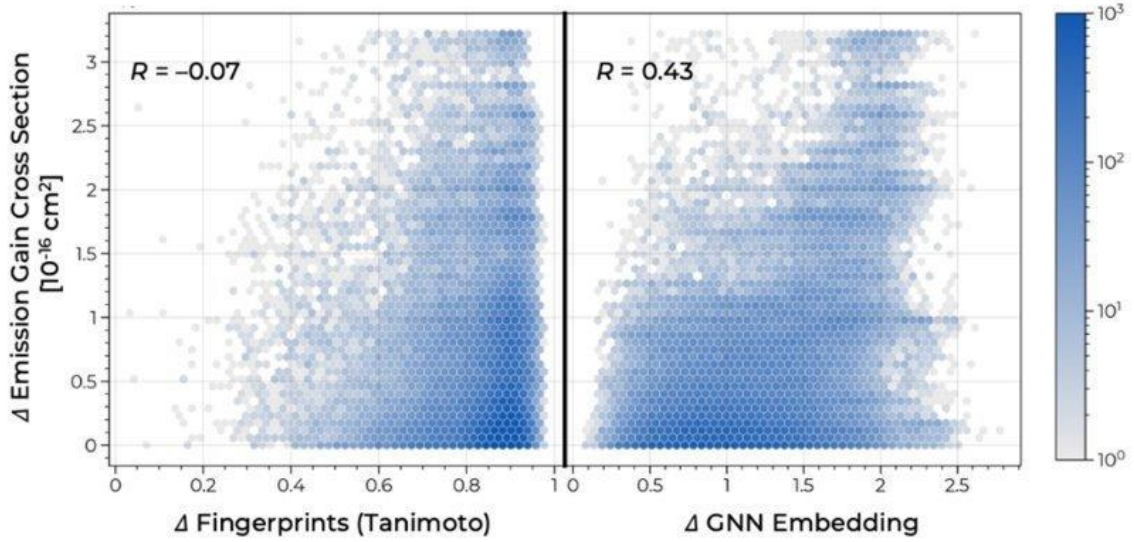




Automated purification
and spectroscopy data in
~30 min

Analyze by interpreting results and feeding it back into the model

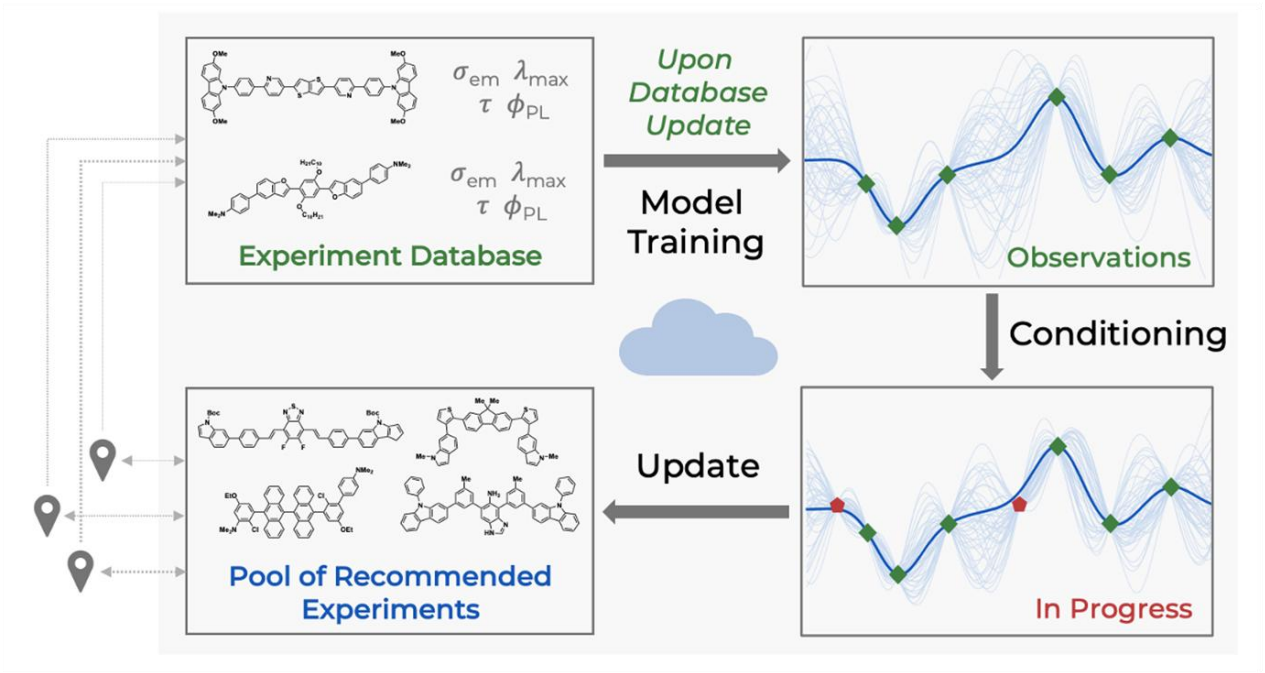




Analyze: find a good model for the experiment

- Transfer learning and incorporating **DFT** knowledge into model
- Features "learned" from *ab initio* data

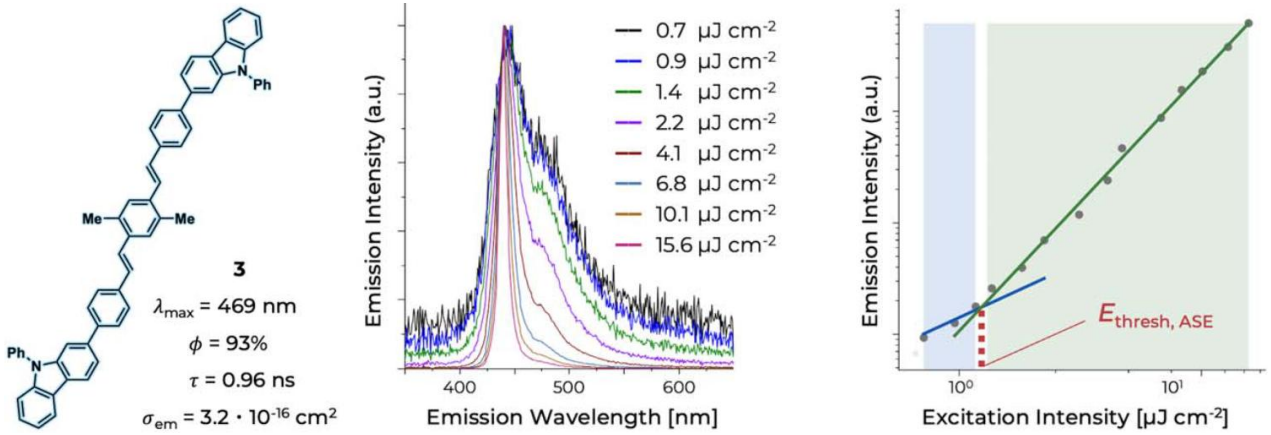
Analyze by interpreting results and feeding it back into the model



Analyze: find a good model for the experiment

- Transfer learning and incorporating **DFT** knowledge into model
- Features "learned" from *ab initio* data

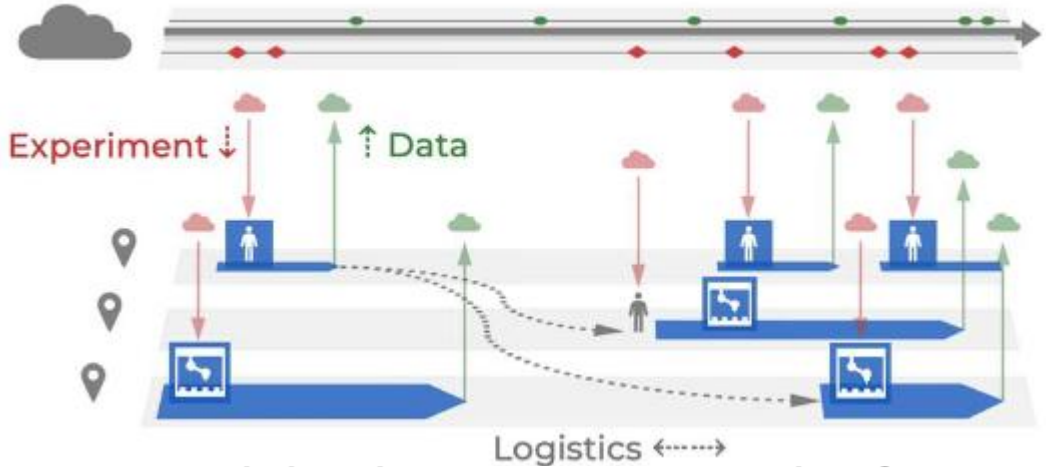
Conclude

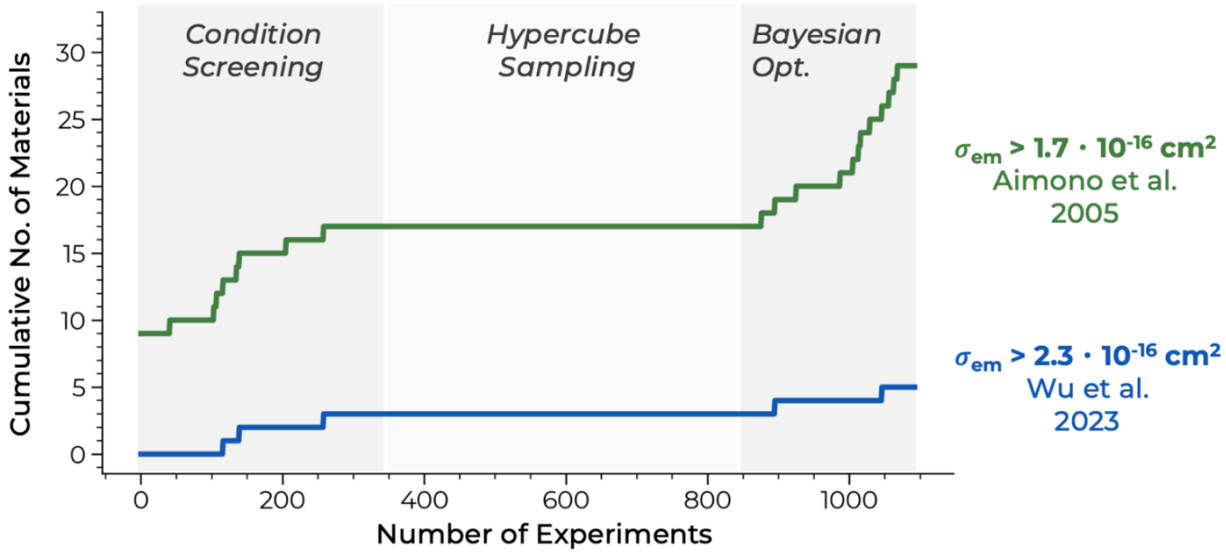


Conclude:

- Emission gain cross section used as a proxy for lasing performance.
- Spectroscopic measurements automated after HPLC-MS purification.
- Photoluminescence quantum yield and lifetime measured (48%). Partial characterization for weak emitters or low yields (52%).
- Thin-film device evaluation after closed-loop discovery shows amplified spontaneous emission (ASE).

Experiments used delocalized and asynchronous orchestration; this allows them to achieve better lasers faster





Details of the orchestration

- Central planner handles experiment scheduling, model updating, and AI-driven suggestions.
- Uses "fantasy values" for incomplete experiments, or experiments assigned to another lab.
- Theoretically shown to be more efficient than synchronous batch optimization.